Project Omega: Liberty University Rocketry Team

Team 32 Project Technical Report to the 2026 IREC

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Omega is Liberty Rocketry's 2026 IREC 10k COTS launch vehicle. Omega uses an AeroTech M1845 solid rocket motor to carry a 2 kg galvanometric laser engraver payload to a target apogee of 3,048 m AGL. The vehicle uses a carbon fiber airframe, fiberglass nosecone, three trapezoidal fins, a boat tail, redundant recovery avionics, and a dual-deployment recovery system. A drogue parachute deploys near apogee, and the main parachute deploys at 365 m AGL to recover the vehicle at approximately 6.0 m/s. Testing included recovery deployment, charge sizing, electronics verification, GPS testing, payload recovery testing, and flight testing to reduce mission risk.

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I. Nomenclature

A_m	=	Area of main parachute (m^2)
b	=	Width of sample (m)
c_r	=	Fin root chord (m)
c_t	=	Fin tip chord (m)
C_d	=	Coefficient of drag
C_k	=	Coefficient of shock
C_P	=	Center of pressure
d	=	Local indentation/Depth of sample (m)
d_{max}	=	Maximum indentation during impact (m)
D_m	=	Main parachute diameter (m)
D_t	=	Drogue parachute diameter (m)
E	=	Modulus of elasticity (Pa)
E^*	=	Effective elastic modulus (Pa)
E_1, E_2	=	Elastic moduli of contacting materials (Pa)
F	=	Force (N)
$F(d)$	=	Contact force as a function of indentation (N)
F_{shear}	=	Maximum shear pin strength (N)
F_{shock}	=	Shock force (N)
g	=	Gravitational acceleration of earth (m/s^2)
G	=	Shear modulus of fin material (Pa)
h_f	=	Fin height (m)
HDT	=	Heat Deflection Temperature
I	=	Second moment of area (m^4)
L	=	Total length / Length of sample (m)
M	=	Mass / Impacting mass (kg)
M_a	=	Molar mass of air (kg/mol)
P	=	Pressure (kPa)
P_0	=	Sea level pressure (kPa)
R	=	Ideal gas constant / Effective radius of curvature
R_f	=	Fin aspect ratio
s	=	Speed of sound (m/s)
S_0	=	Surface area (m^2)
t_f	=	Fin thickness (m)
T_{abs}	=	Absolute temperature (K)
U_0	=	Velocity immediately before impact (m/s)
v	=	Velocity (m/s)
v_d	=	Desired descent speed (m/s)
V	=	Shear force per bolt (N)
v_f	=	Fin flutter velocity (m/s)
v_{impact}	=	Desired impact speed (m/s)
w_0	=	Central deflection / Flexural displacement (m)
$\overline{x}_{CG}$	=	Center of gravity of entire rocket (m)
$\overline{x}_{CP}$	=	Distance from nose tip to Center of Pressure (m)
ε_f	=	Flexural strain
ν_1, ν_2	=	Poisson's ratios of contacting materials
ρ_a	=	Density of air (kg/m^3)
σ_f	=	Flexural stress (Pa)
τ	=	Shear stress (Pa)

II. Executive Summary

The Liberty Rocketry team representing Liberty University designed and tested the launch vehicle OMEGA for the 2026 International Rocket Engineering Competition. Omega is entered in the 10,000 ft Commercial Off-The-Shelf category and is powered by a COTS M1845 solid rocket motor. The primary mission objective is to safely transport an operational 2 kg galvanometric laser engraver payload to the target apogee of 3,048 m (10,000 ft) AGL and recover the launch vehicle in a reusable condition.

Project Omega measures 3.169 m in length, has an outer diameter of 0.157 m, and has a final pad liftoff mass of 24.70 kg. The selected M1845 motor produces a peak thrust of 2,424 N and burns for approximately 4.5 seconds. OpenRocket simulations predict that the vehicle will reach the target apogee while maintaining an acceptable stability margin during ascent. The airframe, fins, nosecone, and boat tail were selected and analyzed to balance structural integrity, manufacturability, commercial availability, and aerodynamic performance.

Omega uses a dual-deployment, dual-separation recovery system. A drogue parachute deploys at apogee with a one-second delay to stabilize the lower airframe during descent. The main parachute deploys from the forward airframe at 365 m AGL, or 335 m AGL in high-wind conditions, to achieve a nominal terminal descent velocity of 6.0 m/s. Deployment sequencing is controlled by redundant altimeters with external arming pull-pins and independent recovery electronics to support safe operation and vehicle recovery.

The onboard payload is an in-flight operational galvanometric laser engraver. The system uses a 4 W laser module, mirror steering, servos, and onboard electronics to direct the laser and engrave a design on the target material during flight. Testing performed for Omega included recovery drop testing, decoupling charge sizing, recovery electronics verification, GPS testing, payload recovery system testing, as well as two test launches to verify all systems. These tests were used to validate subsystem function, reduce flight risk, and support safe operation at the 2026 IREC.

III. Introduction

A. Project Background and Team Structure

Liberty Rocketry is a student-led engineering group based at Liberty University. The organization operates at the capstone level, where students apply mechanical, systems, and aerospace engineering principles to design, manufacture, test, and launch competition-scale rockets. For the 2026 IREC, Liberty Rocketry is competing in the 10,000 ft Commercial Off-The-Shelf category with Project Omega. Omega uses a COTS M1845 solid rocket motor and is designed to reach an apogee of 3,048 m while carrying an operational 2 kg payload.

For this design cycle, the team was divided into five technical sub-teams: Propulsion, Aero-structures, Avionics, Recovery, and Payload. Two additional support teams, Project Management and Media, assisted with system integration, scheduling, budgeting, documentation, and outreach. Each sub-team was responsible for its own design process, manufacturing plan, testing, and technical documentation. The leadership structure for Project Omega is shown in Fig. 1.

The project was completed by student members with support from faculty advisors, Liberty University facilities, commercial suppliers, and outside manufacturing resources when student equipment could not support a specific process. Commercial vendors were used for COTS flight hardware, materials, and several specialized components. The design, analysis, assembly, integration, and most testing were completed by the Liberty Rocketry student team.

B. Mission Objective

Liberty Rocketry's mission for the 2026 IREC is for Omega to reach an altitude of 3,048 m AGL while safely carrying and recovering an operational 2 kg payload. The payload is a galvanometric laser engraver that uses a laser module, mirror steering system, servos, and onboard electronics to engrave a design during flight. This payload was selected to demonstrate a controlled electromechanical system inside a launch vehicle environment.

The main engineering goal for Project Omega was to increase reliability through redundancy, safe arming procedures, and stronger subsystem integration. Omega uses redundant flight computers, external arming pull-pins, a dual-deployment recovery system, and a dual-separation layout. These design choices were made to support range safety, protect the launch vehicle during recovery, and improve the chance of recovering the rocket in reusable condition.

Omega also uses a carbon fiber airframe, fiberglass nosecone, optimized trapezoidal fins, and a boat tail to balance structural integrity, manufacturability, weight, and aerodynamic performance. Each subsystem was selected or designed to support the full mission instead of optimizing one part of the vehicle at the cost of another.

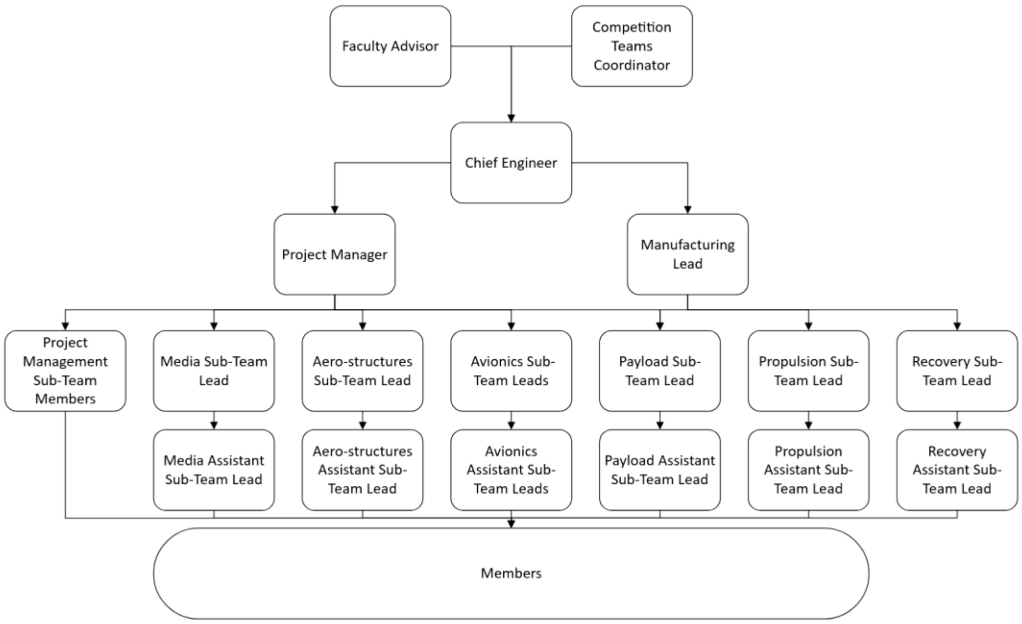

Fig. 1: Leadership structure flowchart

C. Project Management and Report Preparation

Project Omega was managed through subsystem leads, recurring design reviews, shared documentation, and integration meetings. Each sub-team developed its own requirements, design decisions, analysis, and test plan. The project management team tracked schedule, budget, purchasing, integration risks, and reported progress throughout the design cycle.

This technical report was prepared by the student leadership team using a section-based writing process. Each subsystem lead wrote the section related to their area of responsibility. The chief engineer and project manager then reviewed the report for formatting, technical consistency, and alignment with IREC requirements. Supporting analyses, test results, checklists, risk assessments, and engineering drawings are included in the appendices to document the design process and support the final launch configuration.

D. Mission Requirements

To be deemed a fully successful mission, the rocket and team must satisfy the following criteria:

i. *Pad Operations:* Execute all arming and pre-flight procedures safely using the mechanical arming sequence without requiring system resets.

ii. *Ascent:* Maintain structural integrity through maximum dynamic pressure, motor burnout, and the full ascent phase.

iii. *Payload:* Initiate the laser engraving sequence during flight and maintain optical alignment long enough to produce a discernible engraving on the target material.

iv. *Recovery:* Execute the dual-deployment sequence, deploy the drogue parachute near apogee, deploy the main parachute at 365 m AGL, and land the rocket at approximately 6.0 m/s so the vehicle remains reusable.

IV. System Architecture Review

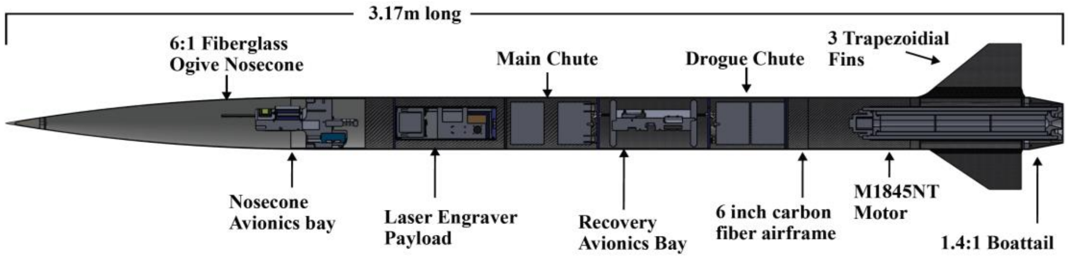

Fig. 2: Integrated cutaway view of the Omega launch vehicle showing the major subsystem layout.

A. Integrated Vehicle Overview

Project Omega is a 10,000 ft COTS launch vehicle designed to carry an operational 2 kg payload to apogee and recover the rocket in reusable condition. The vehicle is 3.169 m long, has an outer diameter of 0.157 m, and uses a COTS M1845 solid rocket motor for propulsion. Omega was designed as an integrated system in which the aerostructures, payload, propulsion, avionics, and recovery subsystems work together to support safe ascent, stable flight, payload operation, and controlled recovery. Figure 2 shows the integrated layout of the launch vehicle and identifies the major subsystem locations.

The forward end of the vehicle contains the fiberglass nosecone, which provides the aerodynamic leading surface and allows radio-frequency-transparent placement of hardware where needed. Below the nosecone is the forward airframe, which houses the payload system. The payload is a galvanometric laser engraver that includes the laser module, mirror steering system, power supply, and onboard electronics. This section was designed to secure the payload during launch while allowing it to remain functional during flight.

Below the payload section is the recovery bay and avionics section. Omega uses redundant flight computers and recovery electronics to monitor the flight and control deployment events. External mechanical arming pull-pins are used to improve pad safety and allow the system to remain in a safe configuration until final launch preparation. The avionics system interfaces with the recovery subsystem, which uses a dual-deployment, dual-separation layout. A drogue parachute is deployed near apogee to stabilize the lower airframe, and the main parachute is deployed at low altitude to reduce landing speed.

The aft section of the vehicle contains the motor tube, fin can, and boat tail. These structures transfer thrust loads, support the propulsion system, and provide aerodynamic stability during ascent. The carbon fiber airframe and fin structure were selected to provide a high strength-to-weight ratio while keeping vehicle mass low enough to meet the target apogee. The boat tail reduces base drag and provides a smooth transition to the aft diameter. Together, these subsystems form a launch vehicle architecture that balances structural integrity, manufacturability, aerodynamic performance, system reliability, and range safety.

B. Aero-structures Subsystems

1. Subsystem Overview

This subsystem consists of the nosecone, the airframe, the fin can, and the boat tail. These components all influence the aerodynamic characterization of the rocket. This is especially important for IREC, where teams aim to design rockets that reach an apogee as close as possible to the target of 3,048 m (10,000 ft). The following sections will outline our team's approach to and execution of the design and manufacturing of the aero-structures subsystem.

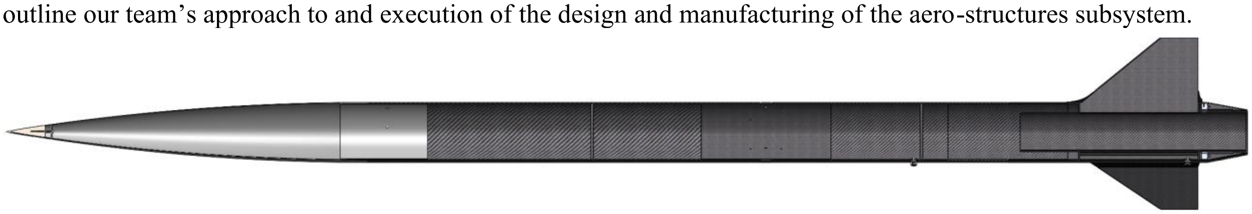

Fig. 3: Aero-structures components

Before proceeding further, several key terms must be defined. For the aero-structure section, constraints can be understood as either quantifiable or Boolean objectives that must be met per competition rules or regulations. Design objectives are targets that are self-imposed and may not be quantifiable, listed in Appendix G, Table G-1. Using this list, the team can assess how each variable affects the relevant constraints and design objectives. Throughout this section, notations such as "(C.1)" indicate compliance with Constraint 1 of Table G-1. Commercial availability is defined as a measure of both the number of manufacturers producing the desired item and the reliability of its supply chain, including consistent stock availability and low risk of discontinuation. Because the manufacturing plan was established in advance and must be executed accordingly, options with low commercial availability are less desirable. Cost, when used as a decision-matrix factor, is defined such that lower financial cost is preferred; accordingly, lower-cost options receive higher scores, with 5 out of 5 representing the most favorable case.

2. Aero-Structures Design Process

Nosecone

The primary variables to consider for the nosecone design process are the aerodynamic shape, fineness ratio, the base outer diameter, and the material. The base diameter of the nosecone is dependent on the size of the airframe to create a continuous surface. The airframe inner diameter was chosen to be 152.4 mm (6 in), the reasoning for which will be discussed under the airframe section. To ensure that there are minimal discontinuities in the rocket body, the nosecone should match the size of the airframe as closely as possible. Therefore, the team only considered 152.4 mm base diameter nosecones.

The next nosecone variable considered was geometry. Because the team does not have the capability to fabricate a custom nosecone, the selection was constrained to commercially available options. The primary candidate geometries identified were tangent ogive, Von Kármán (Haack), conical, and elliptical. On page 2396 of Ref. [1], the author determined the drag coefficients of a common nosecone shape. Since Omega is not expected to exceed the transonic regime, with the most recent simulations predicting a peak velocity of approximately Mach 0.81, the aerodynamic differences between these geometries are relatively small compared to those in supersonic flight. Therefore, the final selection was based not only on aerodynamic performance, but also on commercial availability and procurement cost (D.2). Based on Table G-4 under Appendix G Decision Matrices, the team selected the tangent ogive geometry, which has the best weighted score. In addition to its strong overall balance of drag, availability, and cost, the team has prior experience working with this shape and is therefore more confident in its implementation. The continuously curved surface of the tangent ogive also reduces the likelihood of accidental damage during manufacturing. Tangent ogive tips are less prone to chipping or developing a lip than sharper profiles.

One important note regarding nosecone shape was that the team had been purchasing tangent ogive nosecones for the past few years while assuming it was a von Kármán (Haack) shape. When the nosecone is small, the difference between geometries becomes harder to distinguish and can easily be attributed to vendor tolerance. The team tried to confirm the geometry of the COTS nosecone with the vendor directly, but was unsuccessful. Therefore, an experiment was conducted in which the nosecone was laid on graph paper, and its profile was vertically projected onto the paper using an orthogonal ruler set. The curvature was then converted into x-y coordinates according to the orientation of the nosecone. The sampled points yielded an $R^2 = 0.98$ agreement with a tangent ogive and an $R^2 = 0.93$ agreement with a von Kármán profile. This discrepancy highlighted the importance of verification. Since both von Kármán and tangent ogive profiles are well-suited for subsonic flights, this difference has had no significant impact on prior vehicle performance.

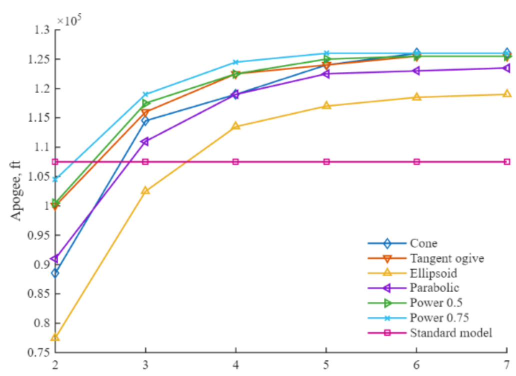

Fig. 4: Apogee as a function of fineness ratio (graph recreated from Apogee Components)

Another aspect of nosecone geometry is the fineness ratio, which is the ratio of the nosecone length to its base diameter. As the fineness ratio increases, the body becomes slimmer, which generally reduces pressure or form drag, but the increased surface area also raises skin-friction drag. For the team's subsonic flight, literature indicated that altitude increases with increasing fineness ratio and remains nearly constant for fineness ratios beyond 6:1 [2,3].

Richard Nakka also recommended fineness ratios between 5:1 and 6:1 for higher-velocity flights [4]. As a result, the team considered a 6:1 ratio favorable because of its low drag and because earlier exposure with a 6:1 nosecone may be useful for higher-speed rockets in future years. Practical availability from the team's trusted vendor also made 6:1 possible. Therefore, the team selected a 6:1 tangent ogive nosecone, see Table G-5 under Appendix G Decision Matrices.

The final consideration for the nosecone was material selection. The primary design objective was radio frequency (RF) transparency. For the rocket and ground station to communicate with the Global Positioning System (GPS) and onboard camera system, some portion of the rocket must be RF transparent. Because the airframe was selected to be carbon fiber, as discussed under the airframe section, the nosecone was required to be constructed from an RF-transparent material (D.4), which eliminated carbon fiber as a candidate. The material also needs to provide sufficient structural performance, low weight, and high commercial availability. As shown in Table G-6 under Appendix G Decision Matrices, filament-wound fiberglass received the highest weighted score because it provides excellent RF transparency, strong structural performance, and good weight efficiency among common COTS nosecone materials. Plastic ranked lower primarily due to reduced structural performance. Kraft paper, while inexpensive and widely available, does not provide sufficient strength in comparison. Therefore, filament-wound fiberglass was selected as the most appropriate nosecone material, see Table G-6 under Appendix G Decision Matrices.

The nosecone must be able to withstand normal operating conditions (C.7), so a structural analysis was performed on the tip. A conservative aerodynamic load bound was obtained from the flight simulation using the maximum acceleration, velocity, and launch-site atmospheric conditions. The dynamic pressure equation yields a pressure of 40.8 kPa. Omega's frontal area is 0.01824 m^2 which resulted in a conservative frontal aerodynamic force of approximately 0.74 kN. The dynamic pressure is still well below the estimated landing impact load of 22.4 kN, which will be derived below. Therefore, landing impact was selected as the governing axial load case for the nosecone tip. For the landing impact, the normal elastic contact relation was obtained from Ref. [5]. The maximum indentation during impact was then derived from the Hertz contact law by equating the initial kinetic energy of the impacting body to the elastic strain energy stored during contact. The governing equations are given below.

$$F = \frac{4}{3} E^* \sqrt{R}\, d^{3/2} \tag{1}$$

$$E^* = \left(\frac{1 - \nu_1^2}{E_1} + \frac{1 - \nu_2^2}{E_2} \right)^{-1} \tag{2}$$

$$\frac{1}{2} m_{top} U_0^2 = \int_0^{d_{\max}} F(d)\, dd \tag{3}$$

Substituting Eq. (1) into Eq. (3) and integrating gives:

$$d_{\max} = \left(\frac{15 m U_0^2}{16 E^* \sqrt{R}} \right)^{2/5} \tag{4}$$

This model assumes elastic deformation, localized contact, smooth surfaces, and normal impact between the nosecone tip and the ground. Plastic deformation, energy dissipation, and terrain variation were neglected. Therefore, the result should be interpreted as an order-of-magnitude estimate rather than an exact prediction. A representative dry sandy terrain condition was used, as informed by the IREC Rule & Requirement. Under these assumptions, the peak landing force was estimated to be approximately 22.4 kN. This value was then used in a finite element analysis (FEA) simulation, in Appendix G, to evaluate the structural integrity of the nosecone under landing impact conditions. The result showed that an extremely sharp aluminum tip will experience deformation or dent if the worst-case situation does occur, but it is not a guaranteed structural failure because the physical metal tip purchased is more rounded compared to the simulated model. However, this simulation result is supported by a small new dent discovered on the nosecone tip post-subscale launch. As a result, the team will bring a replaceable nosecone tip to the competition if excess damage does occur.

Airframe Design Process

The airframe is defined by its inner diameter, thickness, and material. Regarding the inner diameter of the rocket, it is constrained by C.17. To fit the hypotenuse of the 10cm-by-10cm CubeSat form factor, the inner diameter of the airframe would have to be at least 141.42 mm. Currently, the team does not have the capability to manufacture custom airframes, so the team is limited to options that are commercially available. A sensitivity analysis using OpenRocket was conducted to demonstrate that when everything else is held constant, increasing the airframe inner diameter decreases the maximum apogee due to the additional weight. To maximize the apogee of any given configuration, the inner diameter of the airframe should be minimized. However, because commercially available airframes come in discrete sizes, the team selected the smallest commercially available size of 152.4 mm (6 inches) inner diameter. A decision matrix is not applicable here because diameter is a minimization process constrained only by a lower bound and commercial availability.

After surveying the websites of various common vendors, the team chose to analyze a 1.8288 mm (0.072 in) thickness for carbon fiber and a 4.318 mm (0.170 in) thickness for G12 fiberglass because they were the most commercially available. The structural safety of this decision was verified along with the material discussion below.

The team conducted finite element analysis (FEA) using ANSYS Static to compare the performance of G12 fiberglass against carbon fiber airframes with their given thicknesses. Four critical load conditions were investigated. Firstly, the airframe must withstand the internal pressure of 246.15 kPa during black power ignition during recovery. Secondly, the airframe must withstand the maximum flight acceleration predicted by OpenRocket. Thirdly, the airframe must not rupture upon hitting the ground during landing. Lastly, the fins must be able to withstand a man of 1334.5 N (300 lbf) standing on the fins because some judges at competition test fins in this manner. For the accelerated-flight load case, all stresses remained below ultimate compressive strength, with the fin can governing at a factor of safety of 4.44. The structural analysis demonstrated that all primary airframe components satisfied these requirements (C.7). Complete stress and deformation results for the flight and landing-impact cases are provided in Appendix G under FEA.

In addition to the simulation results favoring carbon fiber over fiberglass, material density was evaluated due to its significant impact on apogee. Literature indicates that G12 fiberglass [6] has a higher density, lower strength, and is typically available only in greater thicknesses compared to carbon fiber [7]. Although carbon fiber incurs higher costs, the allocated budget permitted its use (D.2). Therefore, carbon fiber was selected as the airframe material, see Table G-7 under Appendix G Decision Matrices.

Fin Design Process

The fins, the most customizable aspect of the aero-structures subsystem, represent a unique opportunity in a COTS category to showcase a thorough engineering optimization process. Herein, our team will describe the optimization process that was developed to guide the design of the fin geometry for Omega. The six variables considered include the fin can, number of fins, chord lengths, fin height, fin thickness, and material. These six independent variables influence apogee, stability margin, and structural integrity, forming a six-dimensional design-of-experiments (DOE) space. For clarity and simplicity, the design space was decomposed into a series of two- and three-dimensional projections. Although this representation reduces completeness, the team was able to extract relevant information by doing sensitivity analysis between variables that have the most impact on outcomes. Relevant decision matrices can be found under Table G-8 to G-10 under Appendix G Decision Matrices.

The first variable considered was fin cant angle. Based on literature review, the team found that fin cant angle has a negligible effect on apogee and static stability for the standard rockets [8]; although canted fins can induce roll, which stabilizes the rocket in strong wind [8], this advantage was not considered useful because IREC would not make the competition team launch their rockets under extreme wind anyway. Fin cant would also increase manufacturing difficulty, tighten alignment tolerances, and complicate simulations. Therefore, the team selected a 0 deg fin cant angle for the design.

The second variable considered was the number of fins. The team evaluated both common three-fin and four-fin configurations in OpenRocket over a range of fin heights to determine their effects on apogee and stability margin sampled at 0.3 Mach. As shown in Fig. 5, the three-fin configuration produced a higher apogee at every fin height considered, while the four-fin configuration consistently provided a greater stability margin. The results also show that both configurations can satisfy the apogee (D.3) through appropriate fin-height selection, and that both can meet the minimum stability requirement (C.2-4).

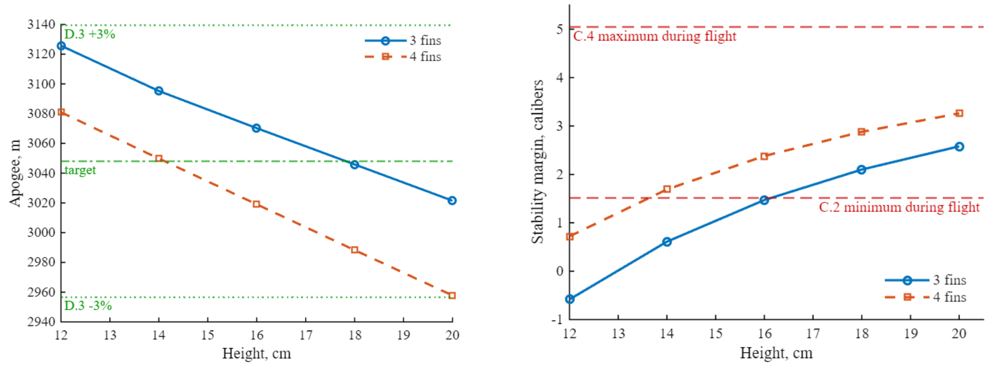

Fig. 5: Apogee (left) and stability (right) response for 3 and 4 fins

Because the three-fin configuration satisfied the design requirements while preserving higher apogee, the team selected the three-fin design. The team also has 3 years of experience with three-fin configurations, which reduces manufacturing uncertainty and guarantees quality. Relative to the four-fin alternative, the three-fin configuration requires less material, fewer fin fillets, and less fabrication time, thereby reducing both cost and manufacturing complexity.

The third variable was the fin shape. The primary goal was to achieve a high stability margin caliber while minimizing aerodynamic drag. Computational Fluid Dynamics (CFD) results comparing fin shapes of equal surface area showed that the trapezoidal shape provided ideal overall balances among the candidate geometries [9]. At a 5° angle of attack, which better represents real flight conditions, the trapezoidal fin produced a drag force of 54.55 N while also offering very high stability [9]. Additionally, this geometry provides a favorable balance between structural simplicity and ease of manufacturing. Therefore, the trapezoidal configuration was selected.

The next variables to consider are those directly related to the side profile of the fin, such as height and chord length. To understand how these variables relate, the team designed the optimization feature included in OpenRocket to aid us this year in determining the optimum fin geometry. The team conducted a design of experiments (DOE) to generate many variable permutations to test together to infer their relationships.

Based on Ref. [10], the most sensitive variables for apogee control are fin height and thickness, and thickness is tied to material selection, which will be determined next. Fin height also dominates approximately 90% when it comes to stability margin changes. The fin height and chord length were the primary variables studied. The DOE consisted of these two parameters at 5 levels each. Stability and apogee were recorded for each combination. Using MATLAB and OpenRocket, the team completed the DOE and summarized the findings in the following figure.

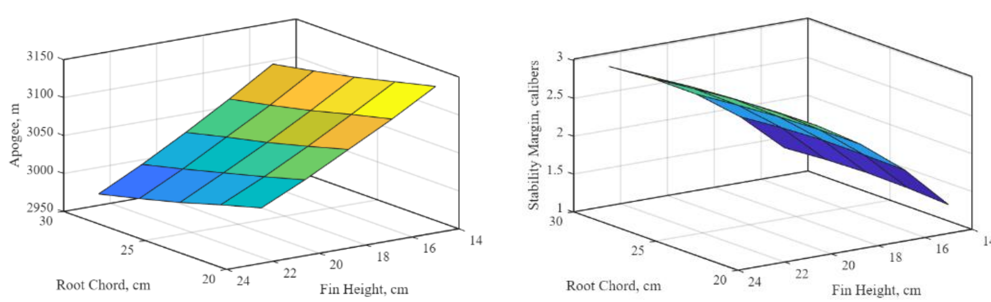

Fig. 6: Apogee (left) and stability (right) sensitivity plot

The results provide critical insights into the relationships between the parameters and the response, allowing the team to draw meaningful conclusions about the fin design. One such conclusion was that fin height has a greater effect on the apogee and stability than the chord length does, as supported by Ref. [10]. Therefore, the team decided to use a chord length of 11.81in (30cm) for all subsequent design iterations.

Using the above information, the team applied the goal seek function in the OpenRocket optimization tool to bring the apogee closer to 10,000 ft while maintaining an acceptable stability margin. This allowed the team to identify a fin design that met all requirements and was straightforward to manufacture. As shown in Fig. 7, a fin height of 17.5 cm (C.1) met the apogee target while providing a high stability margin (C.2-4) at 0.3 Mach.

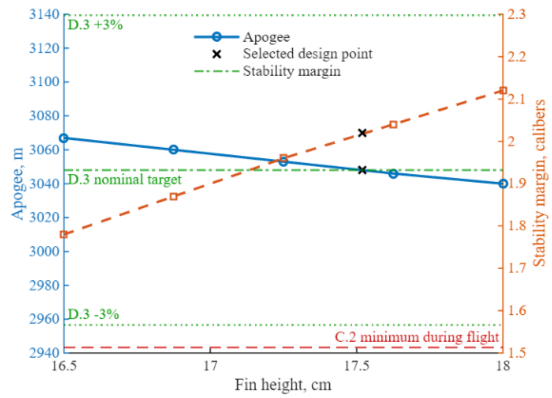

Fig. 7: Chosen fin height with apogee and stability margin

The structural strength of the fins depends on both thickness and material. Based on commercially available parts, the team needed to choose between carbon fiber fins with a thickness of 4.76 mm and G12 fiberglass fins with a thickness of 6.35 mm. Ideally, the team would select the thinnest fins capable of withstanding the required loading cases. Finite element analysis (FEA) was used to compare these options. For the carbon fiber fins with 4.76 mm thickness, the factor of safety was 1.78, whereas the G12 fiberglass fins with 6.35 mm thickness had a factor of safety of 0.28. Therefore, the fiberglass fins were not selected (C.7).

The downside of carbon fiber is the additional cost, but it presents a significant advantage by being lighter yet with stronger mechanical properties. The team chose to use carbon fiber fins because of their reduced weight and additional strength benefits while remaining within the team's budget.

Fin Flutter Analysis

Staying below the fin flutter velocity at all points throughout the flight is critical to maintaining a stable flight. The team calculated this function out of specific fin geometry using MATLAB and equations found in "How to Calculate Fin Flutter Speed" by Zachary Howard, an article in the July 2011 Issue of Peak of Flight magazine by Apogee Rockets [2].

$$v_f = s\sqrt{\frac{G}{\frac{1.337R^3P\left(\frac{c_t}{c_r}+1\right)}{2(R+2)\left(\frac{t_f}{c_r}\right)^3}}} \tag{5}$$

$$R_f = \frac{h_f}{\frac{1}{2}(c_r+c_t)} \tag{6}$$

The shear modulus was approximated as $G = 11.6$ GPa based on the measured in-plane averaged elastic modulus. Details of the derivation of G are in Appendix G under Fin Material Testing. Using Eq. (5) and Eq. (6) with Omega fin data yielded the following graph in Fig. 8, showing that the fins are in no danger of experiencing fin flutter. The maximum velocity is 274.27 m/s; the fin flutter velocity at that point is 609.62 m/s. The fin design had a flutter ratio of 2.223, which is greater than 1.5, a requirement set by C.5.

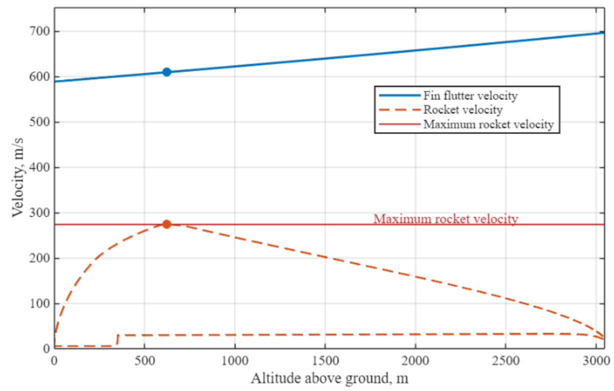

Fig. 8: Fin flutter speed as a function of height above ground

Boat Tail Design

With the addition of a complex payload system, this year's rocket is heavier and would need to be more aerodynamic to reach the same altitude (C.14). To avoid buying a larger casing, a boat tail was used to reduce base drag caused by low-pressure regions behind the rocket. The boat tail guides the airflow smoothly from the rocket body to the freestream air. This causes less pressure difference between the front and rear sections. Boat tail allowed the team to use the same engine that was used last year. The implementation of the boat tail gives the team experience about optimizing the aerodynamics of the rocket, which will be crucial for future years should the team pursue higher altitude or transonic flight.

To further develop the team's ability to manufacture components in-house, the team designed and created our boat tail using only raw materials. There were not many variables that could change, with the forward diameter required to be the same as the fin can's diameter, and the aft diameter required to be the diameter of the motor tube attachment ring for the same reason. This only leaves the material and the length of the boat tail as design parameters. The material of the boat tail had to be a woven tube of fibers that could be laminated in place with epoxy. This left the team with the choices of Kevlar sleeving, Carbon Fiber sleeving, and fiberglass sleeving. These sleeves were quite inexpensive, so we made our decision based on the material properties of the weaves. The manufacturers reported tensile strength in all cases, so the comparison was simple. See Table G-11 under Appendix G Decision Matrices for reasons for choosing Carbon Fiber.

The team validated the acceptability of this strength with a CFD simulation of the pressures that the boat tail might experience, shown in Appendix G under CFD. The length was limited by the flexibility of the carbon fiber weave. If the boat tail was too short, the weave would be forced to change diameters too quickly, which would lead to poor surface quality. The team chose a length of 101.6 mm for the boattail because, from the limited range of 88.9 mm to 101.6 mm, the longest length offered the most favorable stability properties.

Aero-structure screws

This year, the team selected a single bolt type for all aerodynamic structures: 8-32 steel screws with a length of 9.53 mm (3/8 in). Standardizing the bolt size improves modularity between components and reduces confusion during launch operations by requiring only one bolt type. Although the bolt heads are small enough to have a negligible aerodynamic impact, button-head hex screws were selected to further minimize any protrusion into the airflow. To validate the structural integrity of this choice, the bolts were evaluated under a worst-case loading condition using equations given by Ref. [11]. The manufacturer lists the ultimate tensile strength (UTS) of the steel as 70,000 psi. Using a common engineering approximation, the ultimate shear strength (USS) was taken as 60% of the UTS:

$$USS \cong 0.6UTS = 298.6\,MPa \tag{7}$$

The shear force carried by each of the three evenly spaced bolts during recovery deployment was estimated from the OpenRocket-predicted acceleration of the fin can:

$$V = \frac{m * a}{n_{bolt}} = 465.3\frac{N}{bolt} \tag{8}$$

Assuming each bolt is in double shear, the effective area is twice the bolt cross-sectional area. Using the minor diameter of the 8-32 screw, d = 3.302 mm, the average shear stress is:

$$\tau = \frac{V}{2\left(\pi * \frac{d^2}{4}\right)} = 27.2\ MPa \tag{9}$$

The factor of safety (FoS) is then calculated as:

$$FoS = \frac{USS}{\tau} = 10.7 \tag{10}$$

As shown in Eq. (10), the factor of safety is sufficiently high that, even considering the approximation used for the ultimate shear strength, the team has a high degree of confidence that the fin can attachment screws will not fail during recovery (C.7).

3. Aero-Structures Manufacturing Highlights

A more in-depth explanation of the aero-structures design process is in Appendix G under Manufacturing Process. The nosecone was epoxied to the coupler and the 3D printed threaded ring. Holes were drilled into the coupler for the aero screws, pressure holes, and camera holes. The forward airframe was cut to 1 m, and the same holes as the nosecone coupler were drilled into the upper side of the airframe. The team allocated space for the payload in the upper airframe section to give ample space to the payload subsystem (C.16-17). On the lower side, holes were drilled and threaded for shear pins, as well as some pressure holes. A bulkhead was epoxied into place 56.6 cm from the top of the airframe. The recovery bay consists of a 2.7 cm switch band epoxied to a 33 cm coupler. Holes were drilled into both the top and bottom of the coupler.

The aft airframe was cut to 60.3 cm, and at the top, shear pin holes were drilled and threaded. A 30.5 cm coupler tube was epoxied into the bottom of the aft airframe, and holes were drilled for the aero screws. PETG anchors to hold the threaded inserts for the aero screws were epoxied into place behind the holes. A hole was drilled for the upper rail button, and the metal fastener was epoxied to the aft airframe (C.10).

The fin can was cut to 50.8 cm with holes drilled into the upper side. Three fin slots were cut into the airframe, and the motor tube was epoxied into place. Fins were cut using a water jet, then chamfered at a 45º angle and epoxied to the motor tube through the fin slots. Internal fillets were made with epoxy, and external fillets were made from epoxy and chopped carbon and were molded into shape. The fins were laminated to ensure a smooth finish. A hole was drilled for the lower rail button, the second one by constraint C.11, and the fastener was epoxied into place. These rail buttons were able to hold the rocket's full weight per C.12. Holes for aero screws were drilled in the bottom of the fin can to connect it to the boat tail. The boat tail was 3d printed out of glass-fiber-reinforced polyethylene terephthalate (PET-GF). Laminating epoxy and carbon fiber sleeves were used to do a carbon fiber lay-up on the outside of the 3d printed boat tail. The ends were sanded down, and an attachment piece 3d printed out of PET-GF was epoxied to the boat tail.

The aforementioned parts, except the boattail and the fins, received a professional auto-body finish. Pressure holes were drilled in each component to satisfy constraint C.6. Throughout the airframe manufacturing process, no forbidden material prescribed by C.8 was used. 3D printed jigs were used during manufacturing to make sure all separation couplers overlap into the airframes by 153 mm per C.9.

4. Aero-structures OpenRocket Simulation

Weather

To predict the weather conditions used as inputs for apogee prediction, the team used several weather-data collection methods. Apogee was simulated in OpenRocket. OpenRocket requires wind, temperature, and pressure inputs to generate an accurate apogee prediction. Therefore, obtaining accurate forecast values before launch is necessary to improve the accuracy of the simulation. The forecast source also needed to provide high spatial and temporal resolution and be available at least three days before launch.

Wind values were obtained from a website provided by GPSdriftcast called gpsdriftcast.com/orwind [12]. This tool generates a .csv file of wind speeds at different altitudes using weather data from Open-Meteo [13], an open-source weather API that compiles multiple forecast models. The resulting .csv file can be directly imported into the OpenRocket multi-level wind model. The website gpsdriftcast.com/orwind was selected because it is easy to use and more realistic than the single constant wind vector used by default OpenRocket [14]. The website gpsdriftcast.com/orwind is updated hourly, provides 1-km resolution, and can be used for forecasts up to two weeks in advance. An example of the .csv file used to predict the scale-launch conditions in Culpeper, VA, is shown in Fig. 9 below:

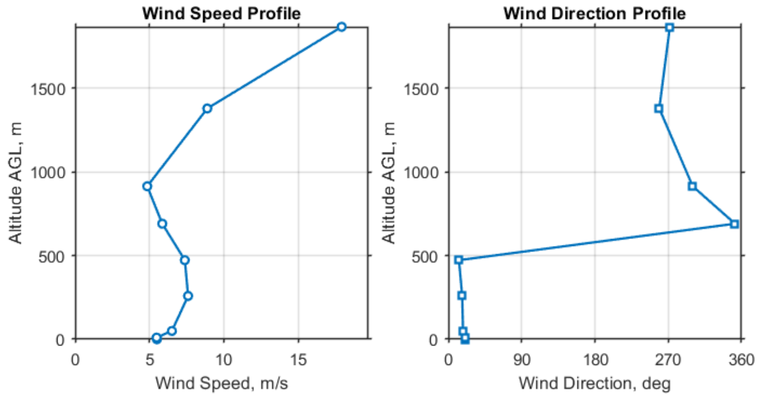

Fig. 9: Wind speed (left) and direction (right) for subscale launch on March 1st at 12:00 pm at N 38.3976º, W 78.0653º

For temperature and pressure inputs, the team used barometricpressure.app [15] and Windy.com [16]. Barometricpressure.app is a web-based tool that displays hourly temperature and pressure data derived from Apple Weather, which aggregates information from multiple forecast models and weather-station sources, including NOAA, ECMWF, and other agencies. The platform provides forecast data up to 7 days in advance; however, the reported values are taken from the surrounding vicinity and may not exactly represent conditions at the launch site.

To supplement this source, the team also used Windy.com with the HRRR CONUS 3 km forecast model. This model was selected because its relatively fine spatial resolution is better suited for estimating local launch conditions. Windy.com also provides access to other forecast models, such as NAM 5 km and ICON 13 km, when additional comparison is needed. One limitation of Windy.com is that exact pressure values are not always directly displayed except at the indicated extrema.

The team compared the temperature and pressure predictions from these sources and selected the values showing the greatest agreement for use in apogee prediction.

Table 1: Launch Conditions

Competition Predicted Weather Conditions	
High Temperature	34ºC
Pressure	91.71 kPa
Average Windspeed	11.58 m/s out of the South
Latitude	31.047º N
Longitude	103.544º W
Altitude (above sea level)	895.88 m

Using the launch conditions provided by IREC DTEG, the team has values for high temperature (33-35ºC) and wind (5.08–5.28 m/s, 77% of the time out of the South). The team used the midpoints of these ranges for the predicted launch conditions and the most likely direction for the wind. The latitude and longitude for the solids launch site were found to be 31.047º N, 103.544º W, respectively. Using this location, the altitude of the launch site was found using the U.S. Geological Survey [17] to be 895.88 m. Then using equation 33b in Ref. [18],

$$P = P_0 * e^{-\frac{g*M*h}{R*T}} = 91.71\ kPa \tag{11}$$

Apogee Accuracy

To get as close to 3048 m as possible (D.3), an accurate OpenRocket file is crucial. Starting this year, the team experimentally measured the rocket exterior to gather surface roughness data. Using the Keyence VK-15000 laser & optical microscope, the surface roughness of the nosecone, airframe, fins, and boat tail was determined to be 5 µm, 5 µm, 30 µm, and 120 µm respectively. Because the standard OpenRocket only allows several discrete surface roughness values for simulation, the team developed an in-house version of OpenRocket that takes any values (D.5).

Last year at the competition, the team's rocket could not achieve the target apogee with the given 6-degree launch angle, so a petition was made to launch with a lesser angle instead. This problem was eliminated this year by overshooting the apogee and controlling it with extensive use of ballast. Stacks of steel plates are secured by the payload's upper bulkhead, which is below the nosecone. This configuration allows space for the use of many ballasts and offers a higher stability margin for the rocket overall (C.2-4). Omega can achieve 3048 m apogee from 0 degrees to a max of 11-degree launch angle, which is greater than the IREC requirement. In addition, each steel plate is intentionally cut so that the combination of them can range from 5 g to 21 kg with a step size of 5 g. This pseudo-continuous range of ballast weight allows the team to fine-tune the rocket's apogee to 3048 m at competition right before flight. Details of the competition flight are in Appendix A.

C. Payload

1. Overview

The payload (Fig. 10) holds a galvanometric laser engraving system that will engrave a small piece of anodized aluminum during the flight of the rocket. The main components of this system are a 4W laser module, two MG90S micro servo motors, two mirrors, a 12.6V Li-ion battery, and a student-designed PCB. These components are mounted on several 3D printed ABS brackets that ensure everything stays in the correct orientation during flight. The payload frame was constructed from aluminum, laser-blocking acrylic, and a 3D printed ABS endcap. Because of the laser module, special care was taken throughout the entire design and testing process to ensure that safety was emphasized.

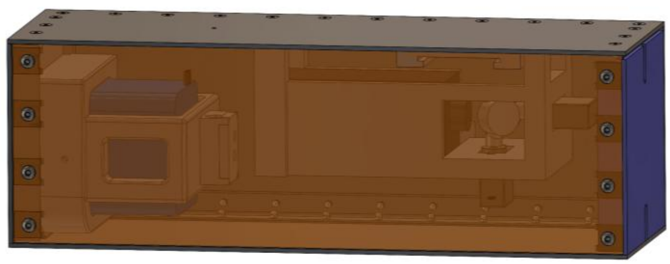

Fig. 10: Full payload assembly

2. Objectives and Requirements

The primary objective of the payload is to engrave a simple pattern onto a piece of anodized aluminum during the ascent of the rocket. The engraving needs to be completed before the payload falls back to the ground to guarantee a safe recovery. Other objectives include implementing safety measures within the system to prevent laser malfunction and designing the system to be viewed while it is outside of the rocket without the need for everyone watching to wear eye protection. A table containing the major payload requirements is shown below in Table 2.

Table 2: Payload Major Requirements

Requirement	Rationale	Verification
Payload mass must be at least 4.4 lbs	Required by IREC Rules 2.3.1	Test
Payload must be completely independent of rocket functionality	Required by IREC Rules 2.3.2	Design
Payload must be removable from the rocket for weighing and evaluation	Required by IREC Rules 2.3.3	Test
Payload must conform to the CubeSAT form factor	Required by IREC Rules 2.3.5	Design
Cannot include more than 1.15 lbs. of non-functional mass	Required by IREC Rules 2.3.5.2	Design
Battery must be LiFePO4, NiMH, Alkaline, Li-Ion, or Lithium primary coin cell	Required by IREC Design, Test, and Evaluation Guide 6.17.2	Design
The system must be able to sit on standby for at least 2 hours	In case of unforeseen delays on the launchpad	Test
System must include a pull pin	To make the launchpad process safer and more streamlined	Design
System must be able to be safely viewed outside of the rocket while it is engraving	So that everyone does not have to wear eye protection to watch it engrave	Design

3. Concept of Operations

The three main phases of the payload are launchpad operations, flight, and recovery. In the launchpad phase, the pull pin will still be in the payload to ensure that the system is powered off and doesn't accidentally turn on when setting up the rocket. As close to the actual launch as possible, the pin will be removed, which will power on the system, including the temperature sensors, but will not begin the laser engraving yet. The system will sit in this standby state until the accelerometer measures an acceleration of 5g, at which point the payload will enter the flight phase and the timer will begin. During this phase, the laser module and servo motors will begin receiving power, and the system will start engraving the piece of metal. This will continue until the full image has been engraved or the timer reaches 100 seconds, whichever one occurs first. The use of the timer in this phase will make sure that the laser is not operating when the rocket reaches the ground. When the rocket reaches the ground, the system will enter the recovery phase and will be fully powered off and in a safe state to be successfully recovered along with the rest of the rocket.

4. Frame

The payload frame, as seen in Appendix F, Fig. F-16, was constructed mainly from 0.08" thick 6061 aluminum panels. These panels were cut out of a larger sheet into the correct dimensions using a water jet to maintain high levels of precision. They were fastened together using 0.0625" 6061 aluminum angle stock and 0.1875"x0.5" blind rivets. This method of construction was chosen because it provides plenty of strength, allows for maximum usable space inside the payload, and is aesthetically pleasing. All the exposed metal surfaces were painted matte black to make everything look cleaner and help hide any small imperfections on the metal. The top side of the frame was 3D printed because it needed to be designed in such a way that it could be removed for easier access and house the wire system that is used to pull the payload out of the airframe. The last side of the frame is made of laser protective acrylic. The laser module has a wavelength of 450 nm, and the acrylic is rated at OD 5+ for 190-532 nm. The decision was made to use this material so that anyone can safely view the laser operating without having to wear eye protection.

5. Laser Module

The payload uses a LASER TREE adjustable-focus laser engraving module with an optical output of 4 W, operating at a wavelength of 450 nm. The laser runs on 12 V and typically draws about 1.6 A, so it can be powered directly from the battery without any additional conversion. Output power is controlled using a PWM signal, which allows the engraving intensity to be adjusted during operation. In this design, the beam is directed using a mirror system instead of moving the laser itself. Because of this, the adjustable focus is important to keep the beam concentrated at the target surface after reflecting off the mirrors. The laser operates at a wavelength of 450 nm, which works well for materials like wood and anodized metals since they absorb this wavelength well. This makes it easier to produce visible marks without needing a higher-power laser. A 4 W laser was chosen because it gives enough power to engrave these materials while not supplying enough power to cut through the payload box and pose a threat to the rest of the rocket.

6. Mirror Steering

A galvanometric system was chosen for the laser engraver to use the small space inside the payload frame efficiently. In this system, the laser module remains stationary while the beam is reflected to specific coordinates by using two rotating mirrors rated at 440-480 nm. In our payload, these mirrors are mounted to MG90S micro servo motors. The rotation of each servo motor and mirror assembly changes the location of the laser beam along one of the two axes on the piece of metal being engraved, allowing for complete control over a relatively large area. The laser module, servo motors, and mirrors are all connected by a 3D printed frame. This frame (Fig. 11) was designed to hold all the components together as rigidly as possible to prevent unwanted movement during flight and to minimize the space between the laser head and mirrors. A more detailed technical diagram is found in Appendix F, Fig. F-17. The laser head has a maximum focal length of 50 mm, so the frame was designed so that the beam does not have to travel more than this length to reflect off both mirrors and reach the metal surface. To offer another level of safety, the design also includes physical stops that prevent the mirrors from rotating too far and reflecting the beam in a harmful direction.

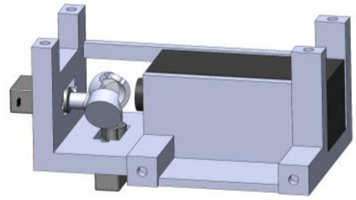

Fig. 11: Mirror steering design

7. *Power Supply*

The payload is powered by a 12.6 V, 7800 mAh Li-Ion battery, which was selected to meet the voltage and current draw requirements of the system while maintaining an adequate safety margin and following IREC's battery requirements. The battery pack is made up of nine 18650 cylindrical cells arranged in a 3-series, 3-parallel configuration, which produces a fully charged voltage of 12.6 V. The battery provides a nominal energy capacity of 86.6 Wh and supports a maximum continuous output of 5 A. Based on the payload power budget shown in Table 3, the payload draws approximately 1.93 A during typical operation and up to 2.69 A in worst-case conditions.

Table 3: Payload Power Budget

Component	Voltage [V]	Typical Current [A]	Max Current [A]	Typical Power [W]	Max Power [W]
Laser Module	12	1.6	1.8	17.76	19.98
MG90S Servo (2x)	6	0.3	0.84	3.33	9.33
RP2040	3.3	0.03	0.05	0.333	0.555
TMP117 (2x)	3.3	0.00001	0.00002	0	0
Total	11.1	1.93	2.69	21.426	29.869

These values are well below the battery's maximum output capability, which ensures reliable operation without excessive voltage sag or thermal loading. Battery life analysis indicates a theoretical runtime of around 4.04 hours under typical loading and 2.90 hours under worst-case conditions, which provides a substantial energy margin beyond the 2 hours desired. Additionally, the battery has many integrated safety features, including protection against over-current, over-charge, short-circuit, and thermal failure, as well as an internal pressure relief valve.

8. *Electronics*

The payload electronics are centered around a custom PCB, *Conkete Mk. 2*, which mainly distributes power and helps reduce wiring complexity. The payload is activated using a pull-pin switch, which completes the circuit and allows current to flow from the battery to the PCB. From there, the PCB distributes power to each subsystem and makes sure that the laser, servos, and sensors receive the required voltage. This helps keep the internal layout more organized and reliable.

A custom PCB was chosen to help improve robustness and simplify integration within the CubeSAT space. With each component requiring a different input voltage and connection, having a fully wired setup would have been prone to losing connections and being more prone to wiring errors. The PCB allows all of these connections to be consolidated into a compact board, which reduces wiring errors and improves reliability under vibration.

Two TMP117 temperature sensors are included to monitor the heat inside the payload during operation. They provide real-time temperature data and are used as a safety feature to shut down the system if overheating is detected. The "brain" of the operation is an Adafruit Feather RP2040, which executes G-code commands to control both the servo motor motion as well as the laser output, allowing the payload to engrave in a controlled and repeatable manner. A full wiring diagram is shown below in Fig. 12:

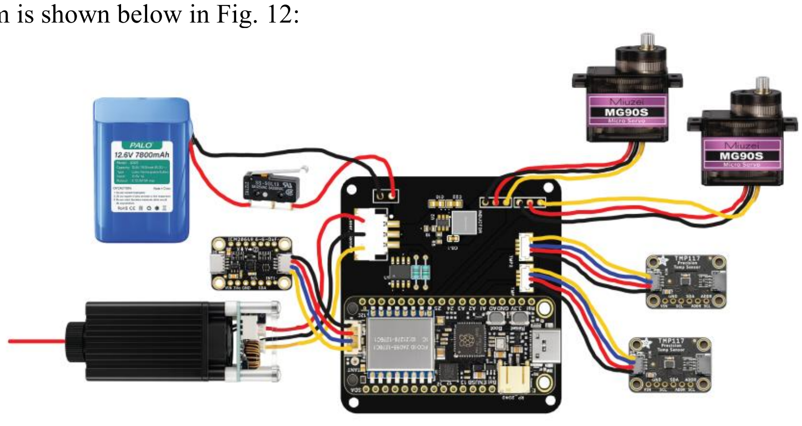

Fig. 12: Payload full wiring diagram

9. *Payload Securing Mechanism*

The payload is secured within the rocket using a 3D printed ABS holster designed to fit the rocket snugly and prevent movement during flight. The holster slides onto steel threaded rods anchored to a bottom bulkhead with nuts, which provides a stable mounting structure. A movable top bulkhead is installed above the payload and screwed down to clamp the payload in place. To allow for easy removal, a steel cable with finger loops is attached to the payload. This makes it simple to pull the whole assembly out once the top bulkhead is removed.

D. Propulsion Subsystems

1. *Subsystem Overview*

The Omega propulsion system is engineered to provide the necessary impulse to reach a target apogee of 10,000 ft. The system utilizes a COTS solid rocket motor housed within a structural motor mount assembly. The primary components include: an AeroTech M1845NT-PS (98 mm) motor, an RMS-98/7680 high power reload kit, precision machined aluminum thrust retention ring, and G10 fiberglass centering rings bonded to a 98 mm Blue Tube motor mount tube.

2. *Motor Selection and Performance Analysis*

The Aerotech M1845NT-PS was selected as the choice motor following a trade study, Table 4. Selection criteria prioritized total impulse, manufacturing reliability, and simulation margin. While the team previously utilized Cesaroni propulsion, supply chain volatility necessitated a pivot to AeroTech to ensure hardware availability. The M1845 was chosen specifically for its thrust profile, which provided the best simulated altitude margin in OpenRocket.

3. *Motor Evaluation*

Table 4: Motor Criteria Comparison

Evaluation Criteria	Aerotech M1845NT	Aerotech M2400T	Cesaroni 6438M1300
Price	$641.99	$641.99	$697.00
Specific Impulse	225 s	213 s	191 s
Total Impulse	8,307.0 Ns	7,716.5 Ns	6,738.2 Ns
Average Thrust	1,875.0 N	2400 N	1,309.9 N
Initial Thrust	2,167.3 N	2,479.6 N	2,492.9 N
Total Weight	6,682 g	6,451 g	5,657 g
Geometry	98 mm diameter 597 mm length	98 mm diameter 597 mm length	75 mm diameter 757 mm length

4. *Discussion*

The propulsion trade study compares three candidate solid rocket motors across performance, cost, and integration constraints. Each motor presents distinct advantages depending on mission priorities.

From a performance standpoint, the *Aerotech M1845NT* exhibits the highest specific impulse (225 s) and total impulse (8,307 Ns), indicating superior efficiency and overall energy delivery. This makes it well-suited for maximizing altitude or payload capability within the Omega Rocket architecture. Its thrust profile is moderate, with a lower average and initial thrust compared to the *Aerotech M2400T,* suggesting a more gradual acceleration profile that may reduce structural loads during ignition.

The *Aerotech M2400T* provides the highest average thrust (2400 N) and strong initial thrust (2479.6 N), making it favorable for missions requiring greater liftoff acceleration or improved thrust-to-weight ratio at ignition. However, this comes at the cost of reduced specific impulse (213 s) and total impulse (7,716.5 Ns), indicating lower overall efficiency compared to the *M1845NT.* Despite this, its identical price point and slightly lower mass improve its attractiveness in thrust-critical scenarios.

The *Cesaroni 6438M1300* offers a noticeably lower weight (5,657 g) and the highest initial thrust (2,492.9 N), but significantly underperforms in specific impulse (191 s), total impulse (6,738.2 Ns), and average thrust (1309.9 N). While its reduced mass could benefit mass-constrained designs, its lower efficiency and sustained thrust output make it less competitive for missions prioritizing altitude or sustained acceleration. Additionally, its smaller diameter (75 mm) and longer length may introduce integration challenges depending on airframe design constraints. From a cost perspective, both Aerotech motors are equally priced and slightly more economical than the Cesaroni option, further strengthening their competitiveness. Overall, the trade study highlights a clear distinction between efficiency-driven performance (M1845NT) and thrust-driven performance (M2400T), with the Cesaroni motor representing a lower performance but lighter alternative.

5. *Test Flight Validation and Risk Mitigation*

To verify flight worthiness and profile of the Omega architecture, the team executed a sub-scale flight test utilizing a 75mm Aerotech L1940 motor and a full-scale flight test using the AeroTech M1845. The test served as a critical Integrated Systems Test, focusing on three primary validation objectives: aerodynamic characterization, avionics and recovery redundancy, and structural load path verification. The sub-scale flight allowed the team to determine the vehicle's drag coefficient. The flight served as a live-fire test for the dual deployment of electronics recovery. The launch validated the integrity of the motor mount centering ring fillets and motor retention under real flight loads. The flight also validated the structural integrity of all systems under actual high-vibration characteristics.

6. *Manufacturing and Integration*

The motor mount serves as the primary load path for the thrust transfer to the airframe. The manufacturing process focused on concentricity and bond strength: (1) Surface Preparation: The G10 centering rings were internally sanded to achieve a minimal tolerance fit with the motor tube; additionally, bonding surfaces were abraded to increase adhesion. (2) Indexing: Ring locations were marked and indexed to avoid interference with the fins. (3) Adhesive Profile: A 1:1 epoxy system combined with chopped fibers, providing increased viscosity and mechanical reinforcement. This mixture allowed for uniform fillets along joint interfaces while minimizing adhesive sag during application. (4) Alignment: The assembly was cured in a vertically oriented jig to maintain proper geometric alignment throughout the bonding process. This setup ensured that the motor tube remained concentric and true to the intended design axis, preventing misalignment that could negatively affect structural integrity and performance.

E. Avionics Subsystems

1. *Avionics Bays*

The rockets' physical avionics components are split between two bays. These are the nosecone avionics bay and the recovery bay. The components are split due to communications with the ground station being interfered with by the carbon fiber airframe. The components that communicate are kept in the nosecone, while all mission-critical avionics that deploy the recovery system are kept in the recovery bay. Reference Fig. 2 for the locations of each bay in the assembled rocket.

The nosecone avionics bay contains our custom flight computer, a GoPro Hero 10, our Altus Metrum TeleGPS, and the necessary batteries. The custom flight computer and TeleGPS use separate protected 18650 Li-Ion batteries in cylindrical casings. Both batteries provide enough charge for 8+ hours of idle operation. The custom flight computer activates the GoPro by providing USB power upon receiving a command from our ground station. The GoPro is running experimental firmware, causing it to turn on and start recording when USB power is applied.

The recovery avionics bay contains both of our altimeters, our previously tried and tested RRC3 altimeter, which we have significant experience using, as well as the Altus Metrum EasyMini altimeter selected as the redundant altimeter due to its analog nature and ease of use. While being simple compared to the flexibility of our backup Featherweight BlueRaven altimeter, that is also a strength, as it is hard to mess up the altimeter's internal logic settings, making it a reliable backup altimeter. Both of our altimeters are connected to pull pin switches, which allow the altimeters to be unarmed when pins are inserted. Upon reaching the launch pad, the pull-pins will be removed, and the altimeters will be armed. The insertion points on the bay itself are very close to the inner coupler wall, and they are chamfered, allowing for easy blind reinsertion of the pull pins should the rocket need to be disarmed on the pad.

2. *SRAD Flight Computer*

The rocket's custom flight computer, Omega, analyzes critical sensor data and manages telemetry communication. It is a six-layer, STM32-based PCB designed entirely in Altium, shown in Fig. 14. Omega features a variety of sensors, including the BNO055 inertial measurement unit (IMU) and BMP384 barometric pressure sensor for orientation and altitude estimates. Additionally, the board uses a flash chip and SD card for data logging, a GNSS module for precise location, and a 433 MHz LoRa radio module to relay information to the ground station. This is all supported by an extensive battery management system to charge, protect, and regulate power from the onboard Li-Ion batteries.

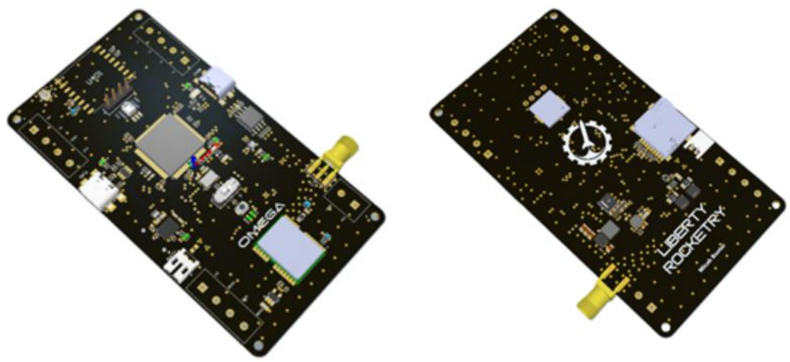

Fig. 13: Omega flight computer PCB (top left, bottom right)

Design Objectives

With each new season, the complexity of our flight computer increases significantly. Early versions utilized a mounted module design, which worked well for understanding our systems, but progress has led to a fully integrated discrete PCB (See Layout Fig. 15). Omega's design parameters are established below in Table 5.

Electrical Requirements

Voltage and Current: The board operates from a 3.7 V battery supply (18650 Li-Ion), which is regulated to 3.3 V and distributed across a dedicated internal power plane for optimized power delivery. The battery power is also boosted to 5 V for camera activation. The maximum current draw does not exceed 1.5 A, but power system components are rated at least 2 A for a safety margin.

Battery Management System: The BMS handles the charging operation of the 1-cell Li-Ion batteries through the main USB-C port, ensuring safe voltage limits, current regulation, and protection against overcharge, excessive discharge, and short-circuit conditions.

Communication Interfaces and Expansion: The system uses multiple communication protocols supported by the STM32F4 microcontroller. SPI manages data transfer between critical components, including LoRa, external flash, and the SD card. These are grouped into two separate buses to avoid data corruption and overloading the interface. I2C is used for the IMU and barometric pressure sensors, which do not require higher update rates. Finally, UART supports GNSS functionality. Expansion headers are connected to each interface along with serial wire debug and extra PWM-enabled pins for modularity.

Features

MCU: The Omega flight computer is built around the STM32F437VGT7 microcontroller, which was selected for its high-performance processing capabilities, extensive peripheral support, and open-source development designs. The MCU features a 32-bit ARM Cortex-M4 processor running at a maximum of 180 MHz with 1 MB of internal flash memory and 256 KB of RAM. It is more than sufficient for its current use and will support advanced progress in the future. An 8 MHz high-speed external oscillator is used as the MCU's primary reference clock source with an internal phase-locked loop scaling the frequency to achieve a system clock speed of 168 MHz. This configuration ensures the necessary processing speed for sensor polling, complex fusion algorithms, and data logging while maintaining stable operation. The supported SPI, I2C, UART, and USB OTG full-speed interfaces function properly as measured by the oscilloscope and timing techniques.

Camera Activation: The flight computer includes a camera activation circuit designed to reliably trigger the rocket's main camera with a command packet sent from the ground station. It features a high-side switching circuit with a P-channel MOSFET to control 5 V power delivery to the camera through a USB source interface. The MCU is protected from excessive current during switching, and the USB interface is configured to detect devices instantly.

Power: The power system manages USB and external battery power delivery for critical components on the board. It includes a BQ29700 protection IC, featuring fault detection and zero voltage charging, and a BQ24232 linear charging IC, featuring overvoltage protection and USB charging. Additionally, a 3.3 V buck regulator feeds the internal power plane, and a 5 V boost regulator supports the camera system. The average current draw for the main components was measured specifically and listed in Table 6.

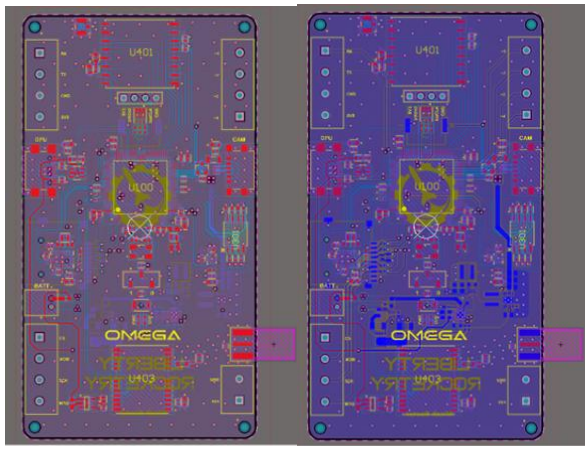

Fig. 14: Layout (top left, bottom right)

Table 5: Custom Flight Computer Component Relations

Power	Sensors and Modules	Controller and Data
2 Cell Li-Ion	BNO085	STM32F437VGT7
BMS and Charger	BMP384	W25Q64 Flash
3.3V Buck Regulator	NEOM8N	Micro SD
5V Boost Regulator	RFM96W	Communication Headers

Table 6: Component Current Draw

Component	Current Measurement (mA)
STM32F437VGT7TR	60
BNO055	15
BMP384	0.5
W25Q64	10
SD Card	30
RFM96W	100
NEOM8N	40
BQ24232	500
TPS61232	2000
TPS6282533	2000
Camera Activation	1500

3. Radio Frequency

At IREC 2025, the team faced communication issues due to interference on the high-traffic, license-free 915 MHz band. To avoid this issue, we are switching the telemetry and GPS link to 433 MHz HAM Radio (70cm). The GPS system is the AltusMetrum TeleGPS paired with the TeleBT ground station unit. Our ground station antenna for GPS is the 6 dBd (~8 dBi) Arrow II Yagi as recommended by AltusMetrum. For our telemetry and camera activation, we selected the Adafruit RFM96W transceivers, along with a 9 dBi directional Yagi antenna at the ground station. Two omnidirectional dipole antennas were selected for the rocket avionics bay (GPS and Telemetry/Activation). To ensure that our antennas were properly tuned, we used a NanoVNA to verify matched impedances and identify any discrepancies.

To further avoid possible interference issues, the following per-packet weighted frequency hopping scheme was developed, but is not being implemented due to competition frequency hopping regulations. Instead of implementing completely random frequency hopping, which is effective but not tailored to the environment, a table of weights corresponding to allowable frequencies is calculated. This is done pre-flight using a Python script (See Fig. 16), pulling data from a Software Defined Radio (SDR) and calculating the Power Spectrum Density (PSD), generating a weighted list of frequencies. The frequencies with lower power values are used more often, and the ones with higher power (interference) are used less often. The weighted frequencies are sent to the flight computer, allowing the ground station and rocket to adjust frequency, using the same starting seed. This allows for pseudorandom hopping, where the frequencies with the least noise (dB) are prioritized over those with greater noise. For this algorithm to work cohesively, we include a verification system in our telemetry data packets, ensuring that the flight computer and ground station both know what frequency they are changing to. This packet verification also accounts for lost packets, for if a packet is dropped, the verification will be incomplete, and the frequency will not change. After a set amount of time, if no packets have been received, both radios will return to the frequency appearing most often in the weighted list. Since the frequency hopping cannot be implemented at the competition, the team is implementing an adaptive spreading factor to improve the signal range. The spreading factor is a characteristic of LoRa radio that determines range by increasing or decreasing the size of the chirps. By adapting the spreading factor according to altitude and RSSI, range can be improved as the rocket reaches a higher altitude, ensuring that telemetry data is still being received.

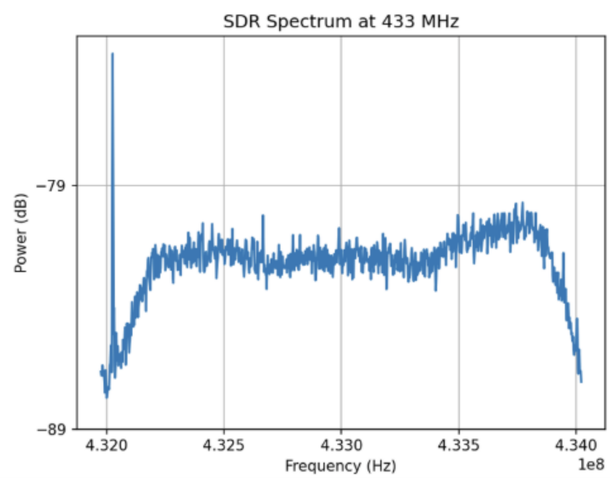

Fig. 15: SDR measuring RF spectrum in Python

4. Ground Station

For our ground station computer, we require a system that is portable, versatile, and weather and heat-resistant. For these reasons, the team chose an Apache 3800 case as the foundation. We then implemented a Lenovo ThinkPad X370 laptop and removed the screen. Using a laptop for our ground station is advantageous for a first design, as the built-in battery allows for long runtime when paired with a battery power bank placed underneath the deck. Paired with the laptop is a bright 16-inch 1920x1200-pixel monitor. Integrated into the ground station is our Adafruit Feather M0 RFM96 433 MHz LoRa ground node, which is wired to several switches for GoPro and live data activation. At the same time, the GPS ground station is also wired to the laptop. With all systems integrated, the computer has the capability to display barometric, orientation, and GPS data live (See Fig. 17).

The T.O.M. (Telemetry Overview & Monitor) system is a multithreaded Python ground station application developed with the PySide6 framework to provide real-time situational awareness during launch operations. The software utilizes PySerial to take in and parse telemetry packets alongside GPS strings across multiple configurable serial connections. To ensure acceptable performance and minimal CPU overhead, the architecture isolates serial I/O from the primary rendering thread, employing PyQtGraph and pre-allocated NumPy circular buffers to generate hardware-accelerated live plots for altitude, multi-axis kinematics, and temperature. The dashboard uses a dynamic 2D geographic map overlay for spatial trajectory tracking, instantaneous altitude taring capabilities, and an interactive apogee meter for real-time flight evaluation against user-defined altitude targets (See Fig. 17).

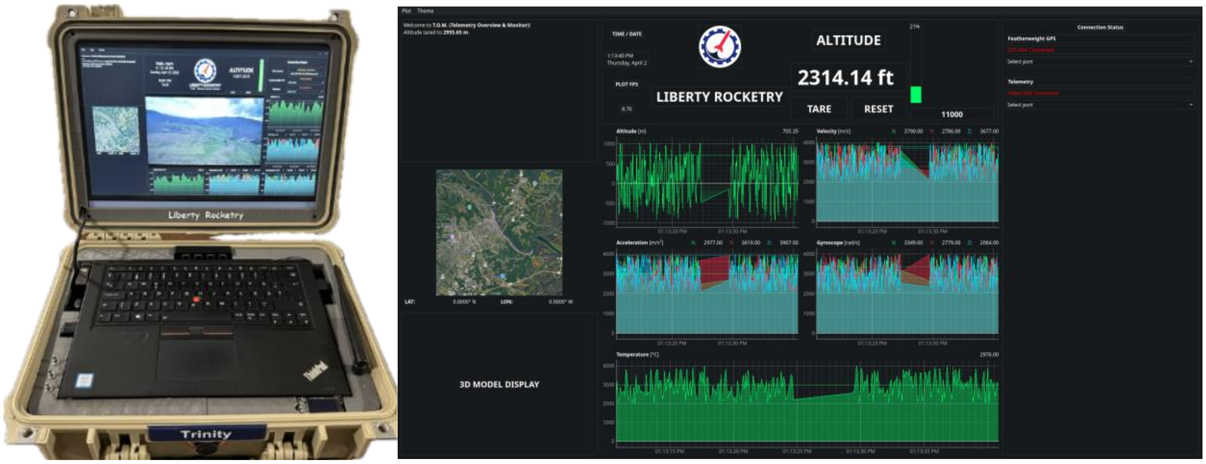

Fig. 16: Ground station computer (left) and GUI (right)

F. Recovery Subsystems

1. Recovery Overview

The recovery subsystem follows a conventional dual deployment, dual separation system. There are two parachutes in this system, a 243.84 mm (96 inch) toroidal main parachute and a 60.96 (24 inch) elliptical drogue parachute. Each parachute is attached via a simple system of Kevlar tethers and quick links. This system allows for accurate ground testing and easy repeatability between loading and unloading the system into the rocket. The drogue parachute will be deployed at apogee, approximately 10,000 ft (3048 m), and the main parachute will be released at 365.76 m (1,200 ft). It is important to note that we consider a later release of the main parachute at 335.28 m (1,100 ft) during higher wind speeds. This height allows our main parachute plenty of room to open fully, while minimizing the potential lateral drifting distance. OpenRocket predicts our lateral drifting distance to be approximately 359 m when accounting for a 5 m/s wind speed and a 6° launch angle. The parachutes are deployed by a barometric RRC3 Sport Missileworks altimeter as the main, and the barometric Altus Metrum Easy Mini altimeter as the redundant. Omega utilizes two of these altimeters, with separate power sources and separate wiring for redundancy purposes. Separation is accomplished via FFFFg black powder charges. The drogue bay will have a charge size of 4 g, with a 4.5 g redundancy charge, and the main bay will have a 4 g charge with a 4.5 g redundancy charge. These charge sizes were thoroughly ground tested to ensure complete separation occurred at these chosen sizes. Figure 18 shows an overview of the recovery system.

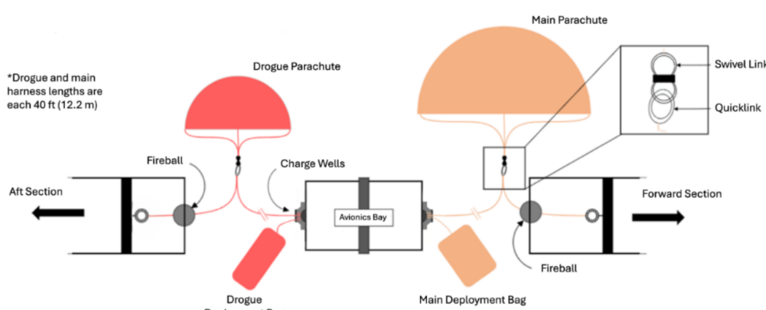

Fig. 17: Recovery system overview

Recovery Design Process

The design for recovery follows a linear step-by-step process. Due to the competition requirements for COTS components, the system is restricted in the customizability of components. Instead, the recovery subsystem can customize the layout of its system. The process detailed below provides a guide to each design decision and insights into the assumptions made.

Target descent velocities of under 26 m/s under the drogue and 6 m/s under the main parachute were established to ensure stable deployment and safe landing, as well as stay within IREC requirements. The corresponding parachute sizes were calculated in Equations (14), (15), and (16) found in Appendix H. Using these constraints, various parachute types were evaluated through weighted decision matrices (Tables H-2 and H-3). This led the team to select the stable 60.96 cm (24 in) elliptical drogue, as well as the robust 243.84 mm (96 in) toroidal main parachute. Recovery hardware was then selected based on predicted shock loads found in Table H-4. This includes two 12.2 m (40 ft) Kevlar shock cords, stainless steel quick links, swivels, and a custom-designed recovery attachment, the recovery "yoke," to ensure a proper load distribution.

The bulkheads comprising the internal anchor for the recovery system are 6.35 mm (0.25 in) thick G10 fiberglass. To validate the use of such thin, lightweight structures, simulations were run in SolidWorks. A detailed analysis of the bulkheads is in Appendix H.

Shock Forces

Based on the sizing of our parachutes, we can then investigate our shock forces. These forces are extremely important to consider when designing an adequate subsystem, as shock forces can easily break components if not accounted for. The following equation is the universally accepted equation used to determine shock force, where the coefficient of shock (C_k). It is the opening shock factor that varies depending on parachute geometry. The drogue and main parachutes C_k were determined by referencing Table 5-1 of the Parachute Recovery Systems Design Manual by Theo W. Knacke [19].

$$F = \frac{1}{2} C_k (C_D S_0) \rho v^2 \tag{12}$$

This allowed our team to gain an idea of how our parachutes will affect the rest of our recovery system. If we deemed these forces too high, then we could revert to our parachute decision matrices and reevaluate our choices. It is important to note that these equations are highly simplified for our recovery system, as they assume all the force generated from the parachute is instantaneously translated throughout the rest of the recovery system. Realistically, with correct black powder sizing and adequate testing, our recovery system should rarely experience the maximum shock force, as these equations do not factor in any drag that may be acting on the body of the rocket. Instead, the equation treats the payload-parachute system as falling with taut shroud lines, at the maximum possible velocity under recovery. It is evident, especially after our team's first test launch, that this is not the case regarding our system. Our calculated maximum shock force from the main parachute is expected to be 8035.44 N.

Additionally, if the opening shock impulse is too high and the shock cords are too thin, a body tube failure known as "zippering" will occur. To prevent this failure, a foam cushion, known as a "fireball", can be placed over the region of the shock cord that passes over the edge of the body tube. The Omega recovery design will incorporate such a cushion on the shock lines for both the main and the drogue chute. Each fireball is wrapped in a Kevlar sheath that will protect it from the black powder charges.

To prevent damage to our main parachute and corresponding subsystem components, Omega's recovery system will utilize a reefing ring to lower the initial shock force transferred to the rocket during the sharp transition from free-fall to drogue fall speed. The reefing ring is placed along the main parachute lines, about two-thirds of the distance from the bottom. This helps control the rate at which the parachute opens, reducing the initial shock and preventing potential damage to the system.

Shear Pins and Black Powder Sizing

As the rocket is coasting, the fins will provide a higher drag than the nosecone, thus the rocket has a potential risk of separation. To prevent this, nylon shear pins are fastened into the sections of the airframe where separation due to recovery will occur. These shear pins provide just enough force to hold the rocket and prevent premature separation, but will shear when the black powder charges are ignited.

When the drogue shear pins are sheared during deployment, it is crucial for the main bay shear pins not to shear and separate. Thorough testing and calculations were made to determine the size of these shear pins and ensure that premature separation of the main bay would not occur during the deployment of the drogue parachute. Our team performed drop tests to help us determine if our initial assumptions were accurate. Calculations, shown in Appendix H, conducted after this test confirmed that six 4-40 shear pins for the main bay, with a shear strength of 1326 N, were adequate to withstand the force that would be exerted during drogue deployment. Using Equation (12), it was determined that the maximum force exerted by the upper section will be 217 N, which results in a shear pin factor of safety of 6.1.

The drogue bay did not need as many screws, as the only force this section will experience is the lower airframe drag. Two shear pins would be the minimum for this section; however, as an added layer of safety due to potential oversights, the count was increased to four shear pins. Calculations and experimental testing data shown in Appendix B provided a required black powder charge size of 4g for the drogue and 4g for the main.

2. Recovery Manufacturing Process

Bulkheads

The five bulkheads in our system were cut from 6.35 mm (0.25 in) thick G10 fiberglass sheets using a waterjet. Hole locations were transferred using a 3D-printed alignment jig and punch tool, then drilled on a drill press. Bulkheads that were epoxied into the airframe were bonded with Aeropoxy and chopped fiberglass, and mating surfaces were sanded before bonding. Each bulkhead serves a different role. The recovery avionics bay bulkheads each carry two threaded rods with a recovery yoke supporting the avionics sled, two black powder charge wells, and two terminal blocks for the e-match connections. These bulkheads also feature a lathed lip that ensures a tight fit against the bay's coupler. The payload bulkheads provide support for the CubeSat and its threaded rods, and the lower payload bulkhead also serves as the upper main parachute bulkhead with an eyebolt. The lower drogue bulkhead only features one drilled hole with a single eyebolt.

Recovery Yoke

The recovery yoke is a critical structural bridge that transitions the parachute shock load from the tethers to threaded rods inside the recovery avionics bay. In previous years, the design only included a standard eyebolt on either side of the bay, and the avionics sled rods experienced permanent deformation due to the "triangulation" of forces pulling the bolts toward the center of the bulkhead. To ensure a purely axial loading on these steel rods, the first iteration of the recovery yoke was developed last year, while this year, a new iteration was made with the support of ANSYS Topology Optimization. This process maximized strength while reducing mass by 58% compared to the previous iteration.

Simulations proved the yoke exhibits a global stress well within the limits of 316L Stainless steel. Additionally, the design satisfies the IREC requirement for steel anchoring geometry and ensures a minimum FoS of 1.5 under peak snatch loads. Given the advanced facilities needed for this process, the manufacturing was outsourced to PCBWay via SLM 3D printing. Further explanation of the yoke design and testing process is provided in Appendix H.

V. Mission Concept of Operations Overview

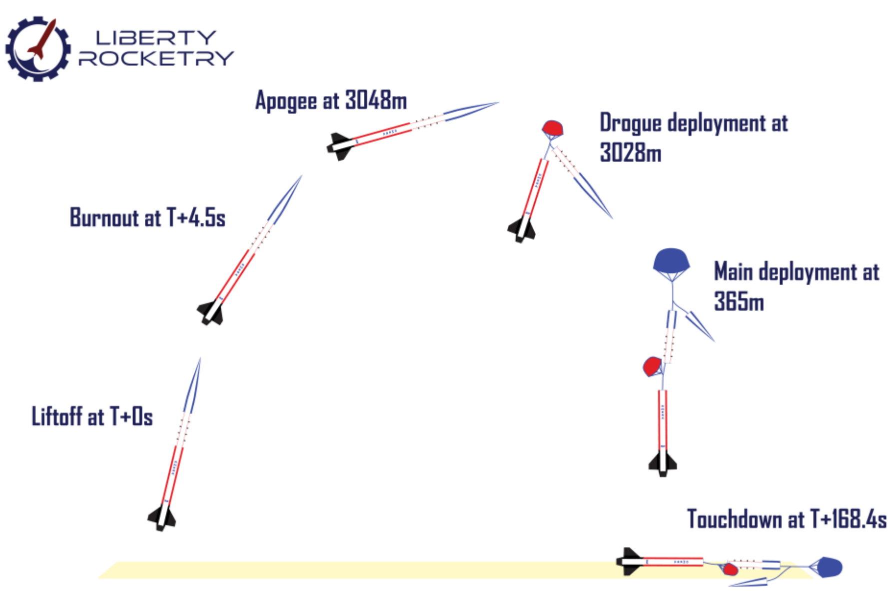

Fig. 18: CONOPS diagram

A. Mission Overview

The Omega mission follows a 3048 m (10,000 ft) AGL flight profile to validate the launch vehicle and operate a galvanometric laser engraver payload. Omega uses a COTS M1845 solid rocket motor, redundant flight computers, independent recovery electronics, and dual-deployment recovery architecture. The mission phases include pad arming, ignition and liftoff, powered ascent, coast, apogee detection, drogue deployment, drogue descent, main deployment, landing, and post-flight recovery. These phases are shown in Fig. 19 and summarized in Table 7.

B. Pre-Launch and Pad Operations

Pad operations begin when Omega is placed on the launch rail, and the team starts final pre-flight checks. During this phase, the aerostructures subsystem supports the vehicle on the rail, the propulsion system remains unignited, the recovery system remains mechanically closed, and the payload remains secured inside the forward airframe. The avionics and payload electronics are armed through a three-pin mechanical arming sequence to support range safety.

The arming sequence is completed as follows:

- Pin 1: Activates the nosecone electronics and laser engraver payload.
- Telemetry Check: Ground station confirms data link with the rocket.
- Pin 2: Arms the primary flight computer, which has an audible continuity check.
- Pin 3: Arms the redundant flight computer, which has an audible continuity check.

The pre-launch phase ends when the payload electronics, telemetry link, primary flight computer, and redundant flight computer are confirmed ready for flight.

C. Flight Profile and Recovery Sequence

Ignition begins when the signal is sent to the motor igniter. Liftoff begins at first vehicle motion and ends when Omega clears the launch rail. Rail departure is predicted to occur at T + 0.38 s at an altitude of 5.18 m AGL and a velocity of 28.4 m/s. During this phase, the propulsion system produces thrust, the airframe transfers launch loads, the fins provide passive stability, and the avionics system begins recording flight data.

Powered ascent continues until motor burnout at approximately T + 4.5 s and 2666 m AGL. During powered ascent, the M1845 motor accelerates the vehicle while the aerostructures subsystem maintains stability and structural integrity. The payload remains secured and begins its flight operation sequence after detecting launch acceleration. The recovery system remains locked out until the required deployment conditions are reached. The coast phase begins after motor burnout and ends at apogee detection. During coast, the propulsion system is inactive, and Omega continues upward under momentum. The avionics system tracks altitude and acceleration to identify apogee. The payload continues its engraving sequence during the flight window while remaining independent of all flight-critical rocket systems.

Apogee occurs at approximately T + 24.3 s and 3,048 m AGL. After apogee is detected, the drogue event fires after a one-second delay. Drogue deployment occurs at approximately 3,028 m AGL. During this phase, the recovery system separates the required airframe sections and deploys the drogue parachute to stabilize the lower airframe during descent. Drogue descent continues until the main deployment altitude is reached. Under nominal conditions, the main parachute deploys at 365 m AGL. Under high-wind conditions, the main deployment altitude may be reduced to 335 m AGL to limit drift. The main parachute deploys from the forward airframe and slows the vehicle to a nominal terminal descent velocity of approximately 6.0 m/s. Landing occurs at approximately T + 168.4 s. At touchdown, the recovery system absorbs the final landing loads, and the vehicle is expected to remain in reusable condition.

D. CONOPS Timeline

Table 7: CONOPS Timeline

Phase	Event	Trigger / Time	Altitude AGL
Pre-Launch	System arming	Manual three-pin sequence	0 m
Liftoff	Ignition and first motion	T + 0.0 s	0 m
Rail Departure	Vehicle leaves launch rail	T + 0.38 s	5.18 m
Powered Ascent	Motor burn	T + 0.0 s to T + 4.5 s	0 m to 2666 m
Coast	Burnout to apogee	T + 4.5 s to T + 24.3 s	2666 m to 3048 m
Apogee	Apogee detection	T + 24.3 s	3,048 m
Drogue Deployment	Drogue event fires	Apogee + 1.0 s	3,028 m
Drogue Descent	Descent under drogue	After drogue deployment	3,028 m to 365 m
Main Deployment	Main event fires	T + 113.5 s	365 m
Landing	Touchdown	T + 168.4 s	0 m
Post-Flight	Recovery and data collection	After touchdown	0 m

E. Post-Launch Operations

Post-flight operations begin after touchdown and continue until the rocket is recovered, rendered safe, and inspected. The recovery team verifies that the vehicle is safe to approach before handling the rocket. The avionics team powers down the electronics and extracts flight data from the redundant flight computers. The recovery team inspects the parachutes, shock cords, airframe sections, yokes, and attachment hardware for damage.

The payload team removes the laser engraver assembly and inspects the target material. The primary payload success metric is whether the system maintained optical alignment and produced a discernible engraving during flight. The team also compares the recorded flight data with the predicted CONOPS timeline to confirm apogee, deployment timing, descent rate, and landing performance.

VI. Conclusions And Lessons Learned

A. Analysis

The development of Project Omega represents a new step in Liberty Rocketry's history in systems integration and testing. We have implemented new, concise risk management protocols and conducted empirical field testing to better support projected tests. The rocketry team addressed our theoretical unknowns miles before the full-scale launch. The final Omega engineering design met all ESRA safety requirements and AIAA engineering standards, demonstrating how a student-led team can competently execute complex aerospace projects within a constrained budget and timeline.

B. Aero-structures

The aero-structures subsystem successfully met all requirements and objectives, including apogee, stability, and structural integrity. Throughout the design process, the team balanced aerodynamic performance, structural integrity, manufacturability, commercial availability, and cost to arrive at a practical and competition-ready rocket.

Last year’s prolonged manufacturing caused the team to not have a competition-ready surface finish for the subscale launch, which rendered the test launch less useful for competition apogee prediction. To streamline and speed up the manufacturing process this year, the team utilized a master document that details the entire process. This allowed any team member to complete the operation correctly, even in the absence of the team lead.

To eliminate discrepancies between the idealized and physical model, emphasis was placed on constantly updating the OpenRocket file. All purchased parts were dimensioned and weighed during inventory. After each aero-structure component was cut or assembled, it was again weighed and dimensioned, and its dimensions were recorded in a master aero-structure sheet, which was then used to update OpenRocket. As a result, the OpenRocket model and the fully assembled Omega had less than a 100 g mass difference, which speaks to the quality of the simulation. The CG location of the OpenRocket Omega also differed by only 5 mm compared to the real rocket. This was a huge improvement from past years, when the team would simply override the mass and the CG location without trying to identify and correct the source of the discrepancy. This new approach to documentation will become standard from now on. This year, all aero-structure files, such as CAD, ANSYS, OpenRocket, and MATLAB files, were uploaded to Microsoft Teams so that technical information is accessible to all current and future members. The team also created and will create more tutorial videos for ANSYS Mechanical and Fluent to train lowerclassmen for technical positions, so that the team can continue to perform effectively despite the graduation of upperclassmen.

C. Payload

The payload this year is a galvanometric laser engraving system inside an aluminum and acrylic box. This design is significantly more complex than previous years, meaning that there were many lessons learned throughout the design and manufacturing process. The primary lesson that the payload team learned is the importance of finalizing design choices as soon as possible so that all necessary components can be ordered and arrive with plenty of time left for testing and troubleshooting. Many of the key pieces of the laser engraving system did not arrive at the team until right before the test launches and the end of the semester, which added a lot of stress to the building process. This was amplified when the team realized that servos were not going to work and had to switch to using servo motors instead. Also, the student-developed PCB had to be redesigned once all the parts arrived, and it was not available for some of the main testing. Next year, the team will work on finalizing a design sooner in the fall semester so that components can be ordered, and a working prototype can be ready for preliminary testing several weeks before the first test launch.

Another lesson learned by the payload team is the importance of communicating well with all the other sub-teams. There were several times when a lack of communication resulted in a rushed change of plans and more work for the team that could have been avoided. This is especially important as the payload designs become more complex and have more specific requirements regarding placement in the rocket, interaction with avionics, and accessibility.

D. Propulsion

The propulsion systems have been improved significantly from previous years. Propulsion has developed in technical rigor, organization, and methodology; this has benefited the propulsion and other sections. Some key things that have contributed to this are the technical onboarding and Tripoli certification launches.

The motor was selected as per the trade study detailed in section D and OpenRocket results; the AeroTech M1845NT was selected for Omega. It provides the best balance of total impulse, specific impulse, cost, availability, and 98 mm motor tube compatibility.

E. Avionics

All Avionics systems have had significant improvements this year. Our custom flight computer was completely redesigned, integrating a non-critical GPS and improved radio hardware for telemetry. Last year’s custom flight computer was the team’s first designed from scratch without using breakout boards. While operational, it had several usability quirks and was somewhat unreliable as a result. This year’s flight computer is much more straightforward, while also providing more capability to enable advanced features, such as airbrakes, to be implemented in future years.

Our ground station computer’s GUI has been completely redesigned, resulting in a more intuitive interface and significantly improved performance compared to last year’s GUI. The ground station computer hardware has also been redesigned to increase usability. At the 2025 IREC, we faced significant interference issues that massively inhibited our live telemetry systems. Thus, one of this year’s core tasks was switching frequencies for SRAD telemetry to 433 MHz and acquiring more sensitive antennas. Combined, this enables more stable and interference-free telemetry. Considerable efforts have been made to upgrade every avionics subsystem, improving core system reliability and expanding our technical capabilities.

F. Recovery

The recovery sub-team had many lessons to learn. First, we now know that it is imperative to be proactive when considering the testing of our recovery systems. Since testing needs to wait until most of our aero-structures manufacturing is done, allowing for an adequate amount of time in between finalizing manufacturing and our first test launch is extremely important. Rushing should never occur with such a mission-critical system, and every precaution, checklist, and safety measure must be considered. Second, thorough documentation has been shown to be a very useful tool to validate our recovery methods, and by having procedures, measurements, data, and written experience, it makes for an easy transition from year to year within the recovery sub-team.

Appendix A: Rocket Vehicle Specifications

Appendix A summarizes the final vehicle measurements, propulsion data, predicted flight performance, and recovery configuration for Project Omega. These values represent the competition launch configuration used in the OpenRocket simulation and the final integrated vehicle design. The flight profile and supporting plots show the predicted altitude, acceleration, velocity, and stability behavior during the nominal 10,000 ft COTS mission.

Table A-1: Rocket Vehicle Specifications

Parameter	Value
Competition category	10,000 ft COTS
Number of stages	1
Length	3169 mm
Airframe outer diameter	157 mm
Number of fins	3
Fin semi-span	175 mm
Fin tip chord	98.2 mm
Fin root chord	301 mm
Fin thickness	5 mm
Weight on launch rail	24.70 kg
Propellant weight	3.79 kg
Empty motor case/structure weight	2.91 kg
Payload weight	2.00 kg
Liftoff weight	24.70 kg
Center of pressure (CP)	2314 mm from the nose tip
Center of gravity (CG)	2044 mm from the nose tip
Launch rail length	5182 mm

Table A-2: Propulsion Information

Parameter	Value
Motor type	Solid rocket motor
Propulsion classification	COTS
Manufacturer and designation	AeroTech M1845NT-PS
Motor letter classification	M
Average thrust	1875 N
Peak thrust	2424 N
Total impulse	8307 Ns
Motor burn time	4.5 s
Propellant weight	3.79 kg
Empty motor case/structure weight	2.91 kg

Table A-3: Predicted Flight Data

Parameter	Value
Liftoff thrust-to-weight ratio	7.25:1
Rail departure velocity	28.4 m/s
Minimum static margin	1.72 calibers
Maximum acceleration	9.46 G
Maximum velocity	282 m/s, Mach 0.81
Fin flutter velocity	609.62 m/s
Target apogee	3048 m, 10,000 ft
Predicted apogee	3048 m, 10,000 ft

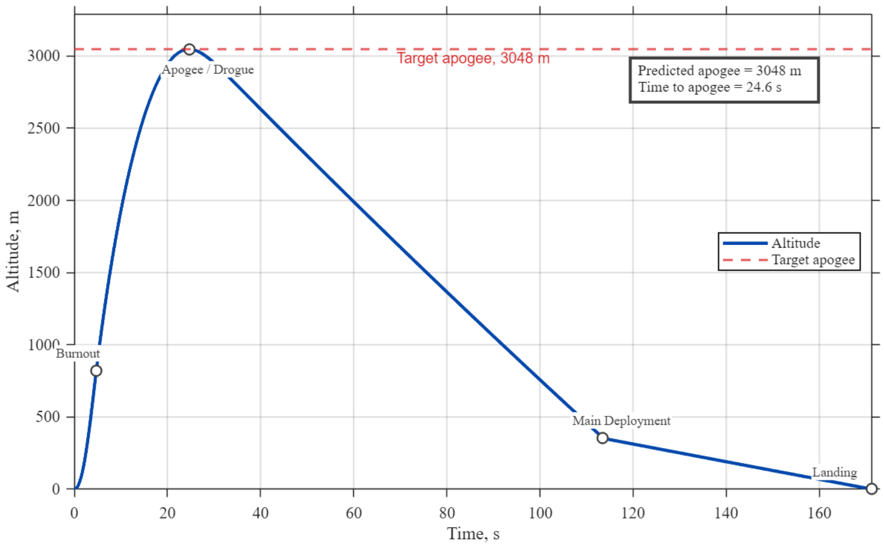

Fig. A-1: Predicted flight profile

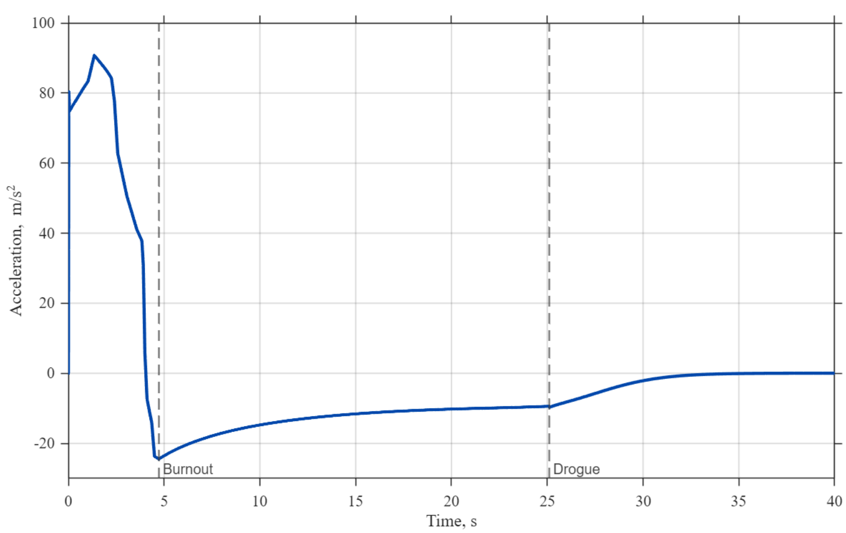

Fig. A-2: Predicted vertical acceleration profile for Omega

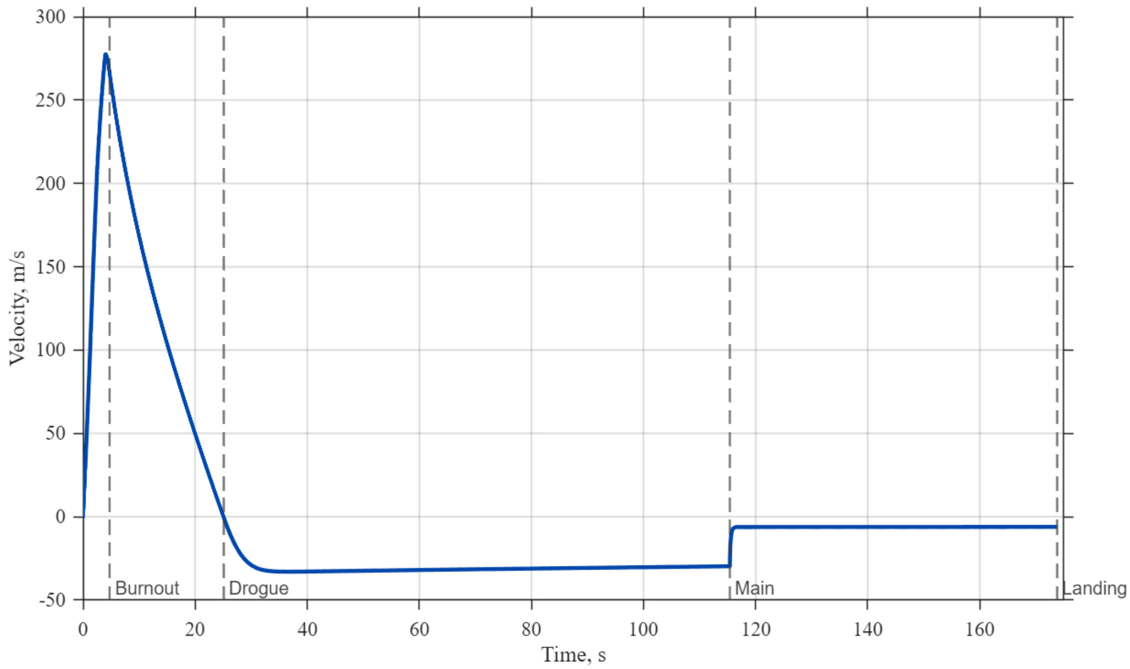

Fig. A-3: Predicted vertical velocity profile for Omega

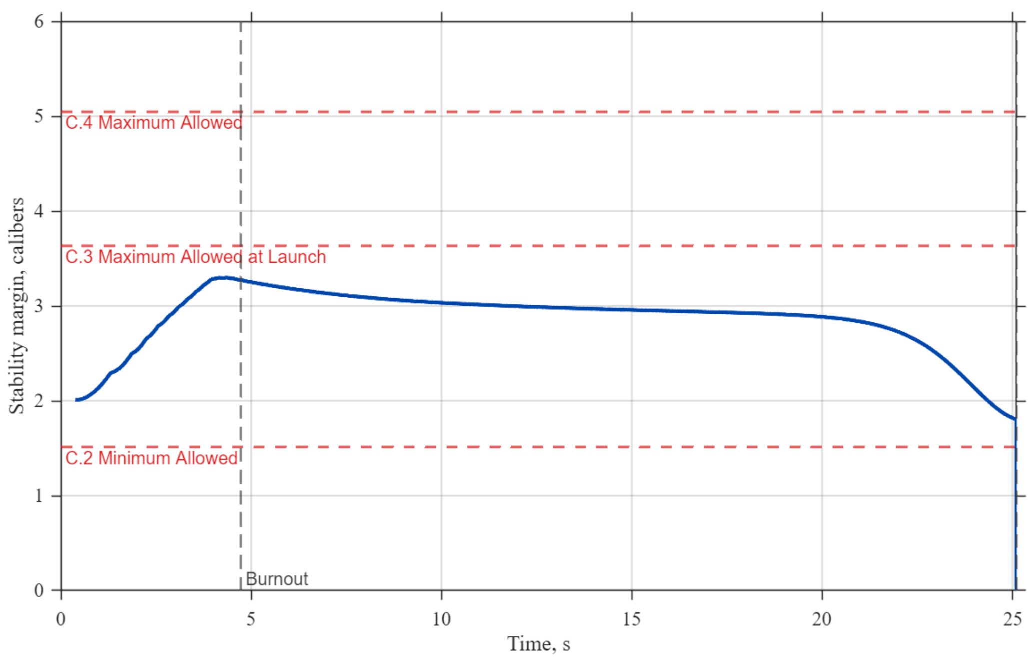

Fig. A-4: Predicted static stability margin for Omega through ascent.

Table A-4: Recovery Information

Parameter	Value
Recovery architecture	Dual-deployment, dual-separation
Primary altimeter	Missile Works RRC3 Sport
Redundant altimeter	Altus Metrum EasyMini
Altimeter power	Separate power sources and separate wiring
Drogue parachute	24 in elliptical parachute
Main parachute	96 in toroidal parachute
Drogue primary charge	4.0 g FFFFg black powder
Drogue backup charge	4.5 g FFFFg black powder
Main primary charge	4.0 g FFFFg black powder
Main backup charge	4.5 g FFFFg black powder
Drogue deployment altitude	Approximately 3028 m AGL, apogee + 1.0 s
Main deployment altitude	365 m AGL nominal, 335 m AGL high-wind condition
Target drogue descent rate	Under 26 m/s
Target main descent rate	6.0 m/s
Drogue harness	40 ft harness
Main harness	40 ft harness
Mechanical links	Quick links, stainless steel recovery yokes, threaded rods, and recovery bulkheads
Main bay shear pins	Six 4-40 nylon shear pins
Drogue bay shear pins	Four 4-40 nylon shear pins

Appendix B: Project Test Report

A. Recovery System Testing

One of the key aspects of recovering a rocket is that the parachutes deploy when the black powder charges are triggered. To ensure this occurred, airframe separation was tested on the ground. Black powder charges are used to separate the rocket airframe. The charges are ignited using an e-match connected to the avionics bay. The ignited charge then builds pressure in the parachute bays. The force from the pressure pushes the airframe apart while shearing the nylon screws that temporarily keep the airframe in one piece. Ground testing allowed us to carefully adjust black powder charge size to ensure separation could occur, without producing an explosion that might harm critical components. To conduct ground tests, charges were prepared by taking the pre-determined amount of black powder along with an e-match and combining them into a latex finger cot. The neck of the finger cot is then zip-tied closed to prevent spilling the black powder and to allow for the gases to initially build for a more complete burn when triggered. The parachutes and tethers were then loaded into their designated compartments. Next, the charges are loaded into the charge wells and taped over to hold them in place during the test. The wires of the e-matches are guided through the rocket, then connected to the terminal block wires and led out through the opening for the pull pin switch. At the time of testing, the recovery avionics bay is empty of electronics and only holds an equivalent weight in its place. The airframe sections are then put together and secured with shear pins. The fully assembled rocket for the ground ejection test is then leaned against a wooden support again to keep it in position during the test. These tests allowed us to determine the ideal charge sizes for each parachute bay. For the drogue and main parachutes, the ideal charge sizes were 4g and 4.5g for the primary and redundant charges, respectively.

Fig. B-1: Main parachute black powder separation test

Without a battery connected, the wires of one of the charge wells are connected to the detonator. With everyone at a safe distance and angle relative to the rocket, the test begins, and the charges are ignited. Finally, the detonator switch is flipped, and the black powder charge is ignited. The pressure builds and forces the airframe sections apart while shearing the nylon shear pins. As the airframe sections are moving away from each other, they also aid the release of the parachutes and tethers, allowing proper deployment in a real flight. Multiple tests were conducted, and both the main and drogue charge sizes were tested to ensure every size was adequate. Each test resulted in the successful deployment of the system, including the release of the drogue and main parachutes. Also, there was no major tangling among the system components.

B. Recovery Drop Testing

One major concern our team investigated was the premature deployment of the main parachute due to an insufficient number of shear pins. When the drogue parachute deploys at apogee, the shock force of the parachute could break the upper airframe shear pins and result in a premature main parachute deployment. To test the maximum impulse the upper airframe shear pins could withstand, a drop test was conducted. The avionics bay was equipped with an accelerometer to accurately measure force on the shear pins. This test allowed our team to determine whether our current number of shear pins was sufficient or if more should be added.

Fig. B-2: Image of drop test

To conduct this test, the upper airframe was assembled with all components, with one exception: a short tether in place of the main parachute tether to act as a failsafe in the event the shear pins broke prematurely. The tether that would be connected to the drogue parachute was tied to a bridge of an off-campus engineering lab. The system was tested by dropping the upper airframe off the bridge and measuring the shock force on the pins. This test was performed at various heights to simulate different velocities at drogue deployment.

The accelerometer showed the acceleration of the rocket body during the drop test. The measured acceleration was then used to calculate the force on the shear pins. Figure B-3 shows the acceleration data for a test that produced 306 N of force. This is more than our anticipated value; however, the shear pins did not break or have any sign of damage.

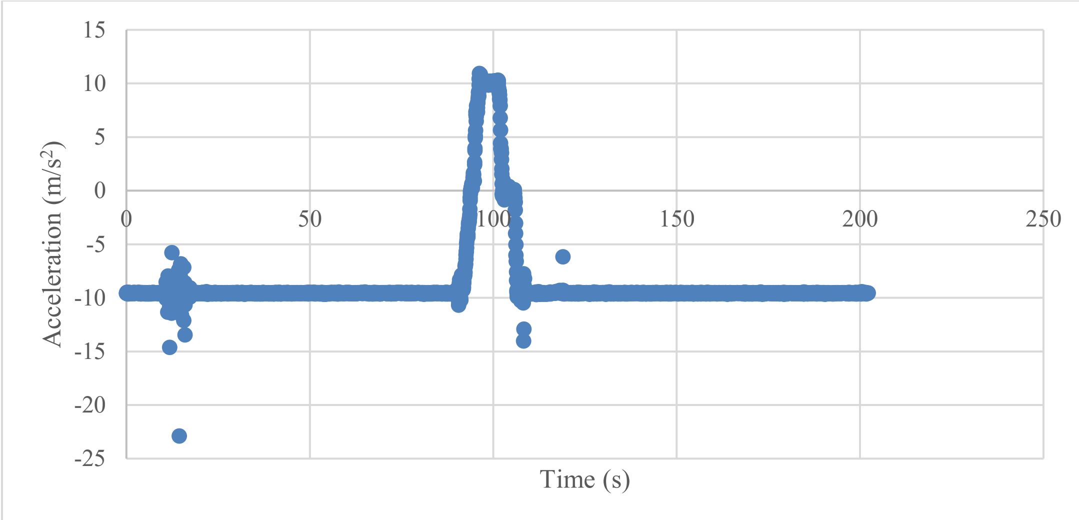

Fig. B-3: Drop yest accelerometer data

C. Decoupling Charge Sizing

Table B-1: Main Parachute Decoupling Charge Size Calculations

Description	Units	Inputs	Outputs
Diameter	m	0.1524	
Length	m	0.31999936	
Bulkhead Force:	N	1334.466	
Volume	m^3		0.005837258
Pressure	Pa		73155.62041
Specific Gas Constant (R)	J/(kg*K)	119.03	
Combustion Temp. (T)	K	1837	
Grams of Black Powder	g		1.952949588

Table B-2: Drogue Parachute Decoupling Charge Size Calculations

Description	Units	Inputs	Outputs
Diameter	m	0.1524	
Length	m	0.159999934	
Bulkhead Force	N	889.644	
Volume	m^3		0.002918634
Pressure	Pa		48770.41361
Specific Gas Constant (R)	J/(kg*K)	119.03	
Combustion Temp. (T)	K	1837	
Grams of Black Powder	g		0.65098423

Table B-3: Scaling From Experimental Results (Main)

Description	Units	Inputs	Outputs
Grams of BP Main	g	4	
BP Charge Scaling Factor	N/A		2.048183949
Bulkhead Force	N		2733.231841
Force of Friction	N		1398.765841

Table B-4: Scaling From Experimental Results (Drogue)

Description	Units	Inputs	Outputs
Grams of BP Drogue	g	4	
BP Charge Scaling Factor	N/A		6.144542092
Bulkhead Force	N		2008.421609
Force of Friction	N		1118.777609

Calculations based on the internal volume of our parachute bays and shear pin sizing yielded charge sizing of approximately 0.7g for the drogue and 2g for the main. The equation used to determine these values can be seen below.

$$Grams(BP) = \frac{Pressure(Pa) \times Volume(m^3)}{R(J/(kg * K) \times T(K)} \tag{13}$$

However, in experimental ejection testing, these charge amounts proved inadequate. After multiple tests, 4g for both bays provided the force needed to separate them. Since these calculations provided a smaller number than what was required to separate the airframe, the charge amounts were scaled to determine the force of friction working against the black powder charge. We can divide the experimentally determined charge size of 4g by the calculated charge size to determine our scaling factor. We can then multiply the expected bulkhead force by this scaling factor to determine the total, real-world force of the charges acting on the bulkhead. The resulting bulkhead force is the required separation force of both the shear pins and the friction from the airframe. These calculations provided a bulkhead force of 2733 N and 2008 N for the main and drogue bays, respectively.

Each parachute bay will have two charge wells, with the redundant charge at least half a gram higher than our primary charges to guarantee deployment if our first charge were to fail. Ground testing was performed with both primary and redundant charges, and each test provided adequate results

D. Recovery Electronics

Both parachutes are deployed using dual deployment-capable COTS altimeters. We utilize a barometric Missile Works RRC3 Sport altimeter as the main, and the barometric Altus Metrum EasyMini altimeter as the redundant. Both are powered by independent 9V batteries, which are replaced prior to each flight, and the voltage is tested as well. In battery drain tests, each altimeter lasted for over 8 hours. Each altimeter operates on a fully independent circuit, including separate power supplies, wiring paths, and dedicated primary and redundant e-matches for both the drogue and main deployment. The wiring schematic is shown below in Fig. B-4.

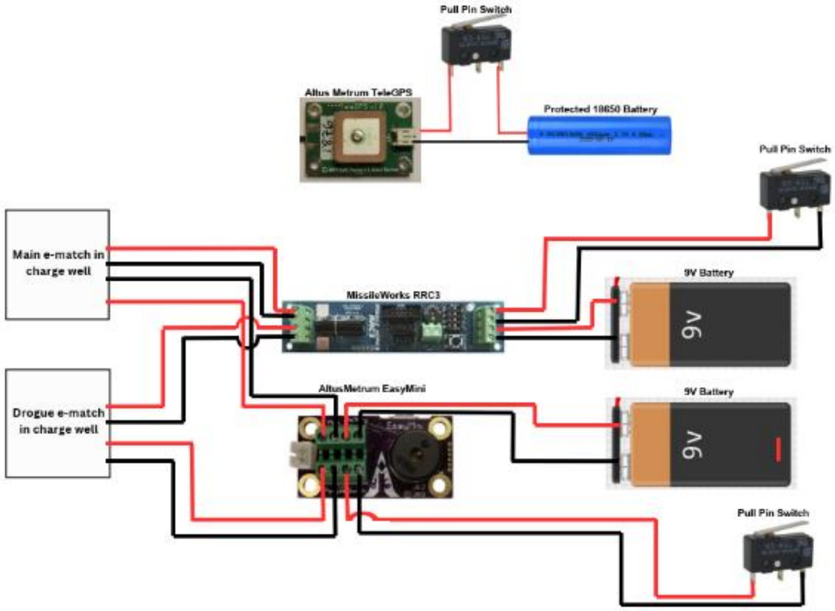

Fig. B-4: Recovery wiring diagram

The GPS chosen was the Altus Metrum TeleGPS, which communicates from the tracker unit in the nosecone of the rocket to an Altus Metrum TeleBT ground station unit over the 433 MHz band. The TeleBT is connected via Bluetooth to a phone application with an integrated map. The TeleGPS tracker unit is powered by a protected 18650 Li-Ion battery in a cylindrical casing.

The FFFFg black powder charges used for our airframe separation are the stored-energy devices in our recovery system. To ensure safe handling, the charges, including their e-match wires, are not loaded into the charge wells and connected to the terminal blocks until the rest of the rocket is assembled and the altimeters are confirmed to be unarmed via their pull pins. This ensures the charges cannot be inadvertently ignited during transport or assembly.

As stated, both altimeters are connected to pull pin switches through the avionics bay. When they are inserted, the power sources to each altimeter are electrically inhibited. The insertion points are chamfered and positioned closely to the coupler wall to allow for fast and easy pin insertion.

Prior to the launch, the pull pins for each altimeter are confirmed to be fully inserted, and neither altimeter is producing continuity across the e-match circuit. Once the rocket is in its final, upright position on the launch pad, the pull pins are removed in sequence to arm each altimeter. Each altimeter's beep-out sequence indicates successful arming, and only after each altimeter is confirmed to be in proper status is the rocket cleared to launch.

E. SRAD Propulsion System Testing

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F. SRAD Pressure Vessel Testing

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G. SRAD GPS Testing

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H. Payload Recovery System Testing

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Appendix C: Hazard Analysis

Table C-1: Aerostructures Hazard Analysis

Hazard	Possible Causes	Risk of Mishap and Rationale	Mitigation Approach	Hazard of Injury after Mitigation
Structural failure of the airframe or fins during launch, causing debris	Manufacturing defects (human error, voids, improper bonding) Insufficient safety factor in design	Medium, composite structures rely on proper fabrication and may fail under high aerodynamic loads	Ensure team members know exact manufacturing processes, procedures, and goals; Ensure proper surface preparation (sanding, cleaning); follow adhesive cure procedures; reduce air bubbles in epoxy; and inspect all bonded joints Design all structures with appropriate safety factors; test expected loading conditions	Low
Fin flutter is causing fin failure and debris	Poor Fin design and material selection	Low, Fin flutter can be easily calculated against	Calculate fin flutter velocity and ensure significant safety factors above maximum velocity, considering fin fluttering during the design process to ensure the fin design meets this requirement	Low
Rail guide or rail button failure at launch, resulting in a non-nominal flight path	Improper installation or insufficient fastener strength Rail binding due to misalignment	Low, good rail buttons should slide on the rail properly	Use standardized rail buttons; inspect rail buttons; inspect attachment and tighten before launch Ensure rail-button holes are one straight line, in line with the center of the rocket; perform rail fit test; ensure smooth movement along the rail	Low
Sharp composite splinters during fabrication and handling, sharp nosecone	Cutting fiberglass/carbon fiber can create splinters and sharp edges A sharp nosecone can stab someone during manufacturing or transport	High, common when working with composites and nosecone	Wear PPE (gloves, safety glasses) During the majority of manufacturing, the nosecone tip was wrapped and stored out of the way; the nosecone tip is wrapped during transport	Low
Composite dust exposure during fabrication	Sanding or machining composites	High, common when working with composites	Use an N95 mask during composite manufacturing. When significant dust is produced, use a respirator	Low

Table C-2: Payload Hazard Analysis

Hazard	Possible Causes	Risk of Mishap and Rationale	Mitigation Approach	Hazard of Injury after Mitigation
Direct eye/skin exposure to the laser	Mirror over-rotation Unintended activation Laser exits enclosure	Medium: limited testing and mechanical misalignment could lead to unintended beam exposure	Physical and software stops in place to limit mirror rotation Pull-pin keeps the system unpowered until final arming Laser-blocking acrylic viewing window and 6061 aluminum box	Low
Electronics or payload casing overheating, causing burns	Prolonged laser operation Insufficient cooling or ventilation Electrical or battery fault	Low; sustained operation can generate heat, but not enough to burn	Laser coded to shut off after 100 seconds, temperature sensors monitor the payload Ventilation holes are drilled, and the aluminum enclosure acts as a heat sink Power-off default state when a fault is detected	Low

Table C-3: Avionics Hazard Analysis

Hazard	Possible Causes	Risk of Mishap and Rationale	Mitigation Approach	Hazard of Injury after Mitigation
Unintentionally powered altimeters	Wrong altitude values, unintentionally powered	Low; could cause altimeters to send a signal for charges to blow	Pull pin switches are inserted before charges are connected, making the altimeters only active when wanted	Low power is no longer delivered to the altimeters to give any signals
Misconfigured altimeters	Switching to backup altimeters or after a previous test flight	Medium; could cause altimeters to fire at wrong altitudes or under unintended conditions	Pre-flight checklists include checking and changing the altimeter configuration before connecting to the charges	None

Table C-4: Propulsion Hazard Analysis

Hazard	Possible Causes	Risk of Mishap and Rationale	Mitigation Approach	Hazard of Injury after Mitigation
The motor is igniting near people	Nearby ignition sources reaching the motor	low; ignition sources would need to reach the interior of the motor	Keeping ignition sources away from the motor until the rocket is on the rail, keeping open leads shorted up until they are needed, and keeping the nozzle as closed as possible until launch	Low

Table C-5: Recovery Hazard Analysis

Hazard	Possible Causes	Risk of Mishap and Rationale	Mitigation Approach	Hazard of Injury after Mitigation
Black powder ignites during assembly	Accidentally connected to a direct power source	Low-power sources such as batteries are secured and kept away from charges	The assembly will only be conducted by informed members	Low
Unignited black powder charge detonation after recovery	"Dud" charges not ignited during flight	Medium; may cause bodily harm in the unlikely event of detonation	The recovery team will inspect charge wells from a safe distance, confirm full charge detonation, or dump extra black powder	Low; only if the recovery team fails to properly identify unignited charges
Parachute ejection failure	Tangled or ripped parachute, improper shock force calculations	High; a rocket will fall out of the air and may cause serious bodily harm if it collides with a person	Thorough inspection of the fabric, rigid folding routine, and mathematical and physical testing are done to ensure safe parachute deployment	Low; proper parachute deployment will slow the rocket to ~6 m/s, which may cause minor bodily harm if it collides with personnel
Main parachute deploys at apogee, rocket drifts to a restricted area	Flight computer improperly set, heavy winds	Medium; potential for high winds is sometimes unknown	Ensure harnesses, attachments, and altimeter settings are all properly armed and secure using checklists, and calculate against varying wind gusts	Low
Parachute ejection during pre-launch or assembly	Flight computer improperly set, pull pins are pulled, e-match prematurely energized	Medium: the rocket is bumped, shoved, or pull pins are knocked out, or if the avionics are not armed properly, the computer may detect apogee	Follow checklists closely during pre-flight assembly, pre-check e-match continuity, ensure pull pins are inside, disengage the recovery avionics, and carefully hold and assemble the rocket	Low

Appendix D: Risk Assessment

Table D-1: Risk Assessment Result Specifications

Level	Probability	Severity
Low	Approx. 1 in 100	No significant effect on the mission
Medium	Approx. 1 in 10	Degrades technical objectives or performance
High	More likely than not	Causes loss of mission success or safe recovery

A. Preflight

Table D-1: Preflight Risk Assessment

Risk	Possible Causes	Risk of Mishap and Rationale	Mitigation Approach	Risk after Mitigation
Ignition of the motor prior to arming the electronics and raising the rocket to the position on the rail	Open flames or heat sources near the motor and static discharge at the ignition leads	Medium, keeping leads not shortened could very well start the igniters	Keeping ignition sources away from the motor until the rocket is on the rail, keeping open leads shorted up until they are needed, and keeping the nozzle as closed as possible until launch	Low

B. Boost

Table D-3 Boost Risk Assessment

Risk	Possible Causes	Risk of Mishap and Rationale	Mitigation Approach	Risk after Mitigation
Airframe fails under motor thrust	Improperly designed and manufactured airframe	Medium; motor exerts high forces on the airframe	Inspections of the airframe during manufacturing, emphasis on accurate manufacturing of the motor tube and fin can, FEA with conservative safety factors	Low
Airframe connections fail from motor thrust	Improperly designed and manufactured airframe connections	Medium; motor exerts high forces on the airframe	Choose screws with a large safety factor and have couples overlap by 6 in (1 caliber)	Low
Fins cause structural failure	Improperly designed and manufactured fins	Medium; fins experience high aerodynamic loads and possible fin flutter	Through-wall fins with carbon fiber-reinforced epoxy fillets; FEA analysis on fins; inspections during manufacturing	Low
Rocket leaves the rail with low velocity	Low off-the-rail velocity	Medium; off-rail speed and wind sensitivity may affect early flight and trajectory prediction	Focus on maintaining adequate off-the-rail velocity	Low
Rail guide/rail button failure on the rail	The rail guide and rail button don't slide smoothly, and the rail button breaks	Medium; unstable rail departure may occur, reducing apogee prediction	Use proven metallic hardware, inspect hardware, and tighten before launch	Low
CATO	Dropping the motor on the nozzle and cracking it	High; if it occurred, the rocket would be destroyed	Careful handling of the motor with only L2 or higher personnel.	Low

C. Coast

Table D-2: Coast Risk Assessment

Risk	Possible Causes	Risk of Mishap and Rationale	Mitigation Approach	Risk after Mitigation
Fin flutter occurs near maximum velocity	Improperly designed and manufactured fins for fin flutter	Low can be designed against analytically	Adequate flutter margin greatly reduces the chance	Low
Airframe structural failure on the coast	Improperly designed and manufactured airframe for aerodynamic load	Medium: aerodynamic loads could cause failure	Ensure the integrity of the airframe during manufacturing	Low
Screws loosen in flight	Screws can loosen due to the vibration of the rocket	Low; screws in inserts are mostly reliable	Ensuring large factors of safety for screws so that the loss of a screw isn't catastrophic	Low

D. Apogee

Table D-3: Apogee Risk Assessment

Risk	Possible Causes	Risk of Mishap and Rationale	Mitigation Approach	Risk after Mitigation
Charge destroys the airframe	Improperly designed and manufactured airframe for charge pressure	Low; carbon fiber airframe should be able to resist the pressure	Ground testing and FEA simulations	Low
Underpowered electronics	Batteries connected for extended periods while not in use, or faulty connections	Medium; faulty batteries lead to electronics not being powered for enough duration or at all	Multimeters are used to check battery charge before flight in checklists and continuity checks conducted	Low
Faulty Connections	Transport of electronics can compromise continuity	Medium: leads to recovery or non-critical system failures	Continuity checks are conducted in the pre-flight checklist on every connection.	None
Altimeters Misconfigured	Switching to backup altimeters or after a previous test flight	Medium; could cause altimeters to fire at wrong altitudes or under unintended conditions	Pre-flight checklists include checking and changing the altimeter configuration before connecting to charges	None

E. Descent / Landing

Table D-4: Descent / Landing Risk Assessment

Risk	Possible Causes	Risk of Mishap and Rationale	Mitigation Approach	Risk after Mitigation
Recovery Misfire	Charges not blown during flight, so they would still be armed on the ground	Low; follow the checklists during the arming process	Immediately disarm recovery electronics when recovering the rocket	Low
Landing damages the airframe	Rough/hard terrain, fast descent rate, improperly designed and manufactured fins for landing	Low; carbon fiber airframe should be able to resist the impact	Design for expected landing loads; maintain acceptable descent rate; reinforce likely impact points; bring replaceable parts to launch in case a part is critically damaged	Low
Parachute Failure	Tangled or ripped parachute, shock force calculation inaccuracy	Medium: follow the checklists during the assembly process	Thorough inspection of the fabric, rigid folding routine, and mathematical and physical testing are done to ensure safe parachute deployment	Low
Payload laser still operational during recovery	Power disconnected and reconnected during flight	Low; all connections will be inspected prior to launch	If a power disconnect is detected, the system will be disarmed for the duration of the flight	Low

Appendix E: Checklists

A. Packing Lists

Recovery Packing List

Check	Item
	(12) Size 4-40 Nylon Shear Pins
	(1) 40' Main Harness
	(1) 40' Drogue Harness
	(1) Main Chute
	(1) Drogue Chute
	(6) Quick Links
	(2) Stainless Steel recovery bars
	(2) E-bay Bulkheads
	(2) E-Bay Stainless Steel Rods
	(1) Main Deployment Bag
	(1) Drogue Deployment Bag
	(1) Black Powder Cases
	(1) Black Powder
	(2) 4.00 -gram Drogue BP charge and 4.50 -gram Backup BP charge
	(2) 4.00 -gram Main BP charge and 4.50 -gram Backup BP charge
	(4) E-matches
	Zip Ties
	Finger Cots
	Screwdriver for Shear Pins
	Masking Tape
	Knife
	Multimeter
	Reefing Ring
	Pliers
	Spare Metal Washers
	Recovery Wadding
	Flashlight
	Sharpie
	Electrical Tape
	Nuts + washers

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______________________________ Assistant Team Lead Signature

Aerodynamics Packing List

Check	Item
	(12) Nose cone screws (8-32 x3/8")
	(3) Threaded inserts
	Nose cone
	Upper airframe
	E-Bay Coupler
	Lower airframe
	Fin Can
	Boat Tail
	Big Forward ballast weights

	Small forward ballast weights
	(20) Shear Pins
	(10) Fin Can Screws (8-32 x3/8”)
	(10) Boat Tail Screws (8-32 x3/8”)
	(3) Airframe Body Hex Screwdriver
	Rail Button Hex Screwdriver
	(4) Rail Button Sets
	(2) Lower Airframe Rail Button
	Flashlight
	Marker

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Propulsion Packing List

Check	Item
	(6) 8-32 hex screw
	Boat tail
	Allen wrench set
	Retainer
	Assembly Grease
	Nozzle
	(3) Ignitors
	(3) 1/8-inch diameter ignitor dowel rods
	Casing
	Aft closure
	Forward closure
	Fin Can
	Motor Retainer Key (Tool)
	Isopropyl alcohol
	Painter’s tape
	Gloves

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Payload Packing List

Check	Item
	Payload box
	TPU payload padding
	Steel balls
	Battery
	Laser Assembly
	Pull-pin
	Anodized aluminum samples
	USB-C cable
	Battery charger
	Laptop
	Ballast
	Bulkhead

	(2) Threaded rods
	(6) Nuts
	(6) Washers
	Metal cable
	(8) 8-32 hex screws
	Acrylic viewing window
	Top payload housing
	Bottom payload housing

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Avionics Packing List

Check	Item
	Go-Pro Hero 10 (Fully Charged)
	Go-Pro Windshield (Should be pre-mounted to the tube)
	Go-Pro Cable (Should be on E-Bay)
	128 GB SD card (in GoPro)
	Spare Micro-SD card
	Micro-SD to SD converter
	Small Apache Case (DAN)
	Li-Ion 18650 Charger(s)
	GPS Antenna (Altus Metrum 433 Yagi)
	Flight Computer Antenna
	SDR Kit
	Ground Station Computer (Fully assembled) The laptop and the external battery are both fully charged
	Full Assembly checklist:
	- USB-C to POWER
	- HDMI angled adapter
	- 4-slot USB Hub
	- (3) installed switches
	- Microcontroller
	- Antenna (Microcontroller)
	- (2) WLAN Antennas
	- 90 deg USB-C adapter (VIDEO)
	USB-C Cables
	USB-A/USB-C outlet
	Backup Battery
	Trinity Green Custom Flight Computer (Not using right now)
	Waymaker Black Flight Computer – (Not using right now)
	18650 Batteries with JST connectors x2
	Spare RP2040 Feather
	Custom BMS Board
	Multimeter
	E-Bay Pre-Assembled
	- RRC3
	- Altus Metrum Altimeter
	- Pull-pin switch (pre-bent)
	- (2) 9V's
	Nosecone Assembly

	- Altus Metrum GPS
	- Omega Custom Flight Computer
	- 433 LoRa antenna
	- GoPro (already charged and QR-coded)
	- (2) Pull-pin switches
	- GoPro cable
	USB-Micro-B Cable
	(4) Extra Blue 18650 batteries
	Extra 9V batteries
	LCD Ribbon Cables
	(2) Altimeter LCDs
	Extra 9v Connectors
	THE Screwdriver
	THE NEW Screwdriver
	The screwdriver kit
	Small Flathead Screwdriver(s)
	A Very Small Flathead Screwdriver
	ZIPTIES (All sizes)
	Screws: M3, M2.5, M2, M5
	Threaded Inserts: M3, M2.5, M2, M5
	Threaded inserts, soldering iron tips
	Soldering-Iron w/ Lead-Solder
	Hot-Glue Gun
	Wire strippers and clippers, wire snippers, needle-nose pliers
	(5) Voice Communication Radios FULLY CHARGED

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Avionics Charging List

Check	Item
	GoPro Hero 10
	(4) Blue 18650 batteries
	Fresh 9V batteries
	Ground station power bank
	Ground station laptop
	Walkie Talkies
	Screwdrivers

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Manufacturing/Tools Packing List

Check	Item
	(3) Folding Tables
	(7) Chairs
	Canopy
	Stepstool
	Tool Box
	Needle-nose Pliers

	Pliers (dup)
	Screwdrivers (small box, 3 flat heads, 4 Phillips, 2 square, 2 star)
	Assorted large screwdrivers (hook)
	Wire cutters
	Assorted Allen wrenches (dup)
	Scissors
	Box knife
	Exacto knife (dup)
	Tweezers
	Channel lock Pliers
	SILVER Sharpie markers (red and blue too) (dup)
	Pencils
	Hammer
	1/4” Wrench / Imperial Wrench Set
	Red Socket Wrench Set
	Black Socket Wrench Set
	3/8 to 3/8” socket wrench 20” extension
	Blue tape socket wrench extensions (backup)
	Roll of masking tape (dup)
	Roll of painter’s tape (dup)
	Roll of electrical tape (dup)
	Calipers
	Adjustable wrench
	Measuring tape
	Plastic Bags
	Trash Bags
	Fin can and Boat tail screw bag (dup)
	LED Light
	Tap and Die set
	Wooden Props
	Sandpaper
	Power drill
	Isopropyl Alcohol
	Shop Towels
	Rubber gloves
	Safety Glasses
	Power drill battery
	Drill bits

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B. Launch Day Checklist

Preliminary Considerations

Check	Item
	Site Rain
	Site Wind Speed
	Site Cloud Cover

	Site Visibility
	Tripoli Insurance
	LU Risk Management Insurance
	Launch Site Approval/Clearance
	FoR signed off
	Give God the glory

Chief Engineer Signature

Technical Advisor Signature

Payload Pre-Flight Checklist

Check	Action Item – * tasks should be performed the night prior
*	Battery is fully charged – check voltage
	Install threaded rods and the payload holder
	Test payload fitment
	Inspect wiring for loose connections and exposed wires
	Test-run code – ensure image is clear and mirrors moving as they should
	Test temperature shut-off
	Insert payload
	Install pull-pin

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Avionics Pre-Flight Checklist

Check	Action Item – * tasks should be performed the night prior
	Turn on all FRS radios and connect to channel 13
	Check to make sure everyone is receiving/transmitting
	Remove the lower bulkhead of the E-Bay Assembly and remove the coupler (This must be done)
	Test the battery voltage levels for all batteries to be used and ensure they are fully charged
*	Connect battery leads to altimeters. Make sure that each 9-volt lead is properly twisted together, and polarities are correct (Will be Preassembled, Verify)
*	Connect Pull Pin leads to altimeters; these are not polarity sensitive. (Will be Preassembled, Verify)
*	Verify that all the connections and wiring for the altimeters are properly connected and secure. (Will be Preassembled, Verify)
*	Make sure that all screw terminals connecting wires to altimeters are tight
*	Ensure no matches are connected to the drogue and main screw terminals
	Perform a tug test on all screw terminals with wires (so far as is possible)
	Are the batteries reading 8.5-9.5V? Always use Fresh Batteries
	Power on the RRC3 and examine beep codes
	Allow RRC3 to boot up. This process should take about 25 seconds. It will give one long beep followed by several short beeps
	Is the RRC3 giving any postcodes? If they are, these must be resolved before launch
	Ensure that at least 10 seconds have passed after powering on
	Power off the RRC3
*	Connect the ribbon connector to the RRC3 and LCD
*	Ensure that 10 seconds have passed after powering off
*	Power on RRC3. The LCDs should beep and turn on, giving you access to the home menu
*	Navigate to the view menu

*	Verify that the RRC3 settings are properly configured for flight and are set to the correct altitude parameters
*	Have you confirmed with recovery the deployment altitudes
*	Are the altitudes displayed on the LCD correct
*	Is the arming altitude correct
*	Is the Main altitude deployment correct
*	Is the RRC3 set to the right deployment modes? One should be mode 1 and the other mode 2
*	Is the drogue delay on the RRC3, denoted with deployment mode 2, correct?
*	Navigate to the diagnostics menu
*	Check for continuity on drogue and main
*	Check the pressure, temperature, and voltage in the PVT menu
*	Power off the RRC3 by inserting the pull pin
*	DISCONNECT THE LCD
*	Power on the EasyMini Altimeter
*	Unscrew EasyMini from the Recovery Bay
*	Connect the EasyMini to the laptop
*	Open the Altus Metrum app
*	Is the arming altitude correct
*	Is the Main altitude deployment correct
*	Is the EasyMini set to the right deployment modes
*	Verify other EasyMini settings
*	Disconnect the EasyMini from the laptop
*	Power off the EasyMini altimeter
	Power up the flight computer relay located on the nosecone assembly by connecting the battery
	Power up the ground station computer and run the GUI
	Ensure the GoPro USB-C cable is connected and secure
	Ensure cables are secure
	Perform a remote power-up check on Go-Pro
	Ensure all nuts and bolts are tight on the E-Bay Sled via a visual check
	Slide on Avionics Bay Coupler casing
	Connect the main chute quick-connectors and lock with a zip-tie and or electrical tape
	Ensure that the camera and pull pins are aligned with the coupler
	Bolt down the upper Avionics Bay bulkhead
	Connect the charged GPS battery to the GPS
	Power on the GPS
	Open the AltosUI app
	Connect to TeleBT-16656
	Press three dots (Top Right)
	Press Select Tracker
	Press Scan
	SCANNING for Live Trackers
	Select 15877 KR4BAA
	Wait for the GPS Locked to turn green
	GPS should begin reading values for soln and view
	Insert Nosecone Avionics Bay 1 (Aero)

	Ensure that the GPS is transmitting and receiving data
	Connect the Adafruit board cable and load up the GUI
	Ensure that the Adafruit board is working with the GUI
	Check to make sure that the GPS is transmitting the correct coordinates
	AERODYNAMICS PRE-FLIGHT ITEMS CAN START

Team Lead Signature   Assistant Team Lead Signature

Aerodynamics Pre-Flight Checklist

NOTE: The airframe is about to withstand significant amounts of stress, so apply a strong but controlled force.

Check	Action Item
	Update the simulation to match on-site wind conditions
	Update the simulation to match on-site atmospheric conditions
	Update the simulation to match the launch site location
	Update the simulation to match the on-site altitude
	Update the simulation to match the on-site launch rail length and angle
	Calculate weight configuration (Large = X, Small =0)
	Ensure correct forward weight configuration
	Slide ballasts on top of the upper airframe bulkhead
	Secure screw connections (Nosecone, Fin Can, Boattail, Rail Buttons)
	Check the camera cover for a tight fit
	Check shear pin count (6 Main, 4 Drogue)
	Check Coupling areas for scuffs, cracks, and dents
	Check Fins
	Check entire body for dents, scuffs, cracks, gouges (use hands to feel)
	Stand back and look at the assembled rocket. Airframe straight/continuous? Anything Missing?
	Shake assembled the rocket. Anything loose or rattling?
	Check the launch rail for the proper angle before mounting the rocket
	Mount with nosecone towards the front of the rail

Team Lead Signature   Assistant Team Lead Signature

Recovery Pre-Flight Checklist

Check	Item
	Drogue: Hold upside down and gather lines together near the base of the chute
	Line up the gores and fold them together
	Once the drogue is flattened, fold the drogue in half long ways
	Z-fold chute to the length of the Kevlar bag
	Invert the inner layers of the drogue parachute
	Z-fold lines
	Insert the parachute into the Kevlar bag, then place the z-folded lines on top. The swivel link should be hanging
	Main: Lay chute flat, with all the gores spread out
	Organize lines in pairs by gores, leaving the thick cord in the middle.
	Fold the pairs of lines on top of one another; the gores should follow naturally and be folded in half
	Fold the chute in half on the long way, check to ensure that this is thin enough to fit within the
	Z-fold chute, each section should be no longer than the length of the Kevlar bag
	Insert the chute into the bag, with the lines facing out of the bag

	Slide a reefing ring about halfway up the lines
	Z-fold lines carefully, ensuring the reefing ring is NOT in the middle of the z-fold, and insert them
	Fold the flap of the bag over, and check to ensure the parachute/lines cannot be seen when the flap is closed
	Shock Cord Folding: Z-fold the shock cords and secure with tape (This should be done with both the main and drogue tethers)
	Loading BP Charges: Take the loaded BP charge and insert it into the charge well. Ensure the drogue and main charges are on the correct bulkheads.
	Cut the e-match wires to size and screw the wires into the terminal block
	Create an X shape with tape, 3 layers thick, and place it over the charge well. Ensure the well is sealed
	Repeat steps 1-3 for all other charges
	Drogue Bay Assembly: Attach quick links to the sewn loop located on the longer side of the 40' tether, and to the middle-
	Attach the Kevlar bag to the quick link located at the end of the long section of tether
	Attach the parachute swivel link to the middle loop of the 40' tether
	Attach the Kevlar bag quick link to the eyebolt in the top payload bulkhead, which is located in the upper airframe.
	Tuck the fireball into the airframe, followed by the bundle of tethers
	Attach the sewn loop to the bar located on the avionics bay. Loop the drogue bag on the same bar, and pack with tethers in front of the bag
	Insert bar over threaded rods and secure using nuts and washers
	Pull out pull-pin
	Check for continuity
	Insert into the rocket
	Push in the pull pin
	Screw in shear pins
	Main Bay Assembly: Attach quick links to the short end of the 40' section of tether and to the middle-sewn loop
	Attach the sewn loop on the long section of tether to the aluminum bar. Loop the deployment bag
	Connect the bar to the two threaded rods and secure with nuts and washers
	Attach the swivel link to the quick link located in the middle-sewn loop
	Attach the short end of the tether to the eyebolt in the forward bulkhead. The fireball should line up with the edge of the airframe
	Tuck the fireball, then the Kevlar bag, then the bundle of tether into the airframe, and pack with tethers in front of the bag
	Screw in shear pins

Team Lead Signature   Assistant Team Lead Signature

Propulsion Pre-Flight Checklist

Check	Action Item
	Slide the motor into the motor tube
	Tighten the slimline over the aft closure
	Cut a section from the nozzle cap
	Shake the boat tail for any loose sections
	Check that the Boat tail is on
	Check that the Fin Can is Attached
	Igniters on side of rocket

Team Lead Signature

Assistant Team Lead Signature

Launch Day Operations

Check	Action Item
	Insert rail guides onto the launch rod
	Rocket vertical
	Activate the Screw Switch
	Correct Avionics sound guess
	Connect the ignitor wires to the power
	Continuity check (ignitor wires)
	Ignitor is in the motor
	Pad clear
	Sky Clear
	GPS Lock
	Faculty advisor "Go."
	Technical advisor "Go."
	Manufacturing lead "Go."
	Recovery team "Go."
	Avionics team "Go."
	Aerodynamics team "Go."
	Propulsion team "Go."
	Chief Engineer "Go"
	Enter the Final Countdown process
	Ignition

Chief Engineer Signature

Technical Advisor Signature

Non-Catastrophic Launch Scrub Disarming Process (Off-Nominal)

Check	Action Item
	Wait as long as 2 minutes
	Re-enter the Final Countdown process
	Re-attempt ignition
	If another ignition failure occurs, disconnect any physical connection that would allow power to flow to the ignitor
	Wait at least 2 minutes before approaching the launch pad
	Remove ignitor
	Deactivate switches (using a magnet and pull pin(s))
	Lower the rocket's horizontal
	Remove the rocket from the launch rail
	Inspect the rocket and launch site for the source of the ignition failure

Avionics Error During Switch Activation (Off-Nominal)

Check	Action Item
	After arming the avionics bay with the pull-pin and/or magnetic switches, either no sound cues are heard, or the wrong sound cue is heard
	Deactivate switches

	Lower the rocket's horizontal
	Remove the rocket from the launch rail
	Inspect the rocket for the source of the failure

GPS Lock Lost After Arming (Off-Nominal)

Check	Action Item
	GPS Lock is lost after exiting the launch pad
	Disconnect any physical connection that would allow power to flow to the ignitor
	Wait at least 2 minutes before approaching the launch pad
	Remove ignitor
	Deactivate screw switches
	Lower the rocket's horizontal
	Remove the rocket from the launch rail
	Inspect the rocket for the source of the failure

BP Charges Do Not Ignite After Landing (Off-Nominal)

Check	Action Item
	Rocket has a successful recovery and lands on the ground safely
	Upon approach, check BP charges for ignition
	If not ignited, deactivate switches (Ensure the loaded BP charge is facing away from the person deactivating)
	Unscrew the e-match wires from the terminal block. (Ensure BP charge is still facing away)
	Remove BP charges from charge wells
	Twist ignitor ends to ensure no ignition
	Dump BP Powder out of the finger cot, save the e-match in the proper storage container

Appendix F: Engineering Drawings

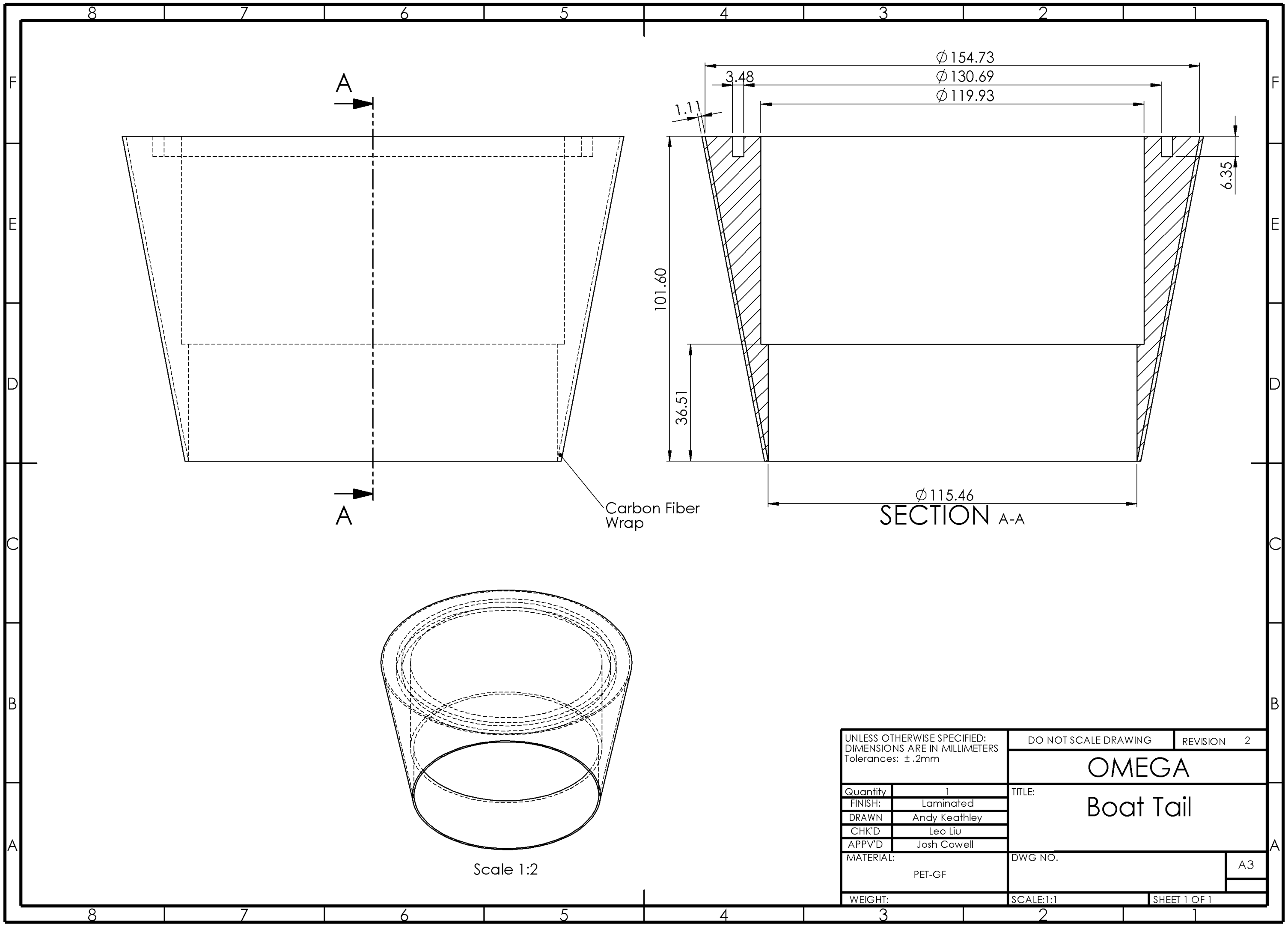


Fig. F-1: Boat tail engineering drawing

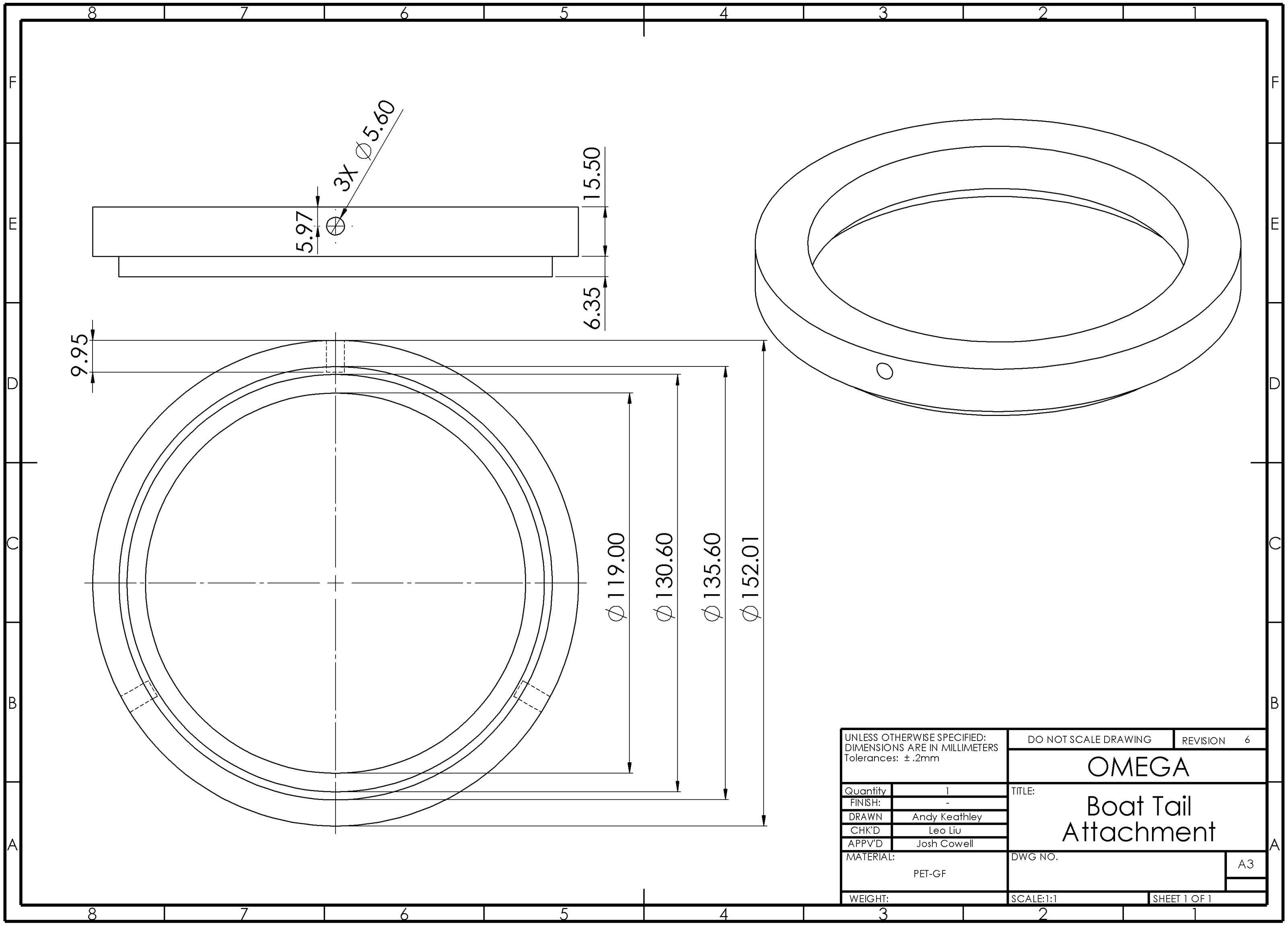


Fig. F-2: Boat tail attachment

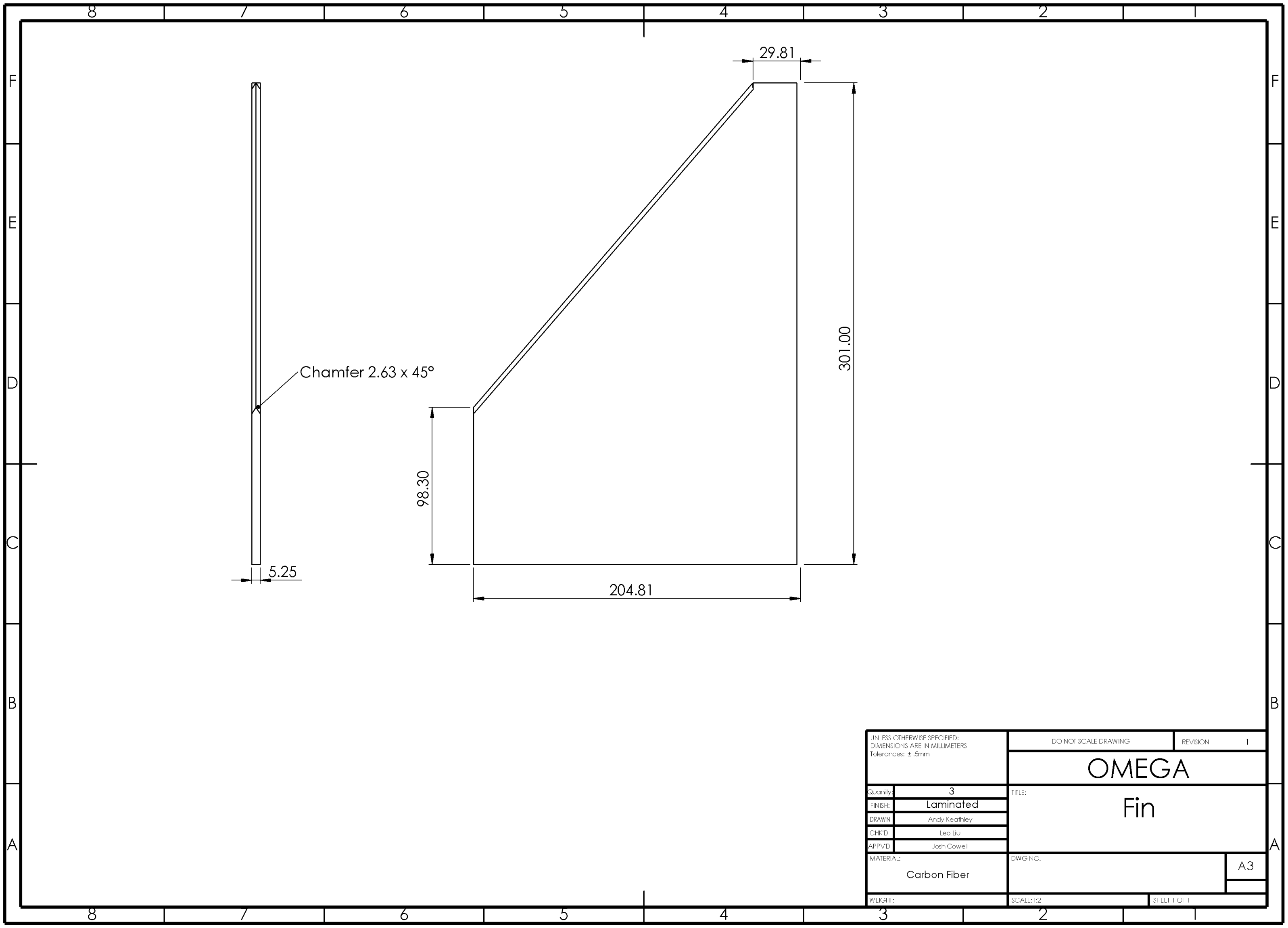


Fig. F-3: Fin engineering drawing

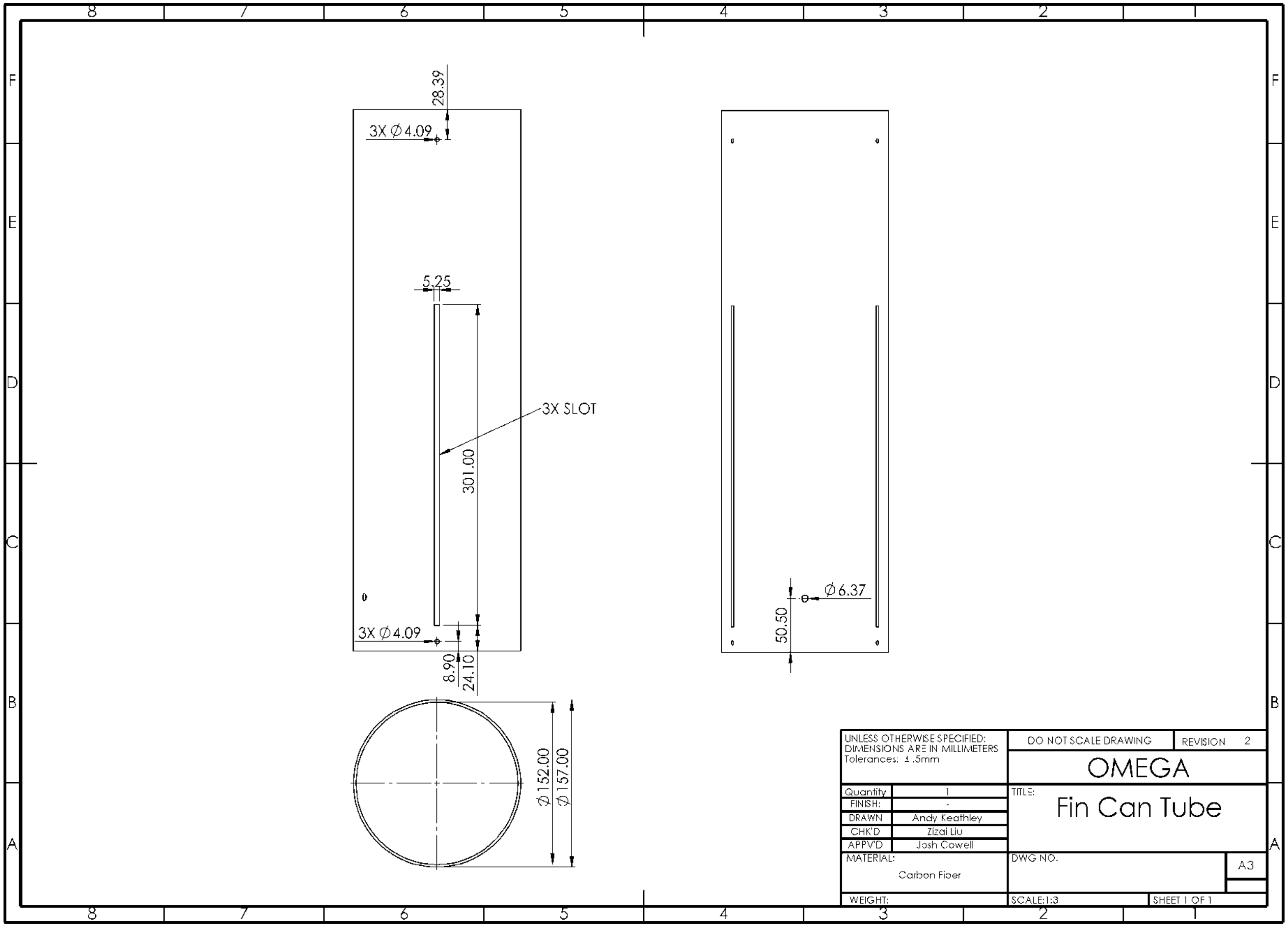


Fig. F-4: Fin can tube engineering drawing

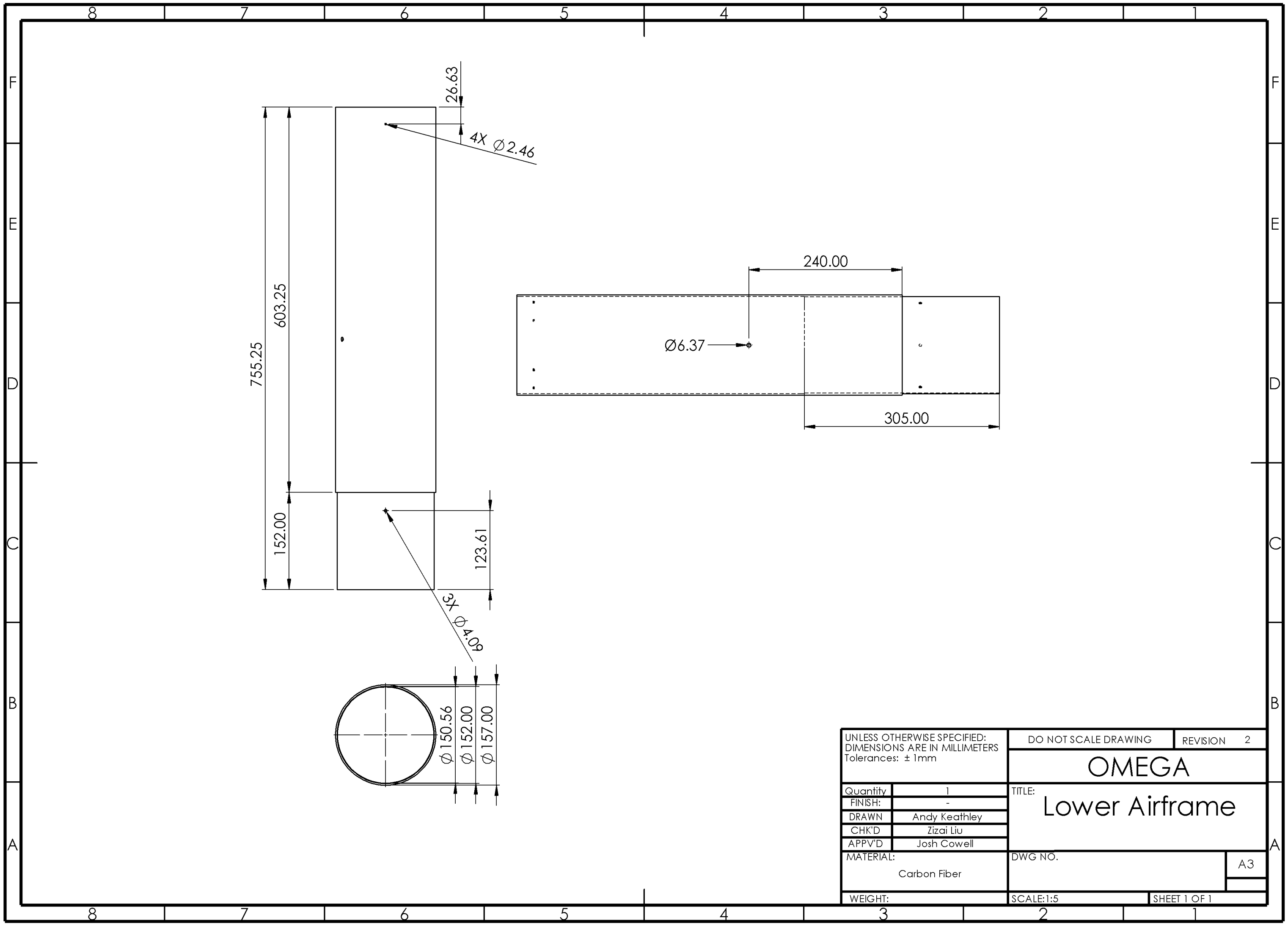


Fig. F-5: Lower airframe engineering drawing

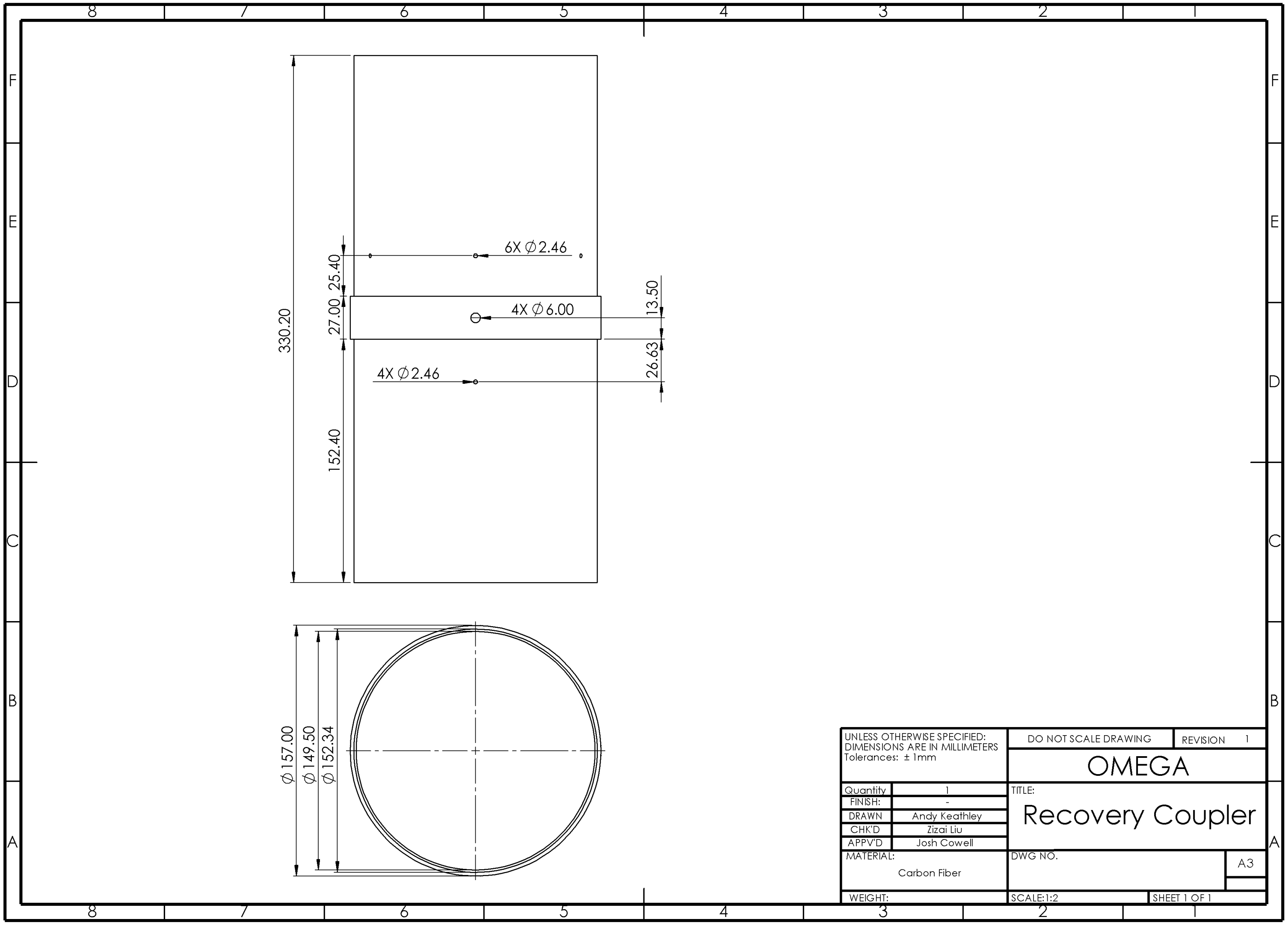


Fig. F-6: Recovery coupler engineering drawing

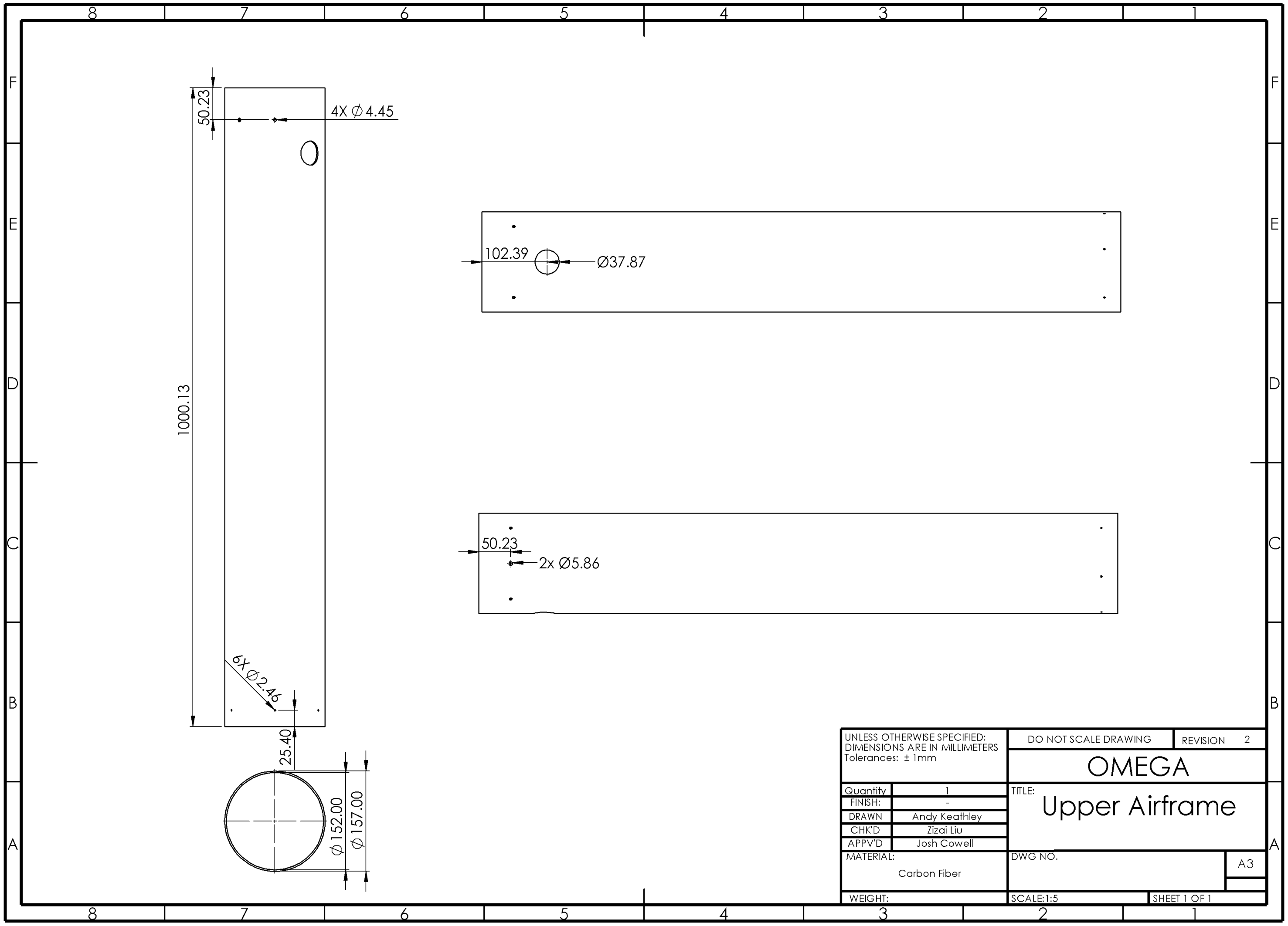


Fig. F-7: Upper airframe engineering drawing

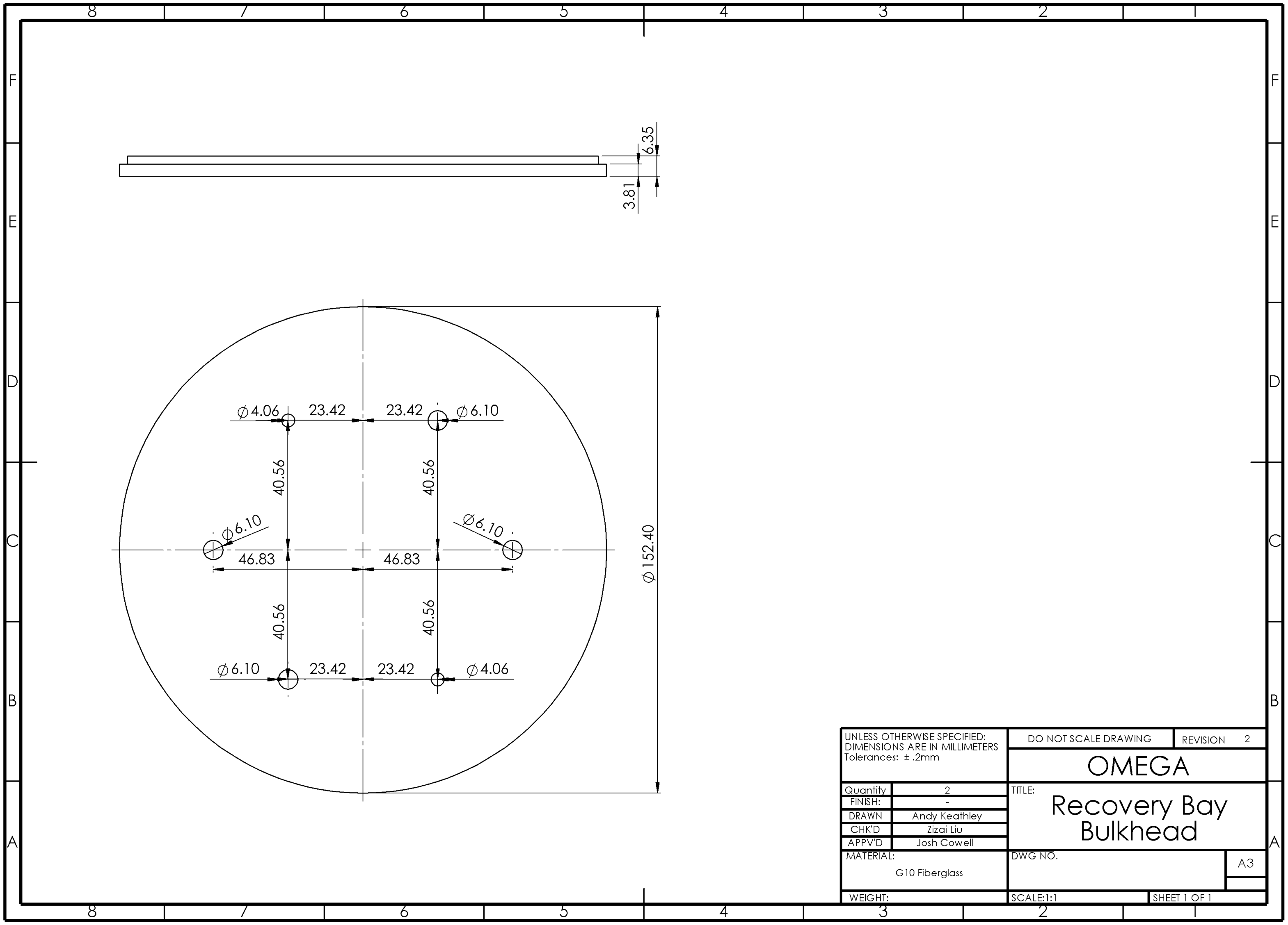


Fig. F-8: Recovery bay bulkhead engineering drawing

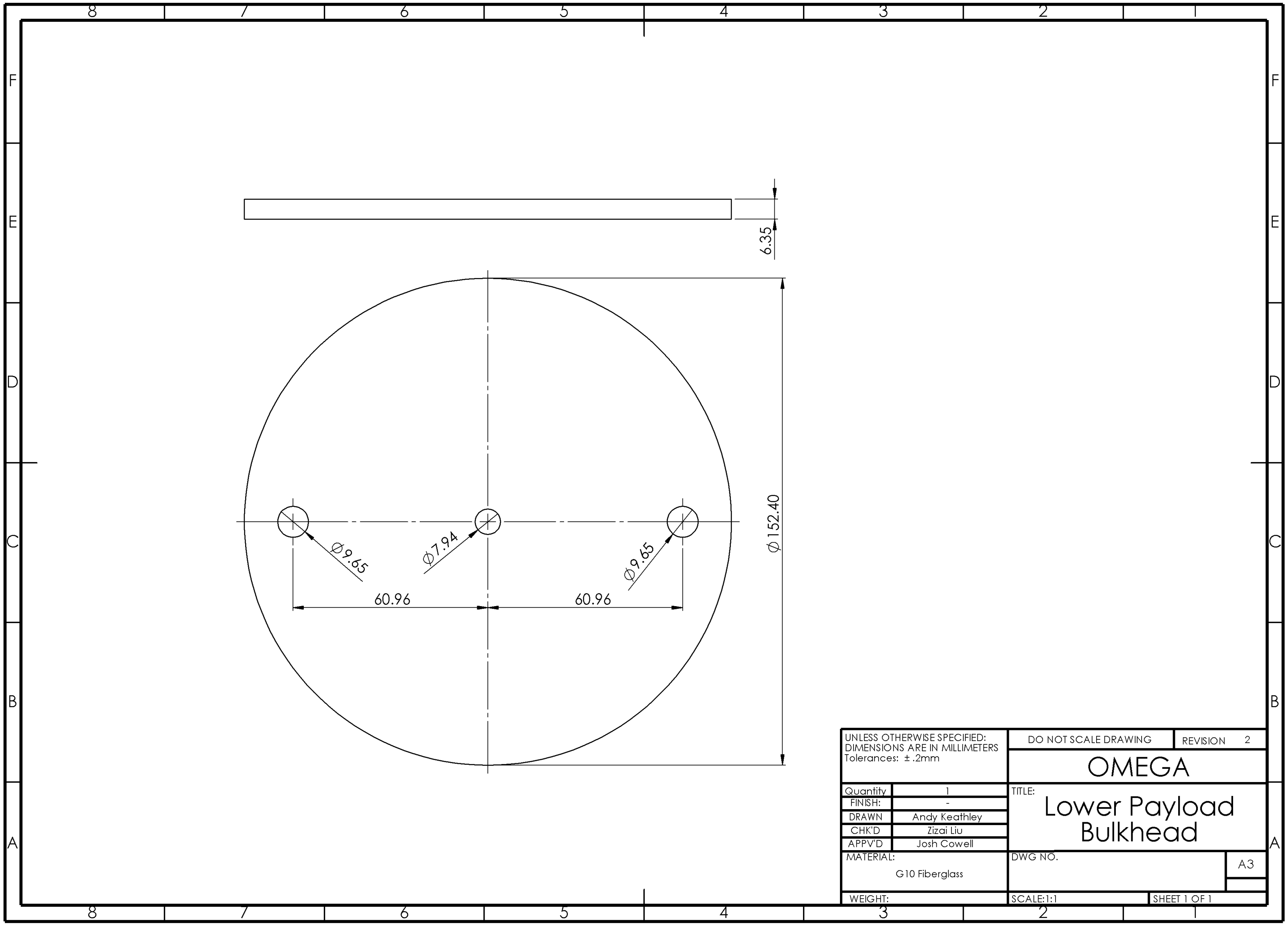


Fig. F-9: Lower payload bulkhead engineering drawing

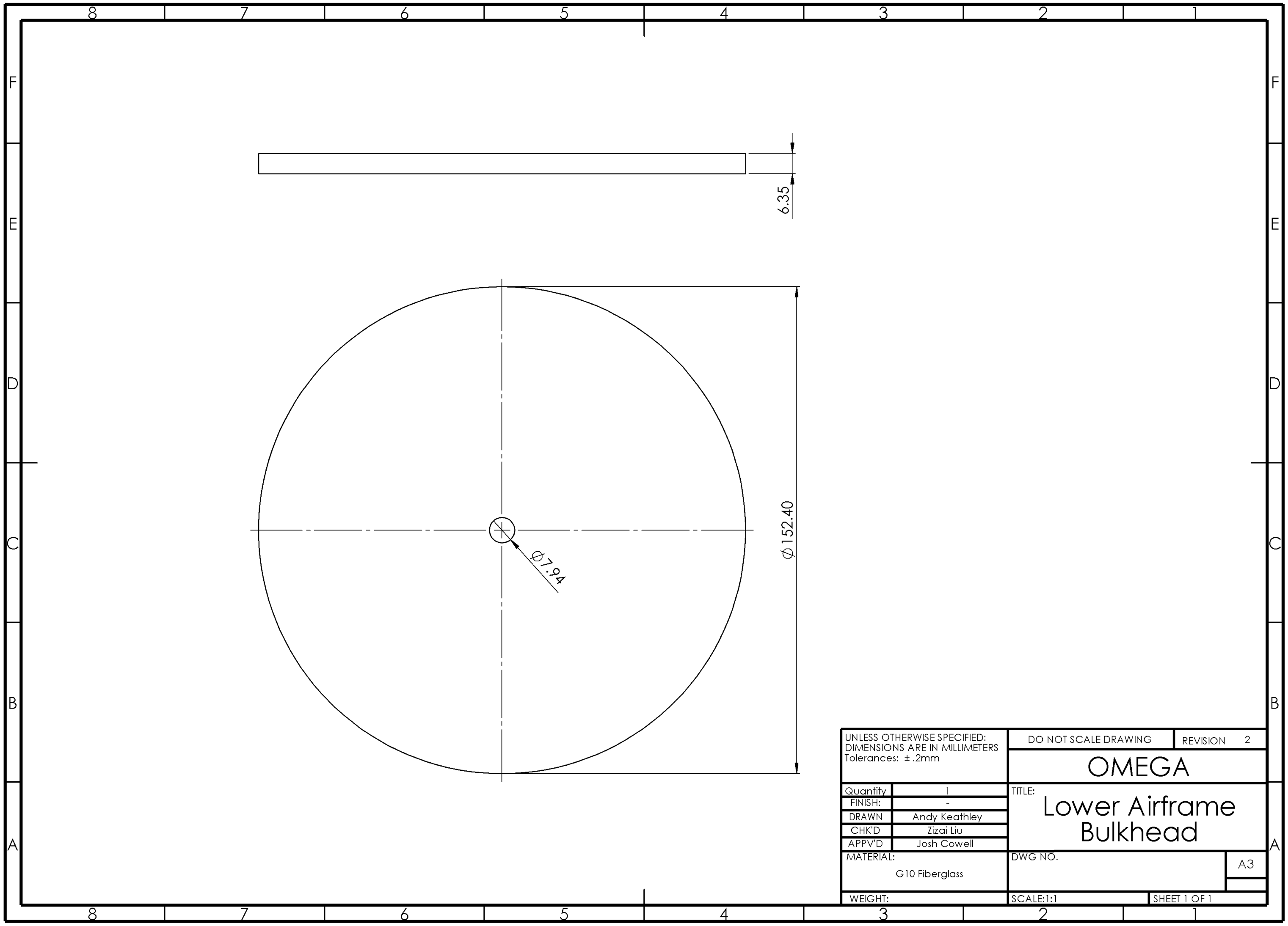


Fig. F-10: Lower airframe bulkhead engineering drawing

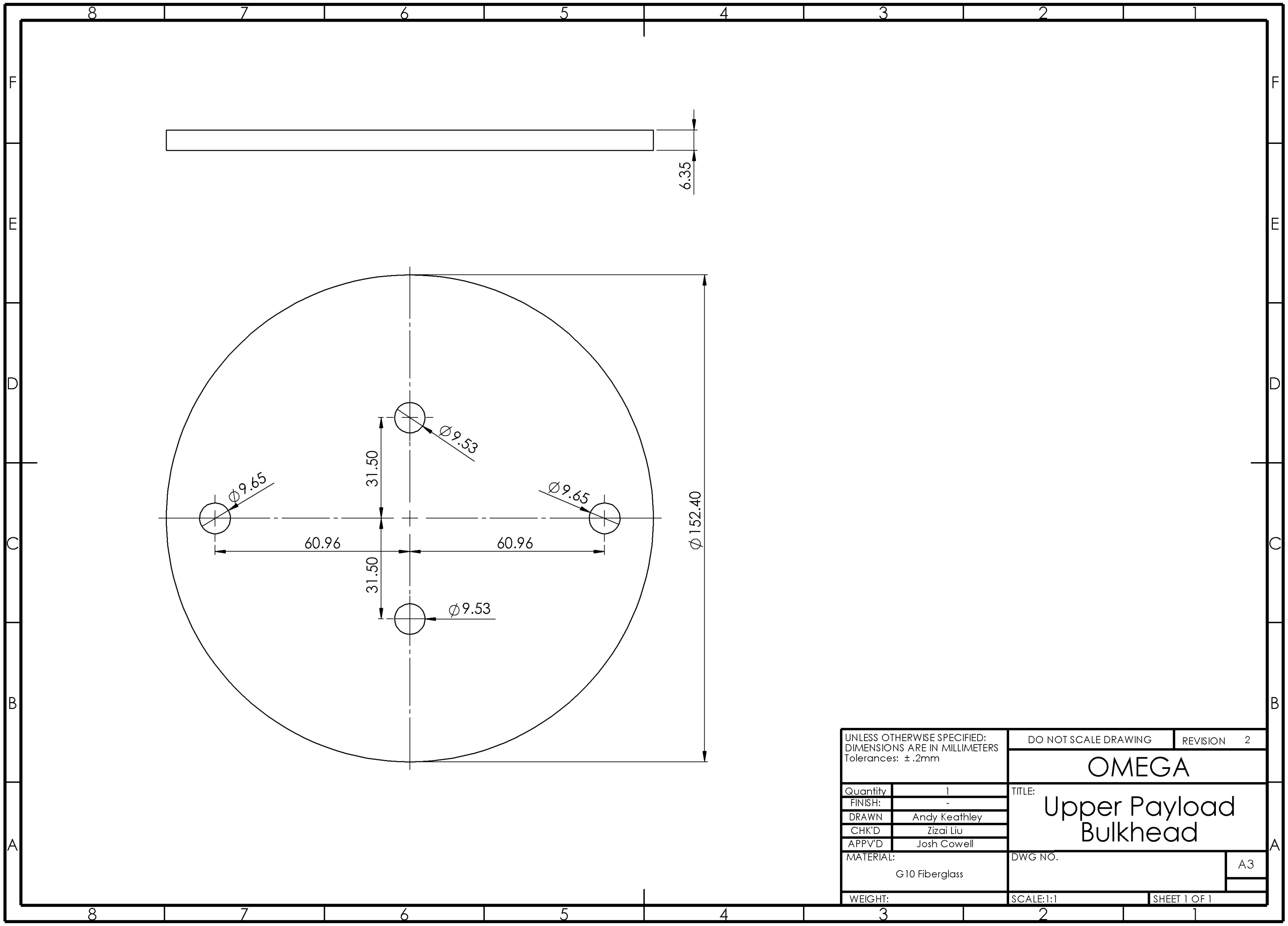


Fig. F-11: Upper payload bulkhead engineering drawing

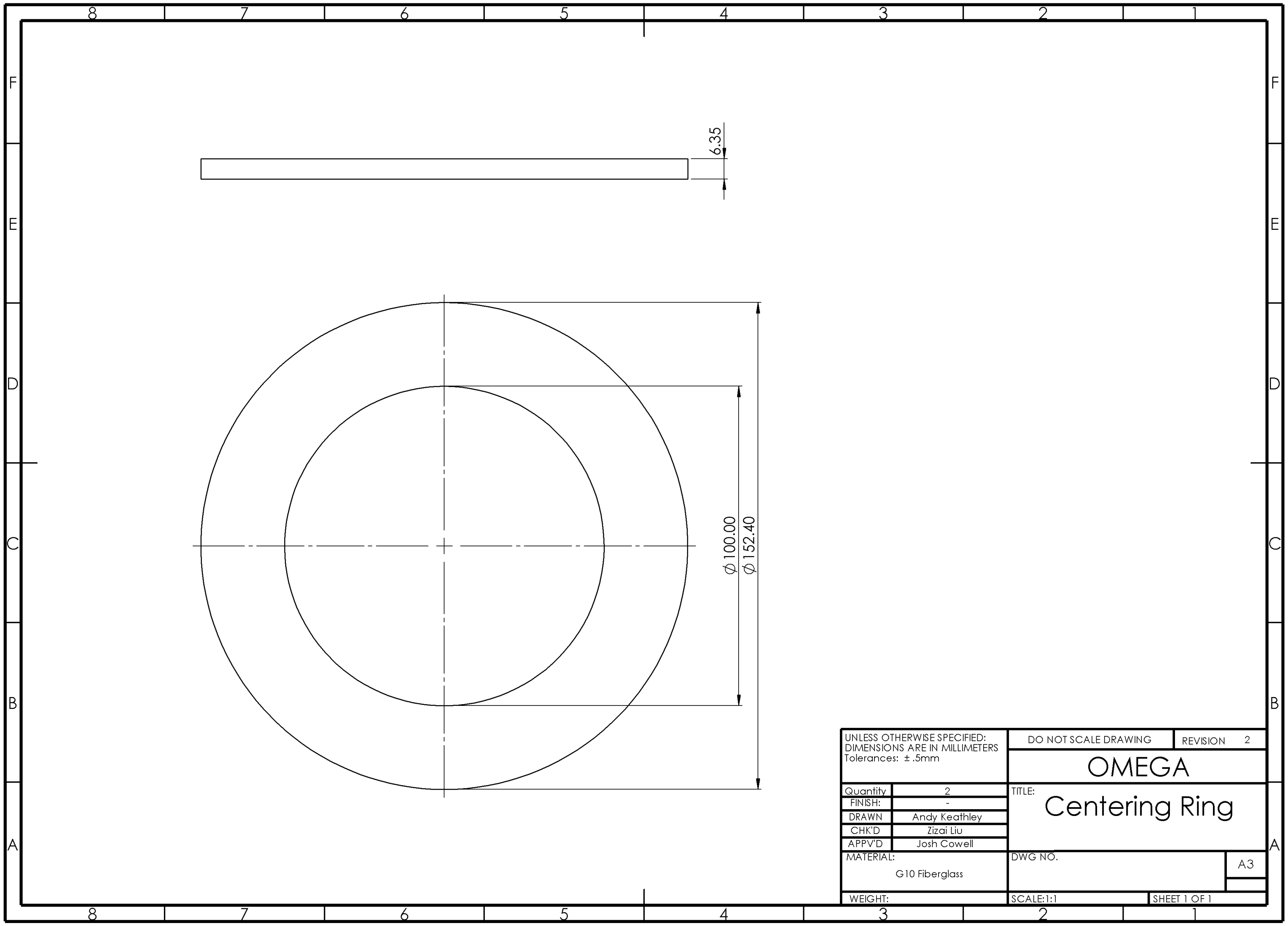


Fig. F-12: Centering ring engineering drawing

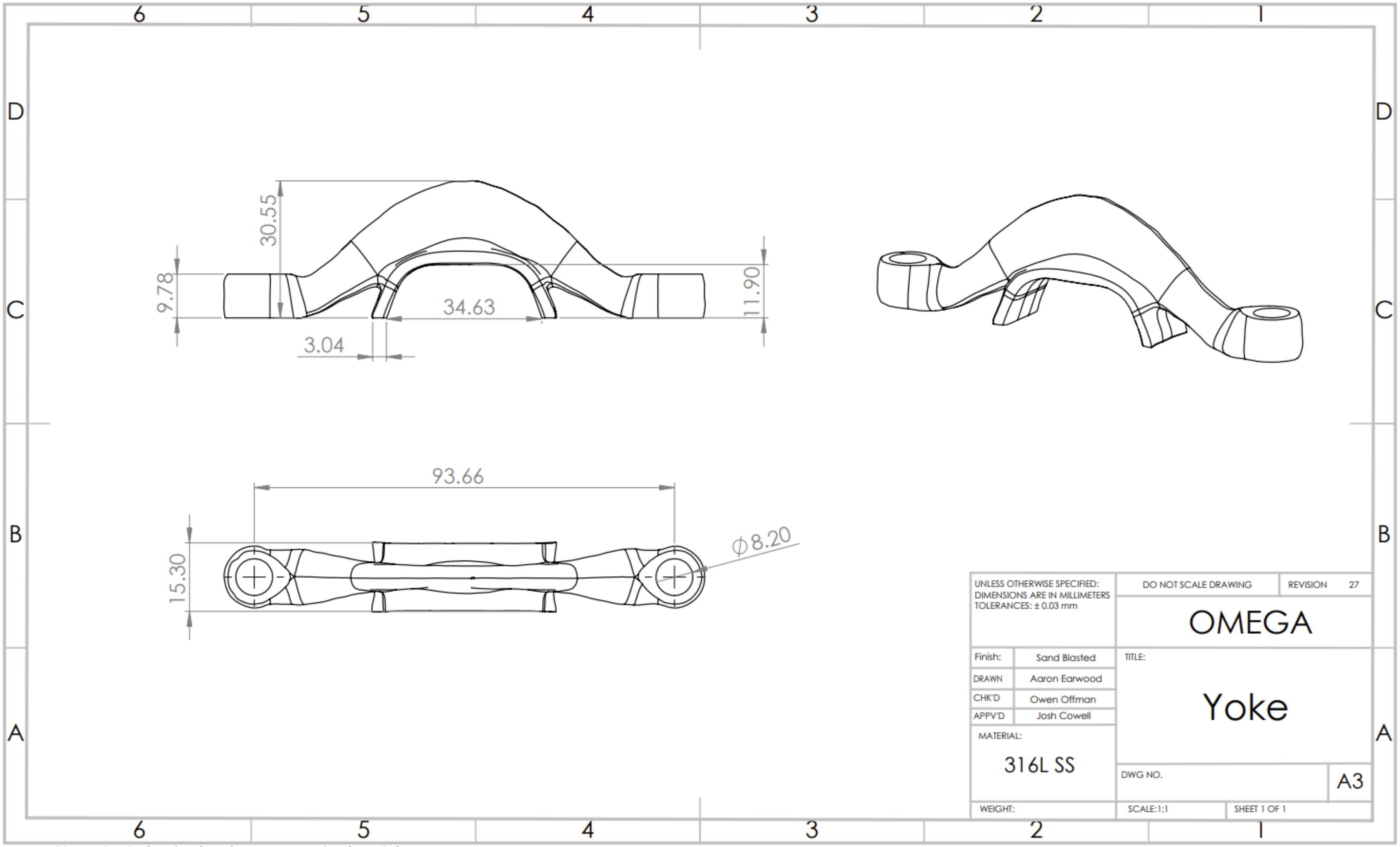


Fig. F-13: Yoke engineering drawing

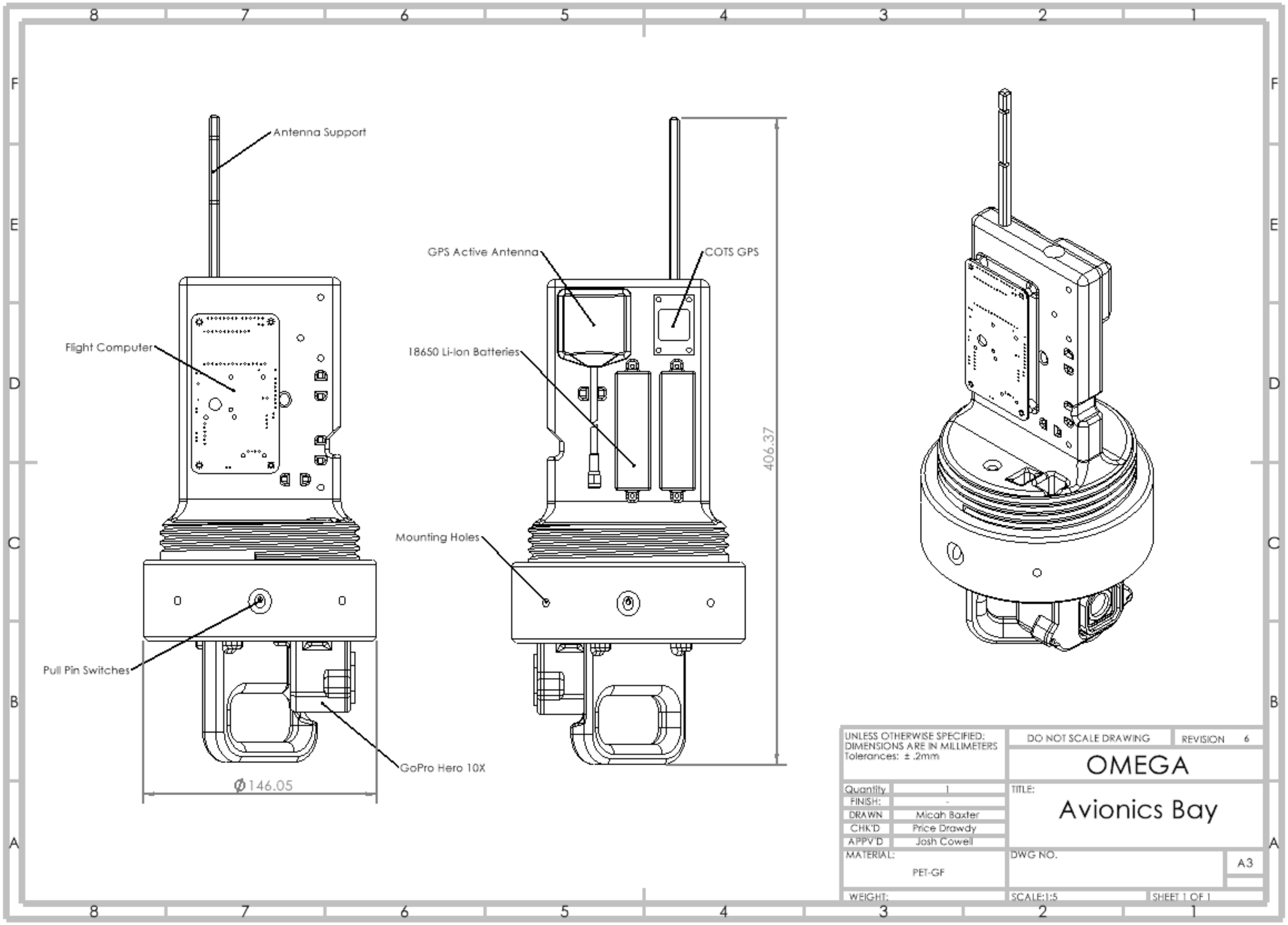


Fig. F-14: Avionics nosecone bay engineering drawing

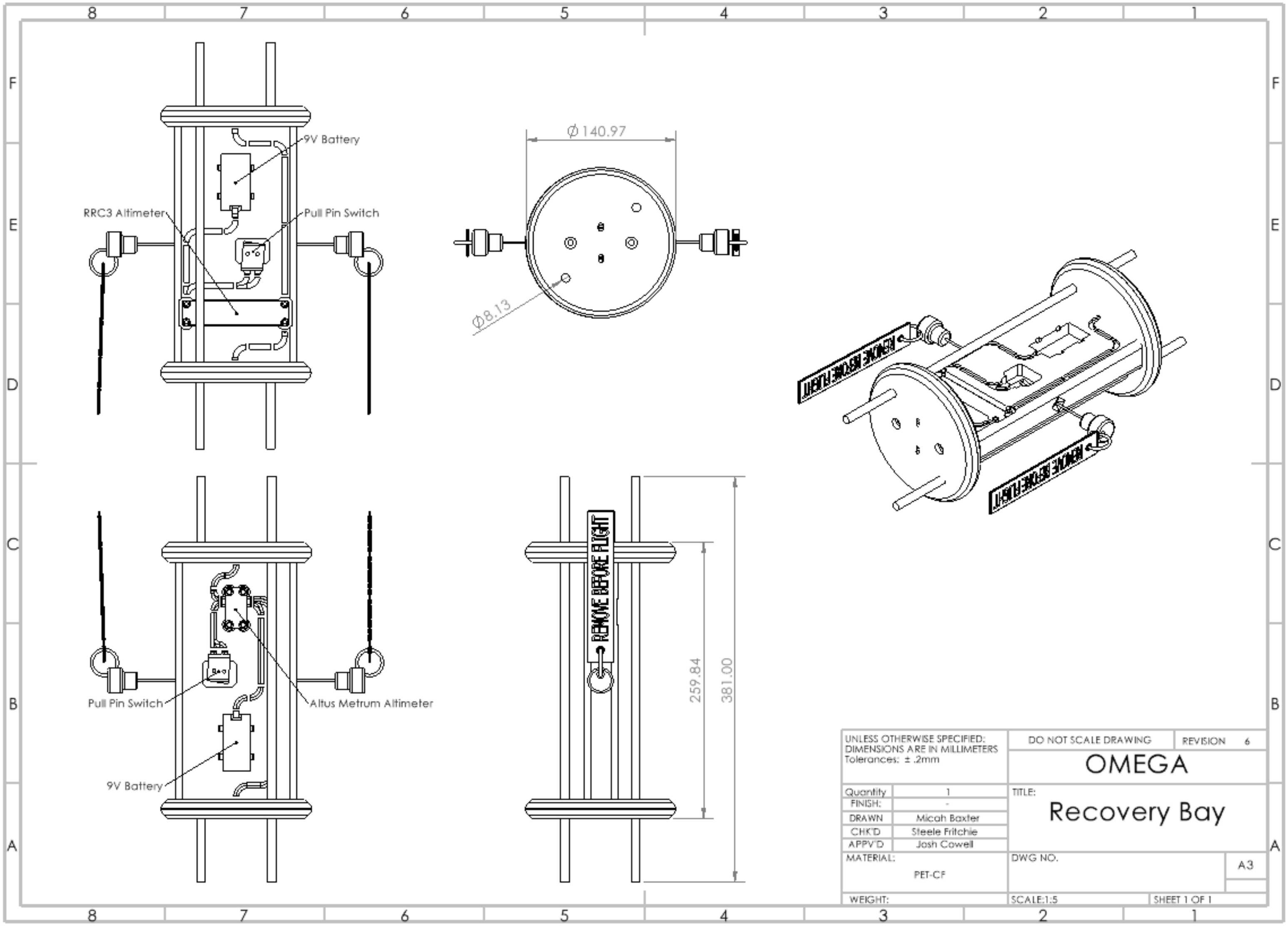


Fig. F-15: Avionics recovery bay engineering drawing

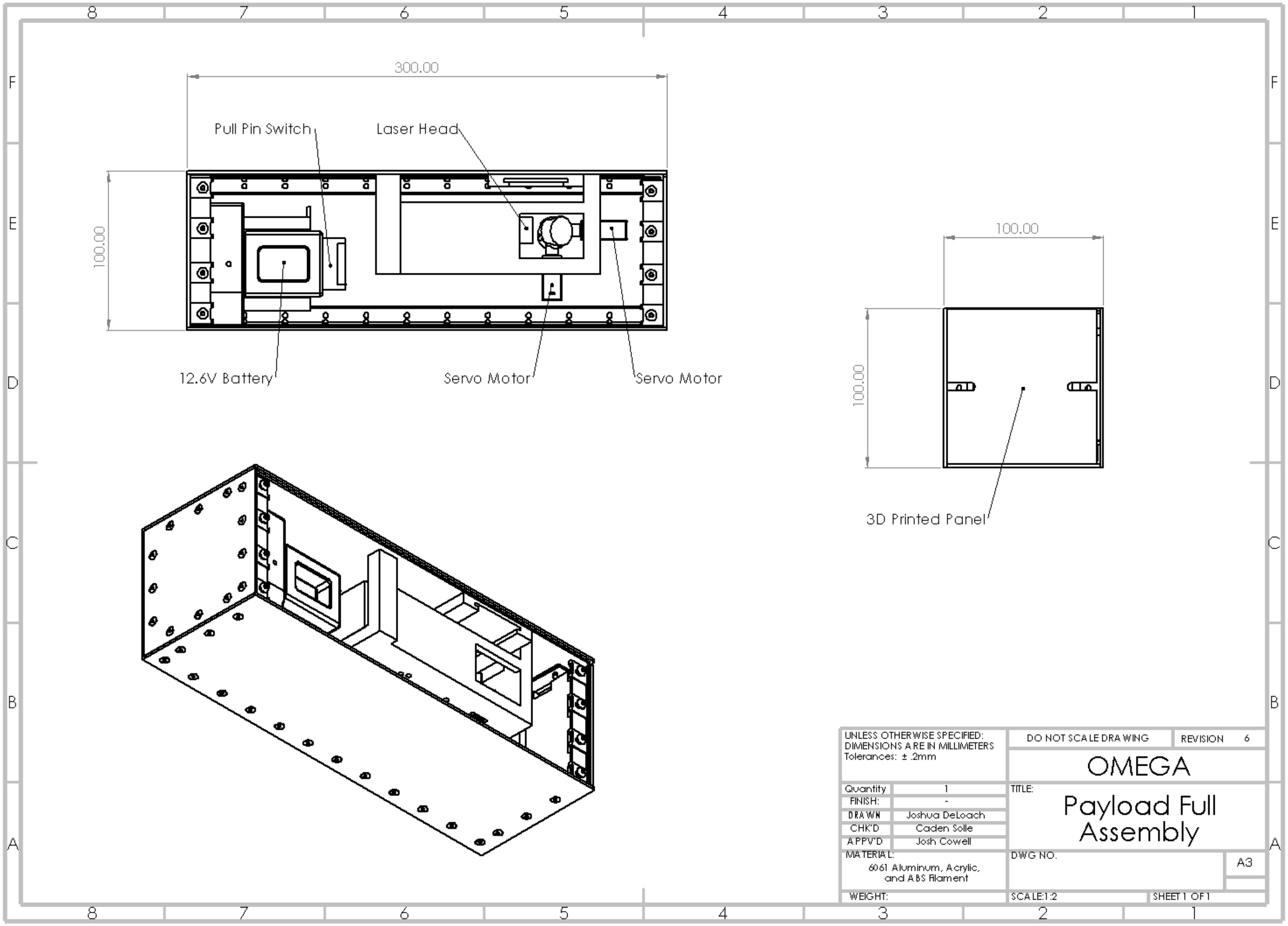


Fig. F-16: Payload full assembly engineering drawing

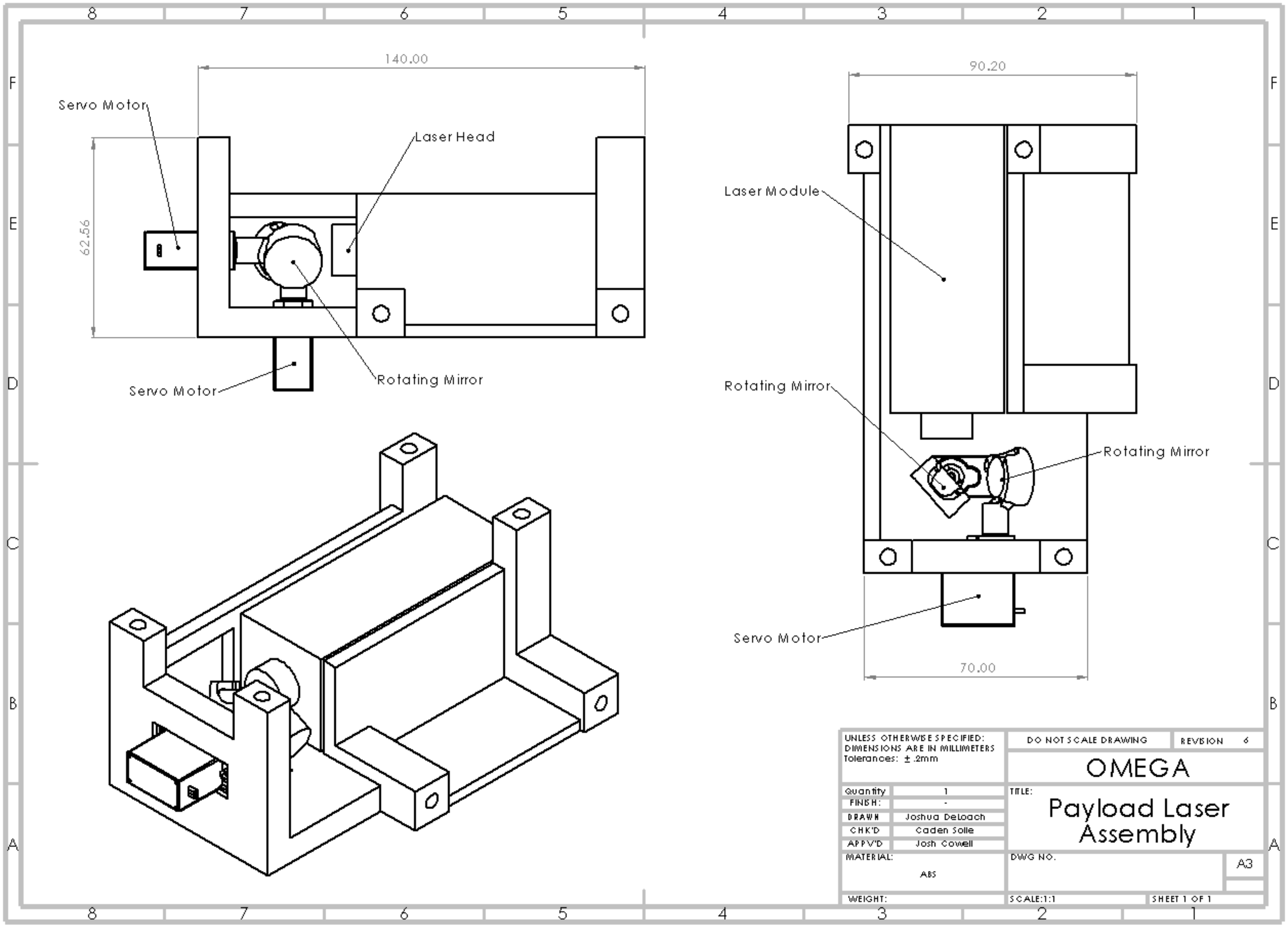


Fig. F-17: Payload laser assembly engineering drawing

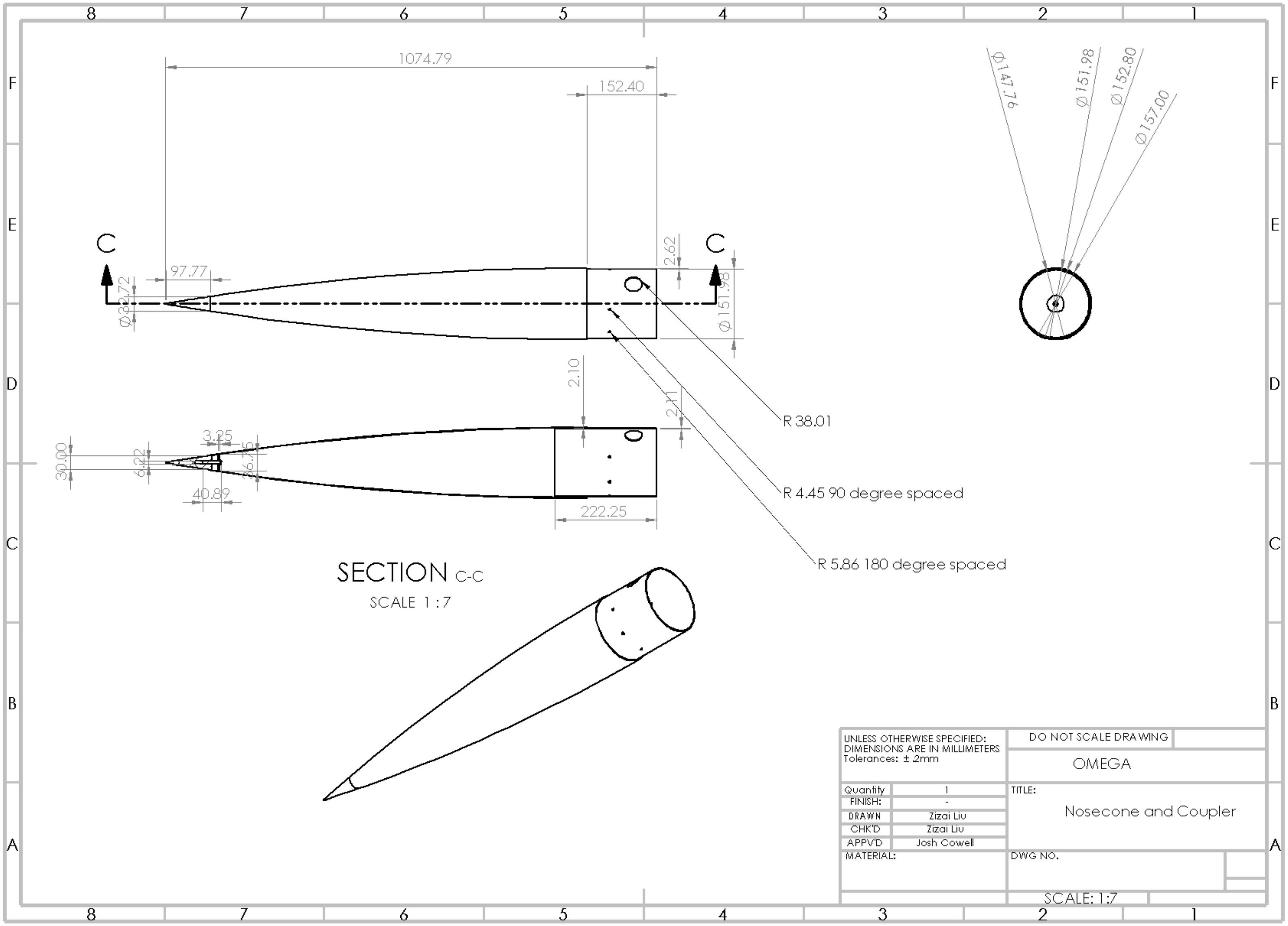


Fig. F-18: Nosecone and coupler engineering drawing

Appendix G: Aero-Structure Supplementary

A. Constraints and Design Objectives

Table G-1: Constraints and Design Objectives

ID	Type	Description	Justification
C.1	Constraint	Fin span shall be no less than 0.8 calibers of rocket diameter (110 mm for Omega)	DTEG 10.2.4
C.2	Constraint	Minimum stability margin shall be ≥ 7.5% of rocket length for subsonic flight (1.513 caliber for Omega)	DTEG 10.2.1.1
C.3	Constraint	Maximum stability margin shall not exceed 18% of rocket length at launch (3.633 caliber for Omega)	DTEG 10.3.1
C.4	Constraint	Maximum stability margin shall not exceed 25% of rocket length during flight (5.046 caliber for Omega)	DTEG 10.3.2
C.5	Constraint	Fin flutter velocity shall be at least 50% greater than the maximum expected rocket velocity (609.62 m/s for Omega)	DTEG 8.2.2
C.6	Constraint	Rocket shall be adequately vented to prevent internal pressure buildup during flight	DTEG 8.1.1
C.7	Constraint	Airframe shall maintain structural integrity under all expected loads	DTEG 8.2.1
C.8	Constraint	PVC, PML Quantum Tube, and steel, including stainless steel, shall not be used in airframe structures	DTEG 8.3.2
C.9	Constraint	Coupler overlap shall be at least 1 caliber for separation joints (152.4 mm for Omega)	DTEG 8.5.1.1
C.10	Constraint	Rail buttons must be mechanically fastened and reinforced	DTEG 8.6.4–8.6.5
C.11	Constraint	A minimum of two rail buttons shall be used	DTEG 8.6.2
C.12	Constraint	Rail buttons must support full rocket weight while the rocket is horizontal	DTEG 8.6.9
C.14	Constraint	Apogee must be within ±30% of 3048 m (10,000 ft) AGL	RRD 2.8.1.4
C.15	Constraint	Rail departure velocity shall be at least 25 m/s	DTEG 5.3.1
C.16	Constraint	Carry a payload of at least 2kg	RRD 2.3.1
C.17	Constraint	If a non-functional payload is used, it shall conform to a 3U form factor (100 mm × 100 mm × 300 mm)	RRD 2.3.5
D.1	Design Objective	Integrate successfully with other rocket subsystems	Crucial for a functional rocket
D.2	Design Objective	Remain within the allotted budget	Work within the given budget
D.3	Design Objective	Achieve apogee within 100±3% of 3048 m (10,000 ft) AGL	Increase competition ranking
D.4	Design Objective	Must use a radio-frequency transparent component for avionic communication	Request from avionics subsystem
D.5	Design Objective	Maintain accurate OpenRocket files for each component's mass and center of gravity	Improve apogee prediction
D.6	Design Objective	Create accurate CAD files of the rocket	Improve ANSYS simulation accuracy

B. FEA

To ensure that the airframe of Omega would be safe from all expected structural forces during flight, the Liberty Rocketry team performed finite element analysis simulations using Ansys 2025 R1 simulation software. The following are descriptions and results from each of the simulations.

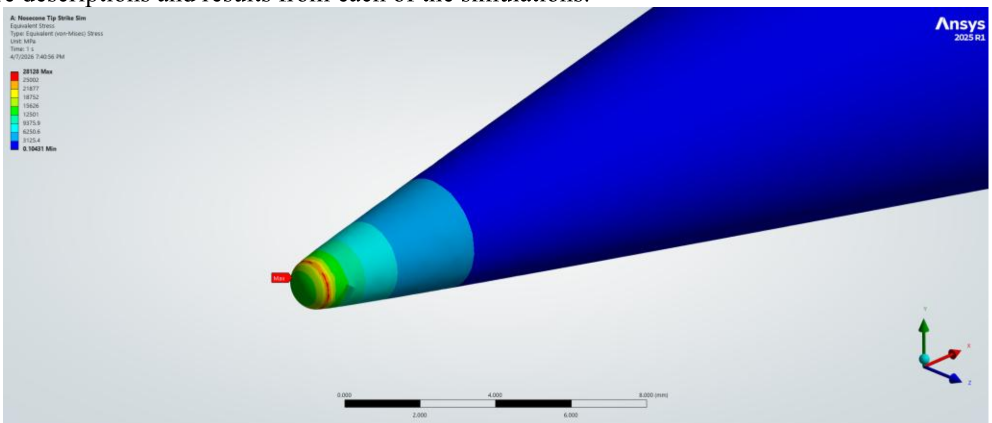


Fig. G-1: Equivalent stress results from the nosecone tip-first ground impact simulation

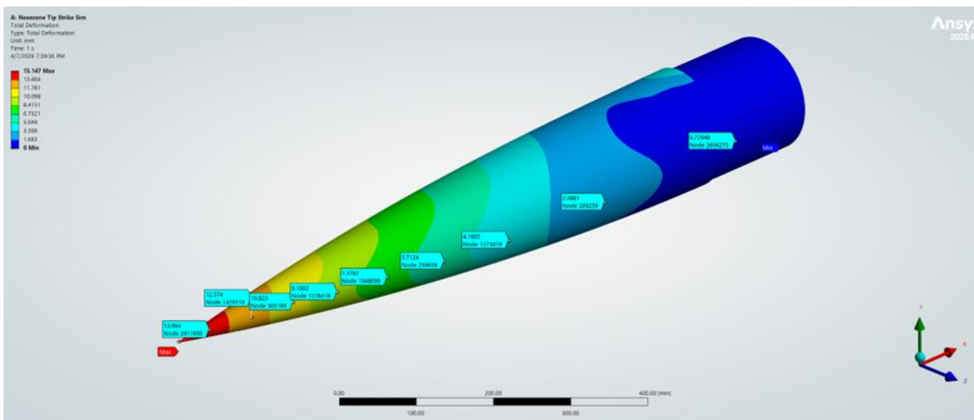


Fig. G-2: Deformation results from the nosecone tip-first ground impact simulation

The first simulation that the team performed was to model the force delivered to the tip of the 6:1 Tangent Ogive Fiberglass nosecone, where the nosecone was to impact the ground tip-first at 6.42m/s (see Fig. G-1 to G-2). The 22.4 kN force calculated for this impact assumed a rigid ground that deformed little during the impact. This assumption generated a large impact force. The deformation of the nosecone is shown in Fig. G-1 with a maximum deformation of 15.147 mm at the tip of the nosecone. The tip experienced the stress shown in Fig. G-2, concentrated almost exclusively at the tip of the nosecone, where the maximum stress is on the order of 30 GPa, far exceeding the aluminum's compressive yield strength by a factor of around 100. Figure G-1 shows a very close view of the tip. The stress at the tip cannot be considered precise due to mesh sizing limitations, but it shows that the aluminum tip will experience damage from extreme loading conditions and will need to be replaced; the rest of the nosecone will remain structurally integral. The physical nosecone purchased has a more rounded tip compared to the ideal tangent ogive profile, which will greatly decrease the stress experienced in the critical section. It should also be noted that all past rockets made have used fiberglass nosecones with the same metal tip, and the team has never experienced serious structural damage to the nosecone.

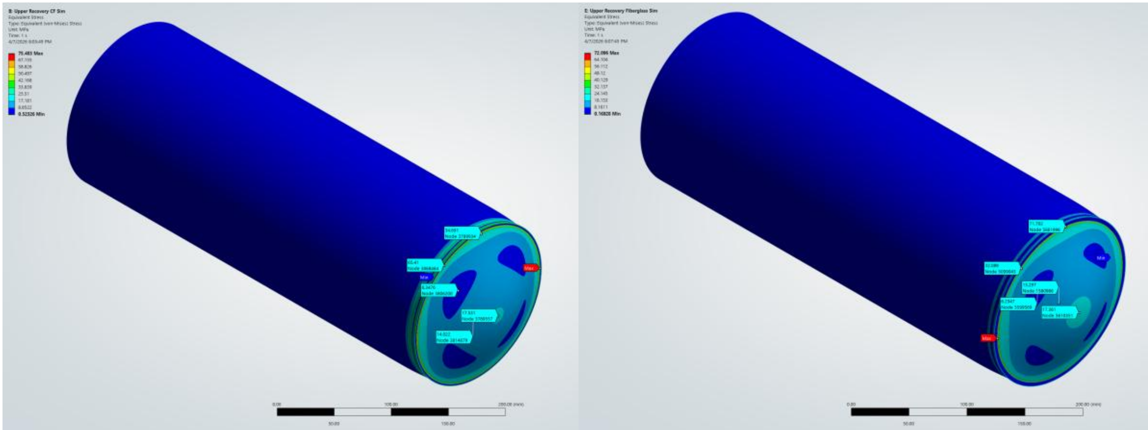


Fig. G-3: Carbon fiber (left) and g10 fiberglass (right) equivalent stress from upper airframe black powder recovery separation

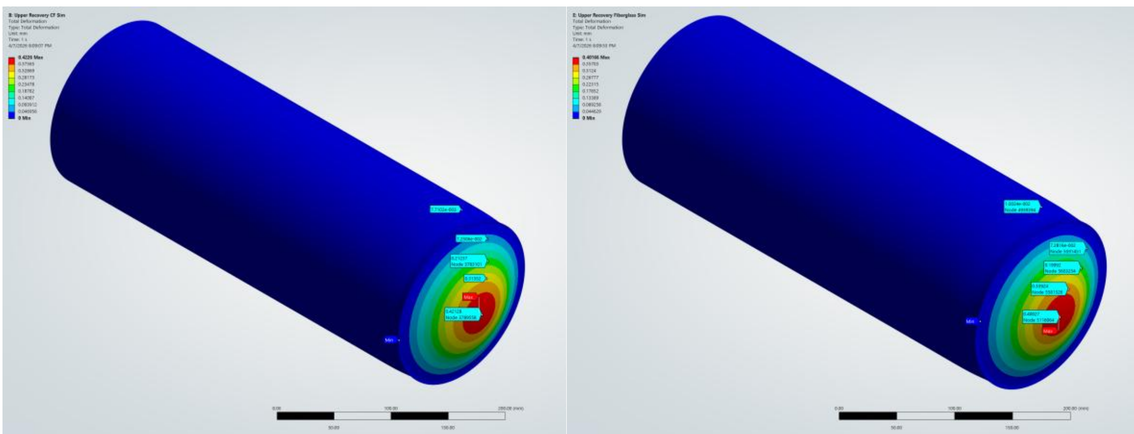


Fig. G-4: Carbon fiber (left) and g10 fiberglass (right) total deformation from upper airframe black powder recovery separation

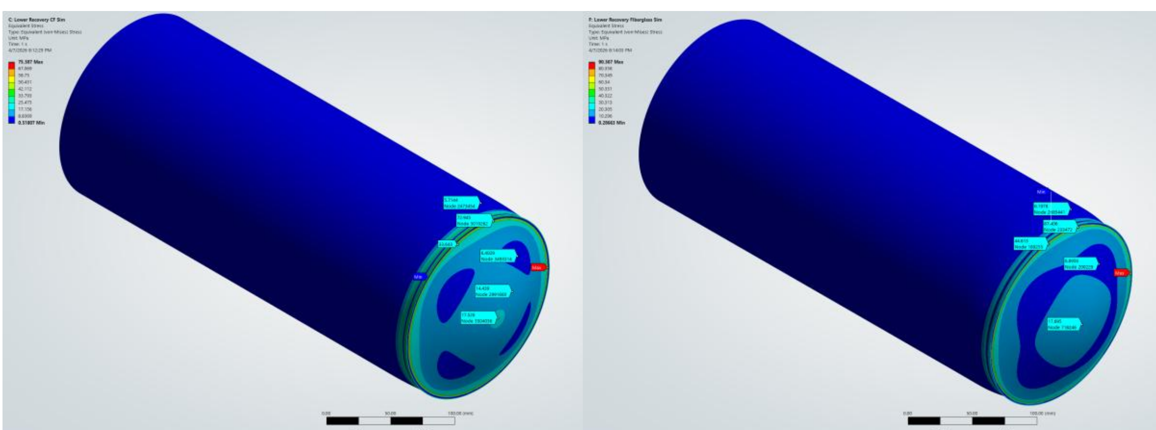


Fig. G-5: Carbon fiber (left) and g10 fiberglass (right) equivalent stress from lower airframe black powder recovery separation

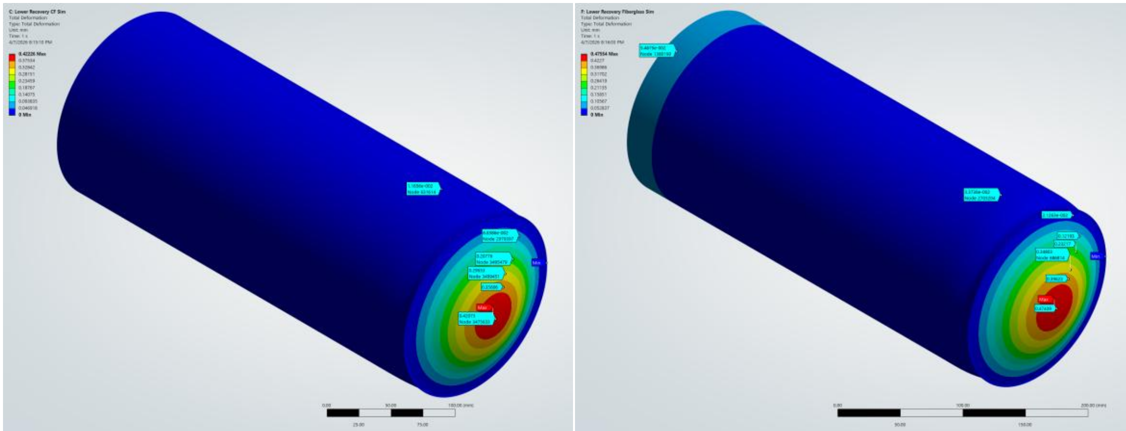


Fig. G-6: Carbon fiber (left) and g10 fiberglass (right) total deformation from lower airframe black powder recovery separation

Next, the team simulated the forces and deformations experienced by the upper and lower airframes when the black powder recovery charges generated the pressure required to shear the shear pins and separate the airframe during recovery. For Fig. G-3 to G-6, the deformation in these images is scaled up by a factor of 45-65. Simulations were performed for the upper airframe and for the lower airframe, with the pressure of 0.24616 MPa from the charges distributed evenly across the inside of the airframe and the bulkhead. Separate simulations were run for a carbon fiber airframe and for a fiberglass airframe made of the same material as the nosecone. A stress concentration ring, likely an artifact of FEA, appeared on the bulkhead and likely skewed the maximum reported stress values, but the bulkhead was not the component of interest in these simulations. Thus, observations were made for the stress and deformation of the airframe, the component of interest. The carbon fiber upper airframe experienced an average stress of 7.4 MPa (Fig. G-3) with an FoS of 6.1 and a maximum deformation of 0.025 mm (Fig. G-4), while the carbon fiber lower airframe experienced an average stress of 7.4 MPa (Fig. G-5) with an FoS of ~6.1 and a maximum deformation of 0.023 mm (Fig. G-6). FoSs for the carbon fiber are based on a 45 MPa radial ultimate tensile strength. For the fiberglass upper airframe, simulations showed an average stress of 4.2 MPa (Fig. G-3) with an FoS of 7.8 and a maximum deformation of 0.030 mm (Fig. G-4), while the fiberglass lower airframe received an average stress of 8.5 MPa (Fig. G-5) with an FoS of 3.9 and a maximum deformation of 0.056 mm (Fig. G-6). FoSs for the fiberglass are based on a 32.75 MPa ultimate radial tensile strength. For a black powder explosion, the airframe is in no danger regardless of the material choice.

Next, the team simulated the forces delivered to the airframe from the maximum acceleration during flight. The simulation was divided into five sections to divide up the computational power. Forces and constraints were applied to maintain the load path of the forces from the bottom of the motor mount tube up through the nosecone. Figures G-7 to G-11 show the relevant simulations.

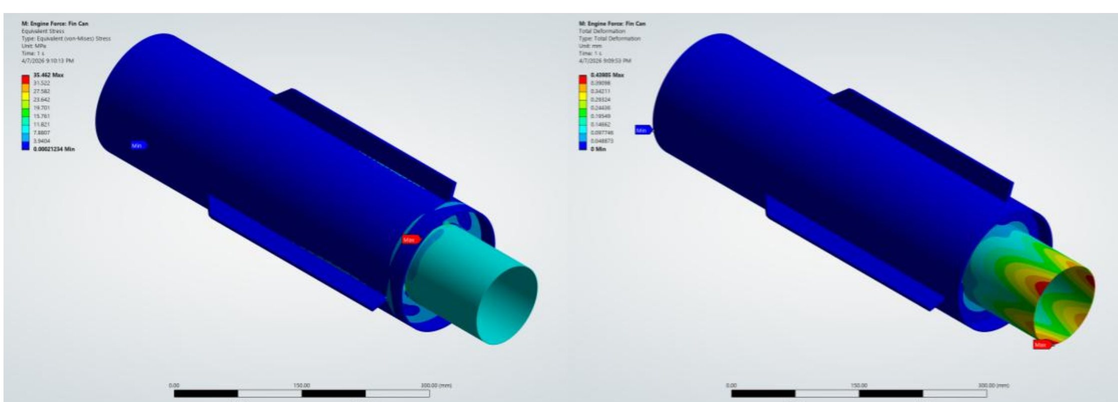


Fig. G-7: Stress (left) and deformation (right) on the fin can from engine forces

2245.4 N of force was delivered to the bottom of the motor mount tube on the fin can (Fig. G-7). The maximum stress was 35.462 MPa on the motor mount tube, below the ultimate compressive motor mount tube strength of 157.5 MPa. This yielded an FoS of 4.44. The maximum deformation was 0.440 mm. Images are true scale. The fin can is thus acceptable.

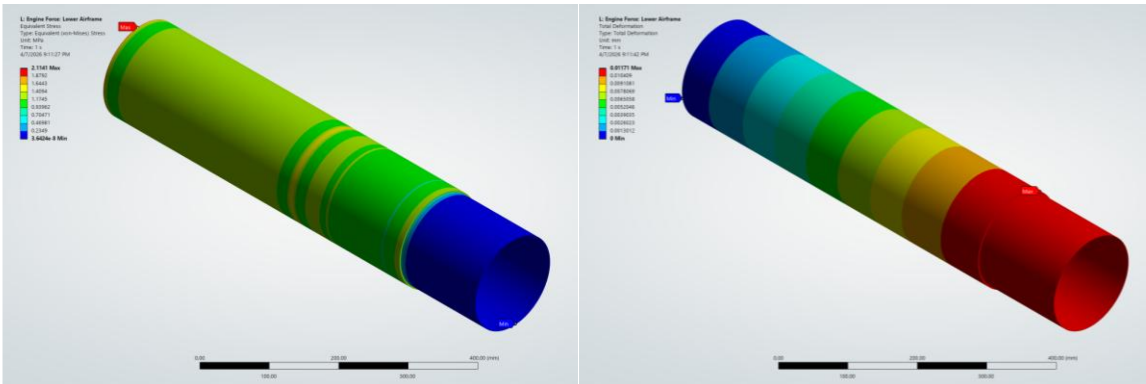


Fig. G-8: Stress (left) and deformation (right) on the lower airframe from engine forces

1494.3 N of force was delivered to the lower airframe (see Fig. G-8). The maximum stress was 2.114 MPa, which is below the body tubes carbon fiber compressive ultimate strength of 387.5 MPa, giving an FoS of 183.30. The maximum deformation was 0.0117 mm. Images are true scale. The lower airframe is thus acceptable.

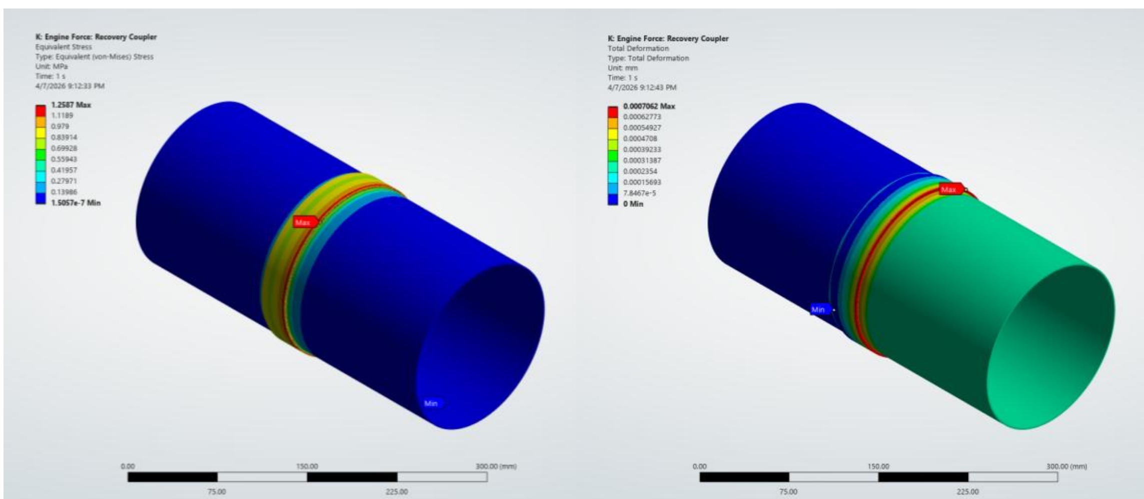


Fig. G-9: Stress (left) and deformation (right) on the recovery coupler from engine forces

1271.7 N of force was delivered to the recovery coupler (see Fig. G-9). The maximum stress was 1.259 MPa, below the body tubes carbon fiber compressive ultimate strength of 387.5 MPa, giving an FoS of 307.78. The maximum deformation was 0.000706 mm. Images are true scale. The recovery coupler is thus acceptable.

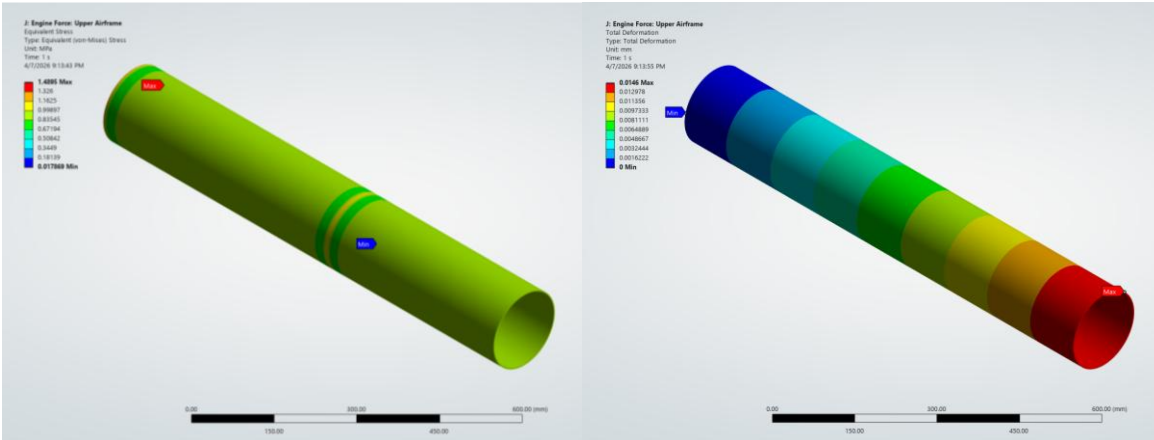


Fig. G-10: Stress (left) and deformation (right) on the upper airframe from engine forces

1067.3 N of force was delivered to the upper airframe (see Fig. G-10). The maximum stress was 1.490 MPa, below the body tubes carbon fiber compressive ultimate strength of 387.5 MPa, giving an FoS of 260.07. The maximum deformation was 0.0146 mm. Images are true scale. The upper airframe is thus acceptable.

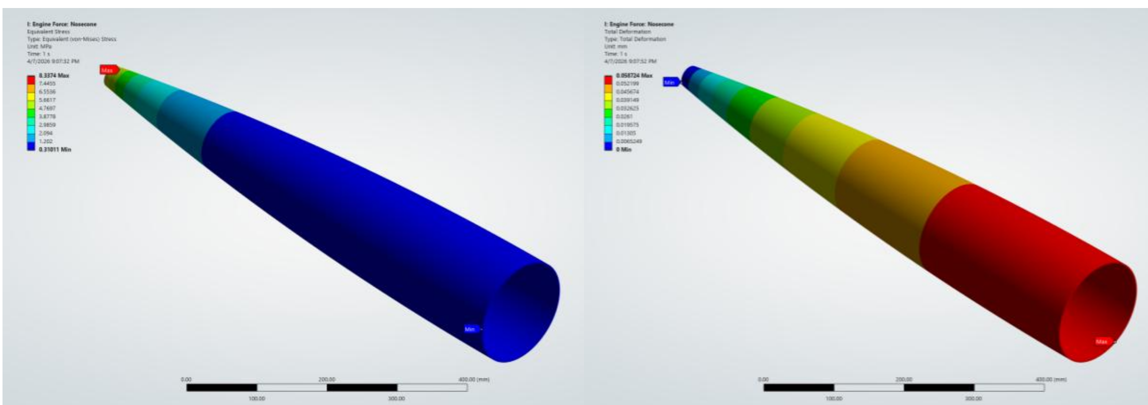


Fig. G-11: Stress (left) and deformation (right) on the nosecone from engine forces

317 N of force was delivered to the nosecone (see Fig. G-11). The maximum stress was 8.337 MPa, below the nosecone fiberglass compressive ultimate strength of 158.33 MPa, giving an FoS of 18.99. The maximum deformation was 0.0587 mm. Images are true scale. The nosecone is thus acceptable. This completes the airframe forces during accelerated flight, showing that the airframe will easily withstand flight forces and remain intact for future flights.

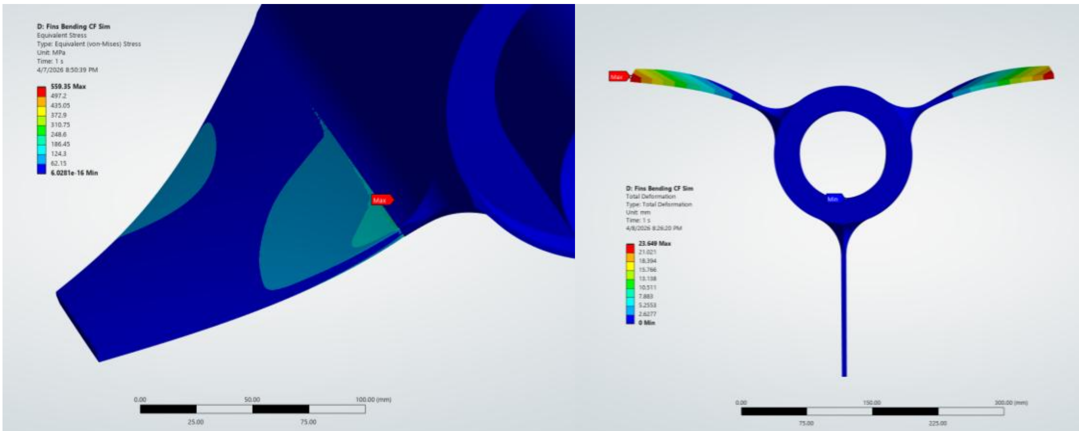


Fig. G-12: Stress (left) and deformation (right) of the carbon fiber fin loading simulation

Figure G-12 (deformations scaled to 1.9) shows the simulation for the force delivered to the 3/16” carbon fiber fins during a loading scenario where one fin is fixed facing downwards and a downwards force is spread across the two upward-facing fins. This loading case is expected when the rocket is evaluated during competition. A downwards force of 1334.5 N (300 lbf) was distributed evenly across the bottom edges of two of the fins, and the resulting stress and deformation were simulated. Ignoring the extraneous stress concentrations formed as an artifact of FEA, the maximum stress on the fins was between 220 MPa and 360 MPa, which gives FoSs of 1.78 and 1.09, respectively. These FoSs are based on an X and Y tensile ultimate strength of 393.67 for the fins' carbon fiber. The maximum deformation of the fins at the tips was 23.645 mm. As 1334.5N is already more than expected for the fins, the respective factors of safety show that the carbon fiber fins used are acceptable and that the fins will withstand the bending forces during competition examination.

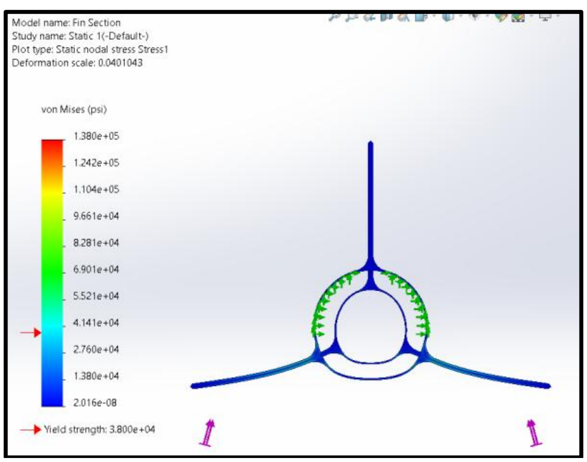


Fig. G-13: Stress of the fiberglass fin loading simulation

The design decisions for the fins' material type centered around the thickness of the fins. The objective was to select the thinnest fins that would withstand the 1334.5 N (300 lbf) force. Considered in previous years were 1/4” thick G10 fiberglass fins compared to the 3/16” fins used and tested in the sim above (Fig. G-13). Previous simulations (see Fig. G-13) showed the FoS for the fiberglass fins to be only 0.28 compared to a previously simulated FoS of 2 for the carbon fiber fins. If the fiberglass were loaded to the worst-case scenario, the fiberglass fins would not survive. With the thickness of the fins presenting increased aerodynamic drag and the strength of the fins threatening the integrity of the structure in worst-case scenarios, the only thing that G10 fiberglass had in its favor was its lower cost. The team’s budget allowed for the carbon fiber fins; however, the carbon fiber was ultimately selected as it made the structure stronger and more aerodynamic.

C. Fin Material Testing

To help validate the fin simulations above, the team worked to develop a custom material model of the carbon fiber sheets that were purchased for the fin material. Particularly, the team was interested in seeing how the angle of fibers affects the properties of the carbon fiber plates and whether the material could be treated as an isotropic material or not. Members of the team used a CNC water jet to cut test specimens out of the fin stock material. Specimens were cut in vertical, horizontal, and 45-degree angles (10 of each) relative to the fiber direction in the material. These specimens were then tested in a 3-point bending experimental testing setup.

Table G-2: Mean and standard deviation results for Yield Strength and Elastic Modulus

Cut Type	Mean Yield Strength, MPa	Yield Strength Standard Deviation, MPa	Mean Elastic Modulus, GPa	Elastic Modulus Standard Deviation, GPa
Horizontal	394.48	28.848	30.118	1.746
Vertical	397.16	38.166	32.426	2.1474
45º Angled	141.96	5.0649	2.9468	.56005

From the results shown in Table G-2, it is clear that the horizontal and vertical specimens behave very differently from those cut at 45 degrees. The numerical findings for the fin material's yield strength and elastic modulus are presented in the table below.

The development of this material model serves to validate our simulation results to better inform the team's decisions. If this experimentation had not been conducted, the team might have falsely assumed it could continue to decrease the thickness of the fin and still maintain a factor of safety above 1. To include these findings in design decisions, the team made sure to cut the fins out so that the weave direction would be in the vertical and horizontal arrangement for greater strength.

D. CFD

The team performed a computational fluid dynamics (CFD) simulation analysis on the nose cone under 0.5 Mach conditions. This simulation produced two contour plots used in this analysis. Figure G-14 displays the contour plot for static pressure along the nose cone, which peaks at 7990 Pa at the tip of the nose cone and decreases along the length of the rocket. This is consistent with published theory, with an article by G. Balaji providing a similar conclusion. In their study analyzing flow along a hemispherical missile nose cone, the flow formed a stagnation point on the tip, causing a high-pressure point that gradually decreased along the length of the missile [20]. Due to the high velocity of the fluid, around 171 m/s, the boundary layer should be appropriately small, on the order of a few millimeters. This is consistent with the results shown in Fig. G-14, validating the simulation accuracy in terms of fluid behavior. Both the pressure and velocity contour plots show a reasonable agreement with the aforementioned article by G. Balaji, lending credence to the simulation and the selected nose cone design.

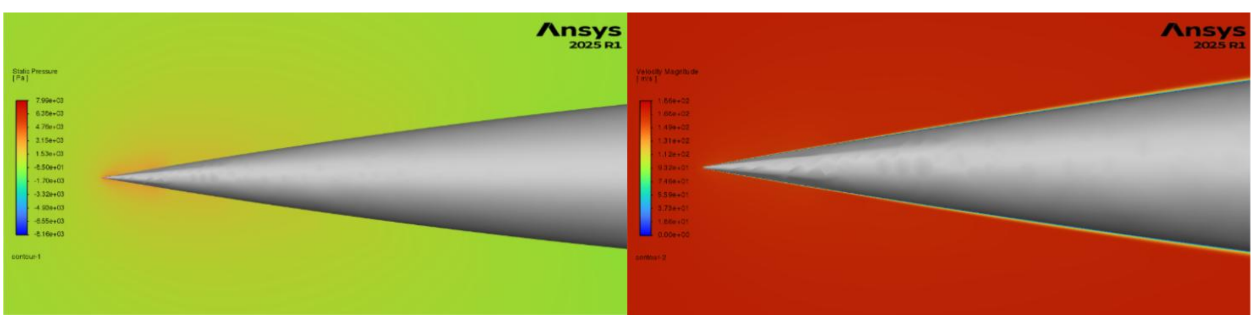


Fig. G-14: Static pressure contour (left) and velocity magnitude contour (right) for tangent ogive nosecone

To better understand the aerodynamics of the rocket, a full-scale simulation of the entire rocket under 0.5 Mach conditions was conducted using ANSYS Fluent. First, the CAD model was defeatured and repaired to reduce computational cost. This included removing the fillets, radially dividing the model into symmetrical regions, and filling any gaps. This refined geometry was then imported into the ANSYS Fluent Watertight workflow, where the mesh was created. The resulting volume mesh can be seen in Fig. G-15. The final mesh achieved a minimum orthogonal quality of 0.1656, and the cell count was 1,208,834, ensuring a reliable, computationally efficient mesh. Any value above 0.1-0.15 is generally acceptable for most meshing applications as it ensures a decent quality mesh

[21]. Next, the mesh was imported into the solver, where the governing equations, boundary conditions, and convergence criteria were defined. The inlet was assigned a freestream velocity of 0.5 Mach, while the rocket surface used a real wall no-slip condition. The outer boundaries were modeled using the far-field conditions, and the symmetry conditions were used for the radial divisions. Lastly, a report definition was created to evaluate the drag coefficient. The setup for this CFD simulation aligns closely with standard methodology, similar to the setup described in the ANSYS Fluent 12.0 User Guide [22].

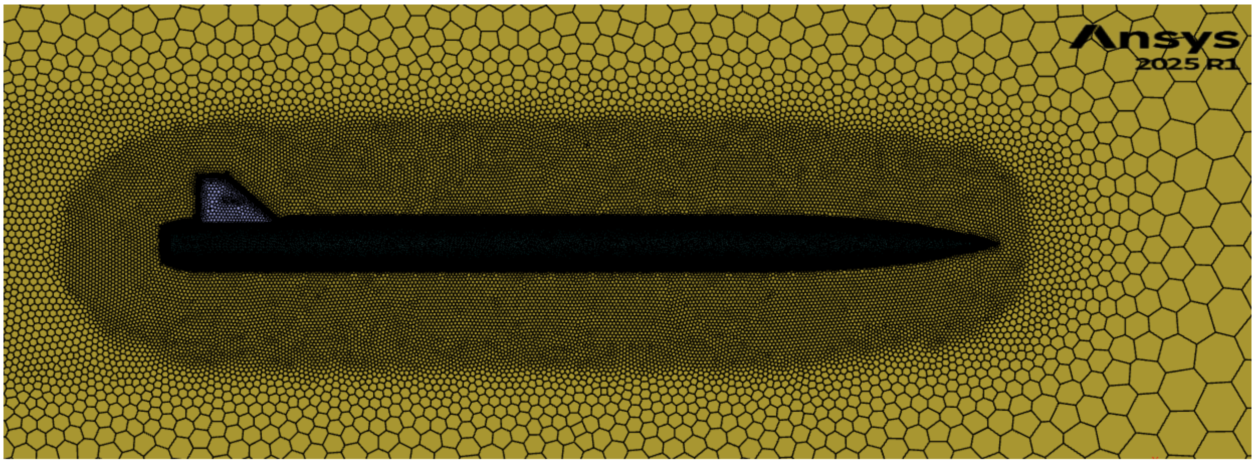


Fig. G-15: Volume mesh of the entire rocket

Post-processing of the simulation provided streamline and contour plots for both velocity and static pressure for the full rocket. Each plot validates the rocket's aerodynamic performance and design. The streamline plots show smooth flow along the rocket with very little separation and no significant chaotic motion. Figure G-16 provides the streamline plot along the back half of the rocket. It is important to note that there is some minimal wake formation at the rocket fin can, but nothing that significantly discredits the aerodynamic performance.

The velocity contour plots produce a gradually developing boundary layer along the length of the rocket, which is expected for compressible, subsonic flow. It is important to note the lack of sharp velocity gradients, leading to a low drag coefficient, providing validation of the rocket's aerodynamic efficiency. The static pressure contour plot shows a stagnation region at the nose tip, but a stable pressure region everywhere else with minimal variation. Again, this contour plot shows a lack of strong pressure gradients, indicating a good aerodynamic performance. The smooth, continuous behavior of these two plots indicates a streamlined model, leading to low aerodynamic resistance. Figure G-17 provides the contour plots for velocity and static pressure.

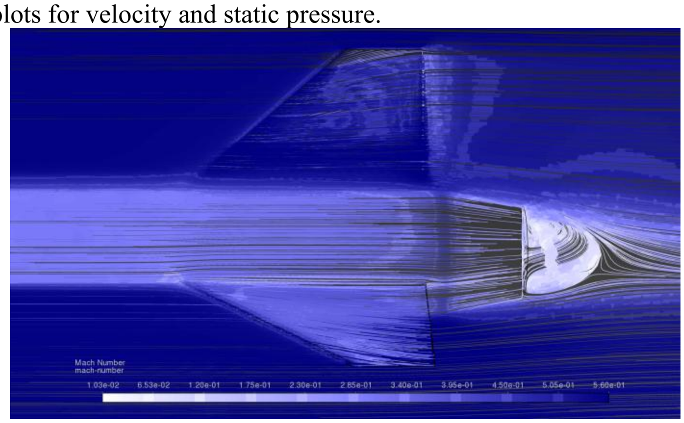


Fig. G-16: A plot of the mach number streamlines on the backend of the rocket

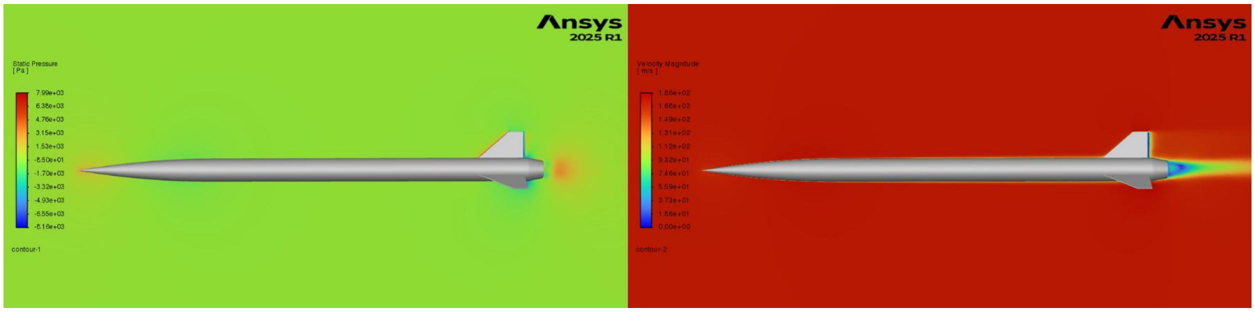


Fig. G-17: Static pressure contour (left) and velocity contour (right) of entire rocket

The CFD simulations provided detailed contour plots to help visualize the flow of the air around the rocket. Additionally, they give data on the drag forces experienced by the rocket. Using these drag forces, the drag coefficient can be found. The drag coefficient is an important metric used for evaluating rocket designs, as it is used to understand and model many complex aerodynamic relationships [23]. As such, a study comparing the drag coefficient and force at different Mach values was then performed to analyze how the rocket will perform at different speeds. A reference area of 0.009675 m^2 was used as it corresponds to half the frontal area of the rocket body. As per Open Rockets Technical documentation, the reference area used to calculate the drag coefficient correlates to the cross-sectional area of the rocket body [24]. At speeds under 0.25 Mach, the solver was unable to converge, likely due to method limitations. Subsequently, the values were omitted from the study [25].

Table G-3: Drag Properties at Varying Mach Values

Mach number	Drag Coefficient (Ansys Fluent)	Simulated Drag Force (½) [N]	Drag Force [N]	Drag Coefficient (Open Rocket)
0.25	0.3822197	16.4	32.8	0.339
0.30	0.3769612	22.9	45.8	0.336
0.35	0.3724713	30.2	60.4	0.336
0.40	0.3670248	38.0	76	0.336
0.45	0.3631613	47.3	94.6	0.338
0.50	0.3601648	65.5	131	0.341
0.55	0.3571979	74.8	149.6	0.345
0.60	0.3545162	86.3	172.6	0.350
0.65	0.3522649	98.7	197.4	0.356
0.70	0.3503704	112.0	224	0.364
0.75	0.3509719	128.7	257.4	0.373
0.80	0.3477589	142.6	285.2	0.385

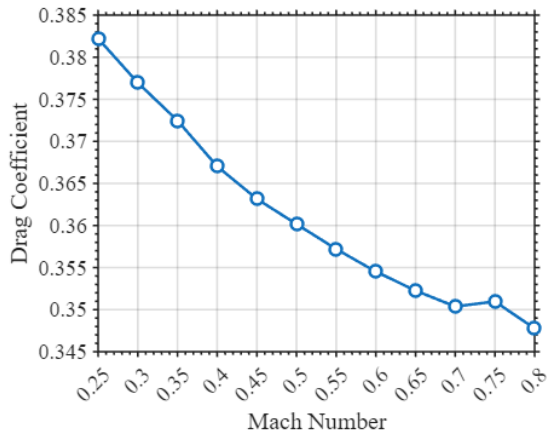


Fig. G-18: Drag coefficient as a function of mach value

Analyzing the drag coefficient trends for both the ANSYS Fluent model and the OpenRocket model yields some important results. In the ANSYS Fluent model, the drag coefficient decreases with increasing Mach number. However, for the open rocket model, the reverse is true. This can be explained by their different methodologies. ANSYS Fluent uses the finite volume method to discretize the region, where each individual volume element is solved using the continuity equations [28]. However, in the process, it often assumes ideal flow and conditions as well as simplified geometry. In this scenario, the increasing Mach value likely reduces flow separation and lowers the drag. In addition, the ANSYS solver struggles more at lower speeds. This is likely because the model used, SST K-Ω It works better at higher Reynolds numbers [26]. A different model could have been used for these lower Mach value tests, but, for the sake of consistency, the same model was used throughout. This accounts for the anomaly seen at the 0.25 Mach speed test. Another discrepancy occurred during the 0.75 Mach value test. This discrepancy has multiple possible causes. First, because it is entering transonic flow, there could be regions of compressible, supersonic flow, leading to a higher drag coefficient. The mesh could be too coarse, leading to errors during the solver process. Lastly, it could be an issue within the model. ANSYS Fluent simulations are powerful, but often very sensitive to the setup.

In contrast, the OpenRocket uses a completely different, simpler methodology. Open Rocket instead primarily uses the Extended Barrowman method to calculate the drag coefficient [27]. Open Rocket is more rocket-focused and is better at including rocket-specific effects, such as surface finish and skin friction. These effects could cause the drag coefficient to increase with the increasing Mach number. At lower Mach numbers, the drag coefficient remains relatively constant [28]. This implies any trend in the drag coefficient is likely caused by the individual program's methodology. However, this study implies that the drag coefficient for the rocket at these speeds will likely fall within the range between 0.340 and 0.385. The results of the CFD simulations help verify the coefficient of drag determined by OpenRocket. It also allows the team to change some OpenRocket values with confidence because of the CFD results. This study also allowed the team to use RocketPy for future rockets should the team pursue airbrakes or have a more accurate simulation should the future rockets ever goes beyond Openrocket's comfort zone.

The CFD results presented in this study provide an independent verification of the drag coefficient predicted by OpenRocket. While OpenRocket offers a rapid and reliable preliminary estimate based on semi-empirical methods, the CFD analysis captures more detailed flow physics. As a result, the CFD simulations increase confidence in the aerodynamic performance predictions and help validate the assumptions used in the lower-fidelity model. Furthermore, the CFD results provide a basis for refining key OpenRocket parameters such as coefficient of drag. The methodology developed in this study allows the use of higher-fidelity simulation tools such as RocketPy. This is particularly important for the implementation of airbrakes or operation in high Mach regimes where Openrocket's underlying assumptions may no longer be sufficient. Therefore, the inclusion of CFD analysis serves as a critical step in validating and extending the aerodynamic modeling capability of the team.

E. Manufacturing Process

1. 3D Printing

The manufacturing complexity of many non-structural aspects of the rocket lent itself towards 3D printing as the best manufacturing method. We chose to print the nosecone E-bay in a combination of PET GF and PCTG. Last year we had a similar nosecone e-bay design that also utilized printed threads for alignment. That design utilized a PA6 CF blend. PA6, however, is notoriously hydroscopic. When it absorbs moisture, it softens. At our test launch, the threads worked flawlessly. However, a month later, at IREC, the threads had softened and were almost unusable. PET GF is still very stiff, very heat resistant (HDT of 130 °C), and crucially, is not moisture-sensitive once printed. PCTG is a lesser-known filament with similar mechanical properties to PETG, but with superior layer adhesion (crucial for printed threads), heat resistance (HDT of 90 °C), and, since it is not fiber-reinforced, it has a much smoother surface.

PET GF was also used for the fin can attachments as well as the boat tail attachment component. PETG (HDT of 65 °C) was utilized for any miscellaneous components. Crucially, PLA is not used for any components on the rocket due to its low heat resistance and tendency to deform under load.

2. *Nosecone Construction*

The nosecone itself was purchased from a vendor. A permanent 3D printed threaded ring was inserted into the nosecone to act as a retainer for the GPS system. This ring was contoured to the inner form of the nosecone for a secure fit and mounted in place using aerospace-grade epoxy. The nosecone was fitted to the rocket using a coupler, which was inserted into its inner surface and extended past the end of the nosecone to fit into the inner diameter of the rocket's upper airframe. Aerospace-grade epoxy was used to attach the coupler to the nosecone. Once the GPS module was screwed into this insert, holes were drilled in the coupler to align with holes printed into the base of the module, along with several drilled for pressure release. These house brass inserts for the screws that run through the coupler and forward airframe to hold the nosecone in place on the rocket. The GPS module also includes a GoPro camera, so a hole was drilled in the coupler to allow the camera to see out. An acrylic film was mounted as a window to reduce drag. To finish the nosecone, a professional auto-body paint job was applied to the outside to be able to customize the aesthetics and ensure a very smooth finish.

3. *Forward Airframe*

The forward section tubing was cut to its desired length, 1 meter. Additionally, a camera hole was cut using a drill press. Next, four screw holes were drilled to match the screw holes in the nosecone coupler to secure the two parts together. On the opposite end of the forward airframe, six shear pin holes were drilled and threaded to allow shear pins to be mounted and hold the main parachute compartment closed until the recovery event. A permanent bulkhead is installed in this forward airframe section that serves as the link to the main parachute tether using Aeropoxy mixed with fiberglass filler. This bulkhead serves as the base connection point for the rocket payload. Small holes were drilled in the airframe to allow for pressure changes. Finally, the airframe received a professional auto-body paint job to create a smooth surface and custom aesthetics.

4. *Recovery Bay Housing Construction*

The electronics bay housing consists of a 2.7 cm switch band section around a 33 cm coupler (see Fig. G-19). The internal components are held between two bulkheads on either end of the coupler. The bulkheads are held in place by a lip that fits into the edge of the coupler section and held in tension by threaded metal rods that also secure the inner recovery bay electronics sled in place. The switch band was then mounted to the coupler, providing a stop in the middle for the forward and aft sections. Vent holes were cut in the switch band section that doubled as holes for the pull-pin switch used in the avionics bay. Shear pin holes were also drilled on either side of the electronics bay to accommodate the necessary shear pins, six on the forward coupler section and four on the aft coupler section. After these steps, the switch band received a professional body-shop finish to match the exterior of the rest of the rocket.

Fig. G-19: Epoxying the switch band to the coupler electronics bay housing manufacturing

5. *Aft Airframe*

The aft section tubing also had to be cut to its desired length of 60.3 cm. This section had four shear pin holes drilled and threaded to allow for a secure hold until the recovery event. A hole was drilled for the rail button to be attached, which was screwed into a retaining plate and epoxied in place. A bulkhead was installed in this aft airframe section to act as an anchor point for the drogue parachute. This bulkhead was secured in place using aerospace-grade epoxy and fiberglass filler similar to the other permanent bulkhead. Next, a coupler section, cut to a length of 30.5 cm, was fitted to the aft end of the airframe to mount the fin can onto. The coupler was permanently mounted to the aft airframe with aerospace-grade epoxy. An anchor ring was printed out of PETG and mounted with Aeropoxy. Threaded brass inserts were fitted into the ring to allow screws to run through the fin can and coupler, securely mounting the fin can to the aft airframe. Finally, the aft airframe was given a professional auto-body finish.

6. *Motor Tube*

The locations for each centering ring were marked on the motor tube. To ensure the rings stayed in place during epoxying, tape was wrapped along the bottom of these lines to create a resting surface. The inside of each of the centering rings was then sanded until they all fit securely over the cardboard motor mount, making sure they were sanded evenly to preserve concentricity. Epoxy and hardener were mixed in a 1:1 weight ratio, with chopped fiberglass epoxy filler added until the mixture reached the desired viscosity to minimize the effect of gravity on the shape of the fillets. Then, this epoxy mixture was used to create fillets on the untaped sides of the centering rings, securely bonding them to the cardboard tubing. This process ensured proper alignment and structural integrity for the motor mount assembly. The front-most centering ring was attached before the motor tube was installed in the fin can. The rear centering ring was attached after the fins were installed and the interior fin fillets poured.

7. *Fin Can*

The interchangeable fin section allows for ease of manufacturing, storage, transportation, and motor integration. First, we cut a 50.8 cm section of carbon fiber tube and drilled holes in the forward section to align with the aft airframe. We also drilled holes on the aft side of the fin to allow for the attachment of a boat tail, which we are implementing to reduce drag induced by the wake of the rocket. Additionally, we cut three fin slots to fit our fins. The fins were made of 5 mm-thick Dragon Plate carbon fiber plates, which were cut to shape using a CNC waterjet according to the engineering drawings included in Appendix F. A router was then used to chamfer the leading and trailing edges at 45 degrees to make the fins more aerodynamic. The motor tube was epoxied into the fin can using aerospace-grade epoxy, and the fins were inserted into the slots and epoxied in place to the motor tube using aerospace-grade epoxy. We applied a fillet on the edge where the fin met the exterior airframe as well as where the fin met the motor tube. Figure G-19 shows an image of our setup. This year, we used 3D-printed molds coated in release wax to shape our fillets (see Fig. G-20). These slid over each fin and compressed and held the epoxy in the desired geometry as it set. After the molds were removed, the fillets were inspected and sanded as necessary to ensure the correct shape, then another thin layer of epoxy was added to bring back the smooth finish. Finally, the fin can receive a professional auto-body finish.

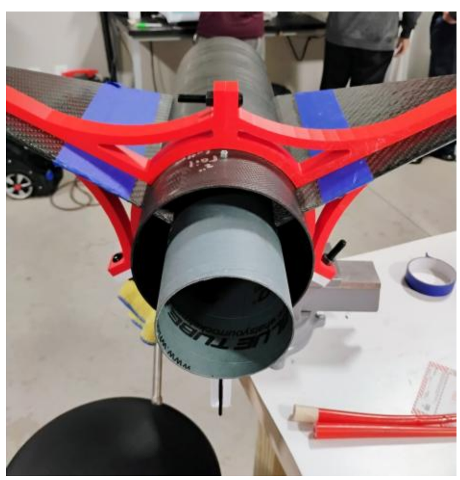


Fig. G- 20: Fin construction setup for epoxying fins to motor tube with internal fillets

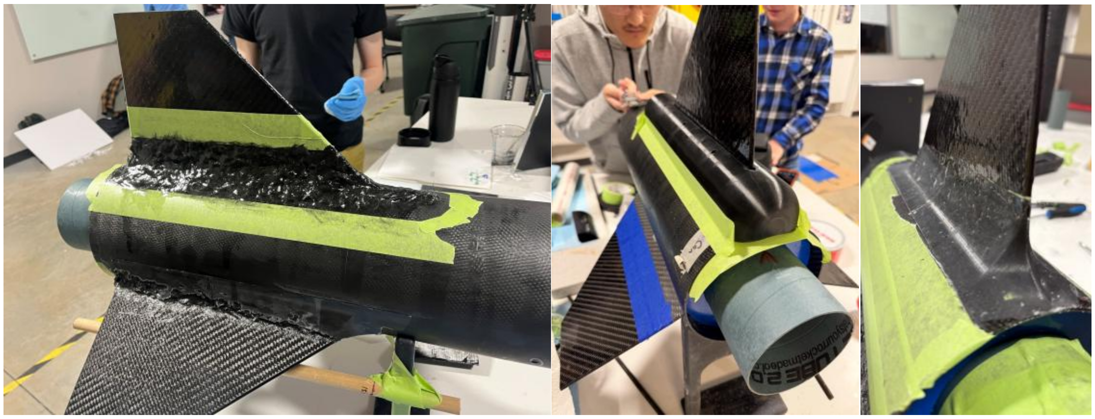


Fig. G-21: Exterior fin fillets before molding (left), with mold on (center), and after the mold was applied (right)

8. *Boat Tail*

The boat tail is designed to decrease aerodynamic drag and is manufactured in-house to decrease financial drag. We printed a boat tail base out of PET-GF15 and used laminating epoxy to lay up several layers of tubular carbon fiber woven sleeving on the exterior surface. This was cured in a custom-designed and fabricated epoxy oven. After the carbon fiber laminate had cured in the oven, we cut away and sanded the excess material, leaving just the desired carbon fiber skin on the printed mount (Fig. G-21). We secured the custom boat tail to a mounting ring with aerospace-grade epoxy and then attached the boat tail to the aft of the fin can using screws from the brass inserts in the mounting ring through the holes drilled in the aft of the fin can.

A ring printed from PET-GF15 with brass heat-set threaded inserts was used to secure the boat tail to the airframe. The ring mount is epoxied to a 3D-printed negative of the boat tail, which is in turn epoxied to the carbon-fiber boat tail. This assembly is secured to the airframe by running bolts through the airframe into the threaded inserts in the upper ring.

Fig. G-22: Carbon fiber wrap for boat tail sleeved over 3-D print

F. Decision Matrices

A decision matrix is often used to quantify the tradeoffs among competing design objectives. In engineering design, a decision often involves multiple desirable properties that may conflict with one another. To evaluate the available options, the team assigns a score to each property of interest for every candidate and applies weighting factors based on the team's priorities. The resulting matrix provides a structured method for comparing alternatives and selecting the option that offers the optimum trade-off.

Table G-4: Nosecone Geometry Decision Matrix

Geometry	Drag Coefficient at 0.8 Mach (Score)	Commercial Availability (Score)	Cost (Score)	Weighted Score
Tangent ogive	0.3088 (4)	High (5)	Low (5)	4.3
Von Kármán (Haack)	0.3082 (5)	Medium (3)	High (2)	4.1
Conical	0.3104 (3)	High (5)	Low (5)	3.8
Elliptical	0.3126 (2)	Low (2)	High (2)	2.0

Weighted scores were calculated using 60% for drag coefficient, 30% for availability, and 10% for cost.

Table G-5: Finesse Ratio Decision Matrix

Fineness Ratio	Altitude (Score)	Commercial Availability (Score)	Experience Benefit (Score)	Weighted Score
3:1	Low (2)	High (5)	Low (2)	3.0
4:1	Medium (3)	Medium (4)	Medium (3)	3.5
5:1	High (4)	High (5)	High (5)	4.4
6:1	Very High (5)	High (5)	High (5)	4.7
7:1	Very High (5)	Medium (3)	Medium (3)	4.1

Weighted scores were calculated using 60% for altitude, 30% for commercial availability, and 10% for experience benefit. Experience Benefit measures how current decisions allow useful experience for future team activities.

Table G-6: Nosecone Material Decision Matrix

Material	RF Transparency (Score)	Structural Performance (Score)	Weight Efficiency (Score)	Commercial Availability (Score)	Cost (Score)	Weighted Score
Filament-wound fiberglass	Very High (5)	Very High (5)	High (4)	Medium (3)	Medium cost (3)	4.6
Plastic	Very High (5)	Medium (3)	High (4)	Very High (5)	Low cost (5)	3.9
Kraft paper	High (4)	Low (2)	Medium (3)	Very High (5)	Low cost (5)	3.0

Weighted scores were calculated using 20% RF transparency, 50% structural performance, 15% weight efficiency, 10% commercial availability, and 5% cost.

Table G-7: Airframe Material Decision Matrix

Material	Structural Performance (Score)	Weight Efficiency (Score)	Commercial Availability (Score)	Cost (Score)	Weighted Score
Carbon fiber	Very High (5)	Very High (5)	Medium (3)	High (2)	4.5
G12 fiberglass	High (4)	Medium (3)	Very High (5)	Low (5)	3.9

Weighted scores were calculated using 50% structural performance, 25% weight efficiency, 15% commercial availability, and 10% cost.

Table G-8: Fin Number Decision Matrix

Configuration	Apogee (Score)	Stability Margin (Score)	Prior Experience (Score)	Material Efficiency (Score)	Time efficiency (Score)	Weighted Score
Three-fin	Very High (5)	High (4)	Very High (5)	Very High (5)	Very High (5)	4.8
Four-fin	High (4)	Very High (5)	Medium (3)	Medium (3)	Medium (3)	3.9

Weighted scores were calculated using 35% apogee, 25% stability margin, 15% prior experience, 15% material efficiency, and 10% time efficiency.

Table G-9: Fin Shape Decision Matrix

Fin Shape	Stability Margin (Score)	Drag, N (Score)	Manufacturability (Score)	Weighted Score
Clipped Delta	High (4)	48.52 (4)	High (4)	4.0
Elliptical	Medium (3)	53.61 (3)	Low (2)	2.8
Trapezoidal	Very High (5)	54.55 (3)	High (4)	4.2
Rectangular	Medium (3)	51.98 (4)	Very High (5)	3.7
Square	Low (2)	47.01 (5)	Very High (5)	3.5

Weighted scores were calculated using 50% stability margin, 30% drag, and 20% manufacturability.

Table G-10: Fin Material Decision Matrix

Option	Structural Performance (Score)	Weight Efficiency (Score)	Commercial Availability (Score)	Cost (Score)	Weighted Score
3/16 in carbon fiber	Very High (5)	Very High (5)	Medium (3)	Medium (3)	4.6
1/4 in G12 fiberglass	Very Low (1)	Medium (3)	High (5)	High (4)	2.3

Weighted scores were calculated using 50% structural performance, 25% weight efficiency, 15% commercial availability, and 10% cost.

Table G-11: Boat Tail Material Decision Matrix

Material	Strength (Score)	Manufacturability (Score)	Surface Quality (Score)	Weighted Score
Carbon fiber sleeving	Very High (5)	High (4)	High (4)	4.5
Fiberglass sleeving	Medium (3)	Very High (5)	Very High (5)	4.0
Kevlar sleeving	High (4)	Medium (3)	Medium (3)	3.6

Weighted scores were calculated using 50% tensile strength, 30% manufacturability, and 20% surface quality.

Appendix H: Recovery Supplementary

A. Custom Recovery Yoke Analysis

The recovery bay is the central dividing section of the rocket that connects both the front and rear sections of the rocket. It is located between two G-10 fiberglass bulkheads and secured together with two 5/16-inch steel rods. When the rocket reaches the target altitude for drogue deployment, the rocket then separates, releasing the drogue chute from the lower aft section. The initial snatch force and drag forces are directly applied to the threaded rods connecting the avionics bay. For rockets, Genesis and Waymaker, the avionics bay was fastened via eyebolts that were threaded on the ends of the rods, and a tether looped between them. The problem with this design is that when a tensile force is exerted on the tether, which is looped through both eyebolts, the tether creates a triangle between the two. That means a component of the vertical tensile force is translated as a horizontal force pulling the eyebolts together, towards the center of the bulkhead. When a great enough force is exerted through the tether, the screws become slightly bent and deformed. This is more concerning when considering the possibility of the main chute prematurely deploying at the wrong altitude or at too much speed. Since the main chute is much larger than the drogue chute, it will exert a higher shock and drag force on the eyebolts and subsequent threaded rods. If the force were applied equally to the screws in an axial manner, then there would be no concern for bending. However, because the eyebolts are pulled horizontally together in a bending/shear fashion, the threaded rods cannot withstand such forces and are permanently deformed.

1. Design Process

To combat the issue of the threaded rods bending, a solution to evenly distribute the load on both screws in an axial manner was devised. The solution to this was to design a load-distributing tether attachment bar, which we call the yoke. The yoke would connect the two threaded rods together with a central attachment point for the tether. Taking the shock and drag forces from the parachutes, it uniformly distributes the load axially between the two threaded rods, preventing unwanted deformation.

The first iteration of the yoke, used on our 2025 rocket Trinity, wasn't designed with specific metrics in mind. The only design goals were that it needed to fit within the bulkhead and be able to withstand the maximum theoretical loads that would be applied without excessive bending or flexing. To achieve this, it was designed with bulky dimensions to withstand the forces. The design was validated using FEA analysis using ANSYS and OnShape. For Omega, we wanted to further refine the design to maximize stiffness, minimize weight, and withstand the maximum calculated snatch forces with a minimum factor of safety of 1.5.

To meet the requirements, it was decided to use ANSYS topology optimization to maximize stiffness, reduce mass, and ensure the desired factor of safety at the given loads. A seed model was designed in SolidWorks, which acts as the initial model for the program to iterate upon. To set up the model in ANSYS, we created a full-body mesh sizing of 1 mm and face sizing of 0.25 mm at critical locations inside and around the holes for the threaded rods. The fixed supports were then defined at the through holes for the threaded rods. The force load location was defined at the middle of the model. Finally, we defined key surfaces to be excluded from the optimization region, including the inner through-hole surfaces and the bottom arch surface. To ensure manufacturability and strength, a minimum thickness of 3mm was defined for the excluded optimization region surfaces.

For the topology optimization setup, it references the previously defined static structural setup utilizing the boundary conditions, mesh, and loading conditions. The objective is set to minimize compliance, which maximizes the stiffness of the model. The response constraint, the amount of material to retain, was set to 42%. The mixable density solver was selected for its ability to adjust to the response constraint percentage after the main simulation was complete. To eliminate the risk of generating internal cavities within the model, a manufacturing constraint, a minimum size control of 8 mm, was selected. A global stress response constraint was implemented and set to 450 MPa to minimize the global stress result within the optimized model. Finally, two design constraints were added to ensure symmetry in the model along the XY and YZ planes.

After completing the setup, the topology optimization was initiated. These processes went through many iterations of making small changes to the optimization parameters to produce a result that resembled what was desired. Once we had an optimized model that looked promising, we began post-processing. The optimized model was exported as a smoothed STL file, which was imported into ANSYS Discovery for further smoothing and refinement. Once we were satisfied with the model's appearance, it was converted from a faceted model to a solid CAD and exported as a PARASOLID. This smoothed CAD model was then imported into OnShape to flatten contact surfaces and cut precise holes for better surface finish and dimensional accuracy. This fully refined model was then imported back into ANSYS for the validation phase.

With the fully refined topology optimized model imported back into ANSYS, we could implement the same meshing, boundary, and loading conditions to see if it met all requirements. The results showed an alarmingly high peak stress value within the model, referring to Fig. H-1. The location of this high stress was at the edge of the fixed surface, where no sharp corners or surface irregularities existed. Because of this, the peak stress was suspected to be a mathematical solver error known as a stress singularity, where computed stress grows without bounds, approaching infinity. A way to confirm whether it was an actual stress singularity was to conduct a mesh convergence study. This study gradually increases the refinement of the mesh and plots the resulting stress to observe whether it converges. Under normal conditions, the stress value will plateau at a certain mesh refinement. Conducting this study revealed that stress exponentially rose as the mesh became further refined, as shown in Fig. H-2. Probing just outside of the stress singularity gives a better idea of the expected stresses in the region, as displayed in Fig. H-1.

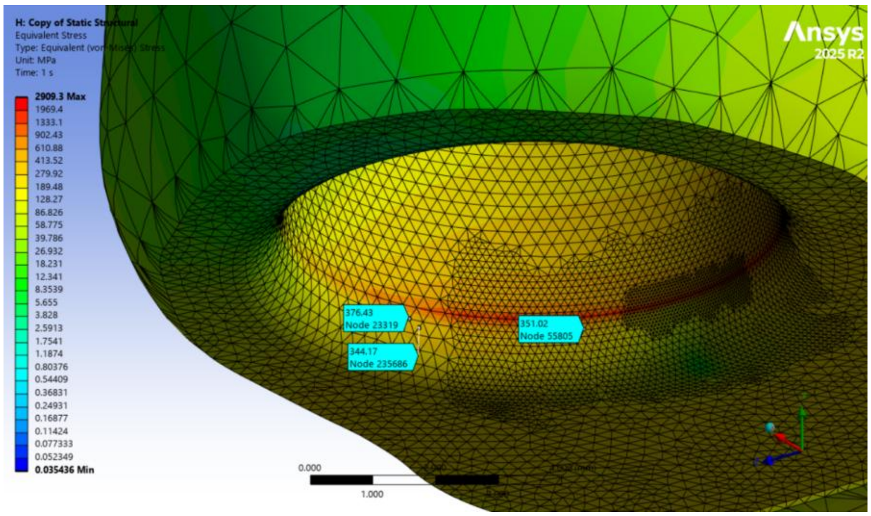
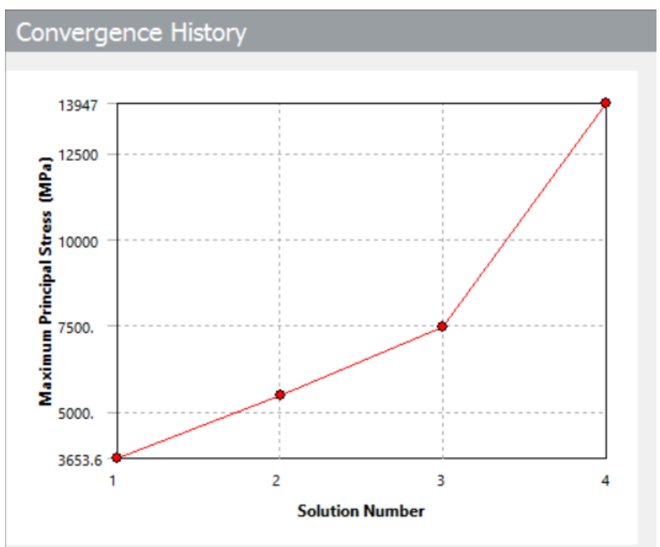


Fig. H-1: Stress singularity boundary probing



	Maximum Principal Stress (MPa)	Change (%)	Nodes	Elements
1	3653.6		410448	267099
2	5485.8	40.095	563117	374877
3	7478.7	30.745	1234566	857169
4	13947	60.38	2639804	1874346

Fig. H-2: Mesh convergence study

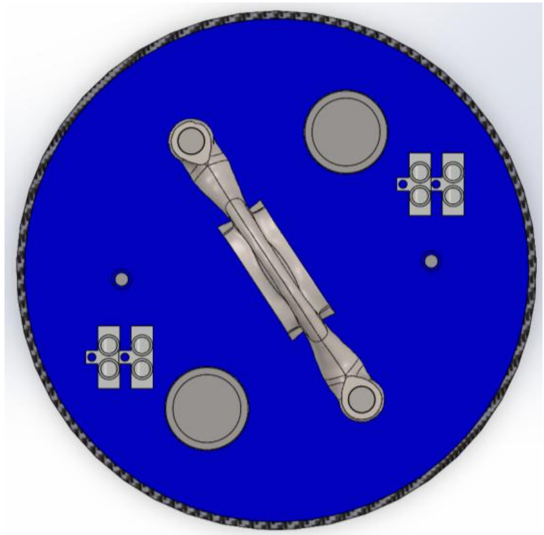


Fig. H-3: Top-down view of bulkhead with the yoke attached

Because of the organic nature of topology optimization, one of the few ways to manufacture such a model is through 3D printing. There were two main constraints in material selection for the yoke: material availability and competition rules. Because of competition rules, steel has to be used for anchoring geometry. Knowing that steel is a requirement, we were only limited by what our manufacturer offered. The two steel types offered by PCBWay for SLM 3D printing were tool steel and 316L stainless steel. Ultimately, 316L stainless steel was chosen due to its being half the manufacturing cost of tool steel without significant performance loss.

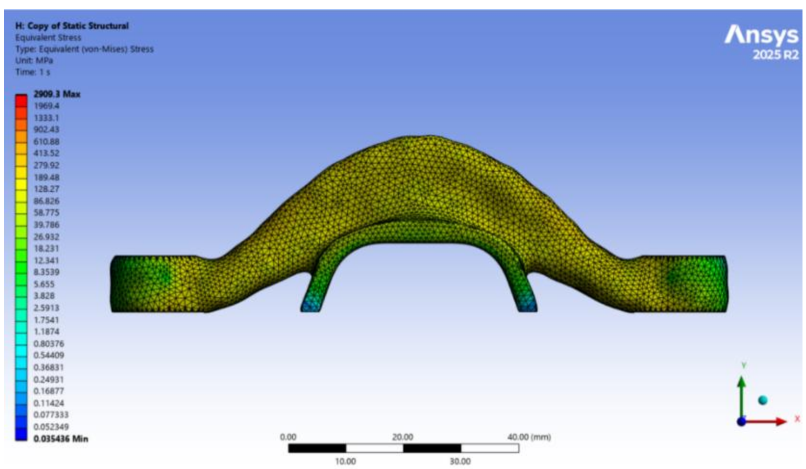


Fig. H-4: Ansys static stress simulation of the yoke

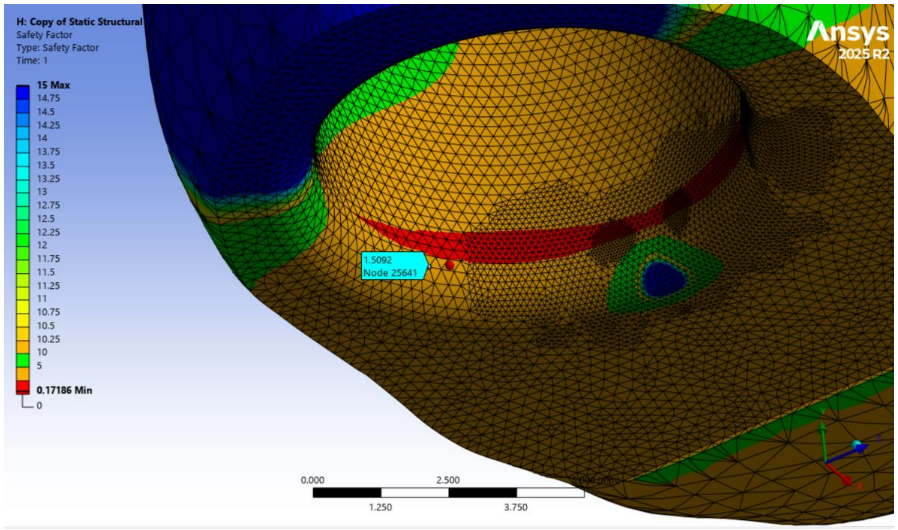


Fig. H-5: Factor of safety probe at stress singularity boundary

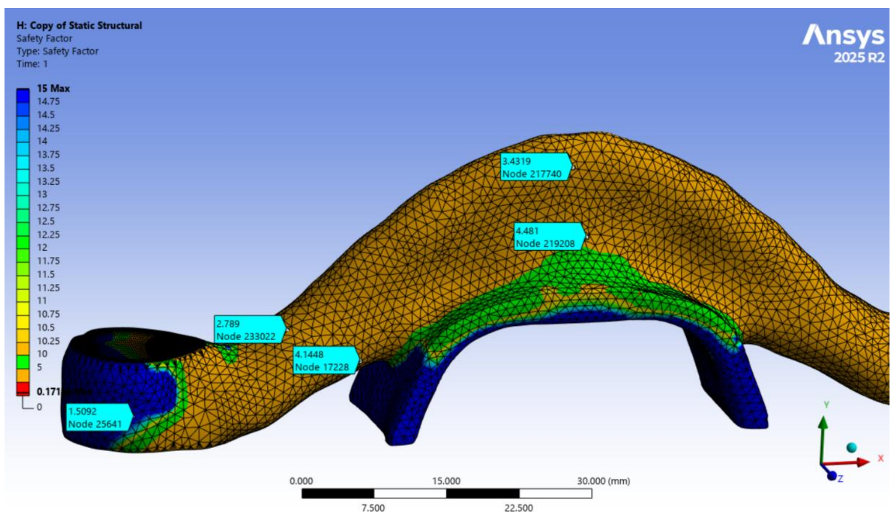


Fig. H-6: Factor of safety probes

2. Structural Testing

Objective

The objective of this test was to experimentally determine the ultimate load capacity and failure behavior of the SRAD recovery yokes under a load condition representing parachute shock forces, as well as to ultimately validate the finite element analysis simulations above. The test aimed to replicate the in-flight load path, where tensile force is applied through the yoke and reacted through the threaded rods.

The test article consisted of two specimens of the latest yoke design, and the assembly matched all critical dimensions of a real flight configuration.

Experimental Setup

Testing was conducted using an INSTRON 6800 Series Universal Testing System configured for axial tensile loading. A custom multi-part rig was developed to replicate flight conditions. The bottom fixture featured a grip holding a spare piece of Kevlar tether that attached to the bottom yoke. It is important to note that the Kevlar tethers used in this test are rated to withstand over 25.8 kN of force. The upper fixture was set up the same way, and the two yokes were joined together in the middle using the same washers, nuts, and steel threaded rods going through a spare bulkhead as pictured in Fig. H-7. Essentially, this setup tests the worst-case scenario without using the exact parts set aside for flight.

Fig. H-7: Tensile test setup

Method and Control Parameters

The test was configured and executed using INSTRON Bluehill Universal software in displacement control mode to ensure stable loading through the worst-case calculated parachute shock force. A customized method was used to measure force versus displacement up to a defined limit of 11 kN to compare against the absolute worst-case 8.035 kN calculated shock force value. This ensured stable loading through the region of interest without inducing unnecessary damage to the test article. Table H-1 includes the listed parameters of this specific test.

Table H-1: Test Parameters

Parameter	Value
Frame	Dual Column Floor Model
Load Cell	100 kN Capacity
Grips	Wedge Grips with V-Faces
Software	Bluehill Software
Control Mode	Displacement Control (Quasi-Static)
Output Channels	Load (N), Crosshead Displacement (mm)
Test Speed	10 mm/min
Preload	100 N
Test Limit	11,000 N
End Condition	Load Limit Reached

Results

As a result, the recovery yokes were tested in tension up to an 11kN load. The force-displacement curve remained as predicted, where the Kevlar cords would initially stretch and show a uniform curve up towards the load limit, with no plastic deformation shown in Fig. H-8. Data was acquired continuously throughout the test at a sufficient sampling rate to capture any slip or settling events, also seen through very minor force drops in the curve.

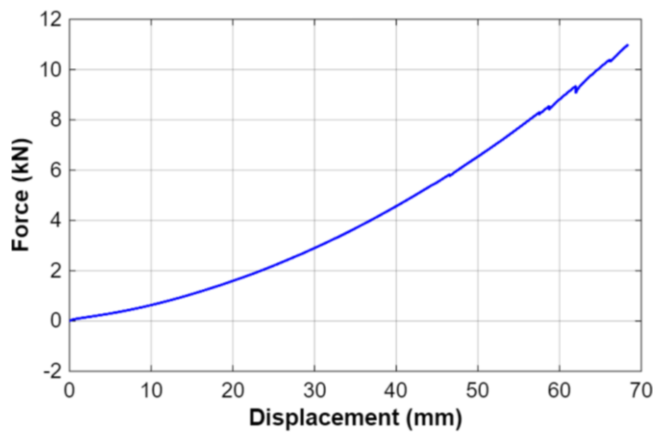


Fig. H-8: Recovery yoke force vs. displacement curve

Discussion

The custom fixture successfully reproduced the primary tensile load path experienced during parachute deployment. Each boundary condition was carefully assembled to provide the same realistic stress distribution that each yoke iteration aims to uphold. These results confirm that these new 316L stainless steel yokes remain fully within the elastic region at 11 kN with no yielding, giving a minimum FoS of 1.4. The small force dips are consistent with tether slip and settling and are not indicative of yoke deformation or damage. These findings correlate very well with the Ansys finite element simulations and validate our predicted elastic response.

Conclusion

Flight-representative tensile testing of the SRAD recovery yokes was conducted on an Instron Universal Testing Machine using a customized Bluehill Universal displacement-controlled method up to 11 kN. The yokes exhibited fully elastic behavior with no plastic deformation at a force over the expected shock load. These results validated the Ansys simulations, ensured the team stayed within the IREC requirements, and confirmed an exceptional structural performance for Omega's safe recovery.

B. Parachute Selection

Many factors were weighed in the selection of our main and drogue chutes' types and sizes. Following standard guidelines for a rocket of its category, the desired initial and final descent speeds were 26.0 m/s for the drogue and 6 m/s for the main chute for its final descent and impact speed with the ground. Knowing the descent speeds along with the atmospheric conditions, the theoretical main and drogue chute sizes are determined (refer to Equations 14, 15, and

16). With these theoretical values, a standard size and style of parachute can be chosen that meets these requirements. Refer to Tables H-2 and H-3.

$$A_m = \frac{2mg}{\rho C_D v_{impact}} - \frac{\pi D_r^2}{4} \tag{14}$$

$$D_r = \sqrt{\frac{8mg}{\pi r C_d v_d^2}} \tag{15}$$

$$D_m = \sqrt{\frac{4A_m}{\pi}} \tag{16}$$

Table H-2: Decision Matrix for Drogue Parachute Design Selection

Options:		**Annular**			**Parabolic**			**Elliptical**		
Wants Criteria	Weight Factor	Rating	Value	Weighted Score	Rating	Value	Weighted Score	Rating	Value	Weighted Score
Cost	4	7	$80	28	8	$56	32	7	$80	28
Low Pack Volume	4	8	8.6in^3	32	8	9 in^3	32	8	12 in^3	32
C_d	6	7	~1	48	7	0.97	42	9	1.5	54
Durability	7	7	Nylon	49	7	Nylon	49	7	Nylon	49
Stability	8	7	-	56	6	-	48	7	-	56
Total Score:	-	-	-	213	-	-	203	-	-	219

Table H-3: Decision Matrix for Main Parachute Design Selection

Options:		**Toroidal**			**Elliptical**			**Iris**		
Wants Criteria	Weight Factor	Rating	Value	Weighted Score	Rating	Value	Weighted Score	Rating	Value	Weighted Score
Cost	5	6	$420	30	7	$378	35	8	$260	40
Low Pack Volume	3	6	140 in^3	18	7	121 in^3	21	7	92 in^3	21
C_d	6	7	2.2	42	8	2.7	48	7	2.2	42
Durability	7	9	Ripstop	63	8	Ripstop	56	6	Nylon	42
Weight	6	7	25 oz	42	5	30 oz	30	7	18 oz	42
Total Score:	-	-	-	195	-	-	190	-	-	187

To narrow our options further, each design decision was weighed against another using decision matrices. For our parachutes, the matrix consisted of six styles of parachutes: annular, toroidal, disk gap band, ringsail, elliptical, and SkyAngle. An example of the top three options is shown in the matrix above. The list consists of various criteria (durability, stability, packing volume), each multiplied by a weight factor.

Durability is a measure of the parachute's ability to withstand wear and was designated as the most important criterion because if the parachute breaks, every other significant element of the chute is sabotaged. Additionally, failure prevention of this category is most optimal. Since we still opted to purchase a conventional chute with drogue dimensions, the style of the drogue chute needs to maximize its durability. Certain parachutes are designed for certain requirements. For example, disk-gap band parachutes were developed for high-altitude rocket applications and planet re-entry. On the other hand, parachutes such as ringsail often serve as escape system parachutes. For our system, we need a small parachute with high durability. The Toroidal and SkyAngle chutes ranked highest for main chutes, while elliptical chutes had the highest score for drogue chutes. Following the process, the Omega system would be determined to use a 96 in. (243.84 cm) Toroidal Rocketman High Performance Parachute for the main and a 24 in. (60.96 cm) Elliptical Apogee Parachute for the drogue.

C. Recovery Hardware

Based on the shock force, each component of recovery hardware was chosen. The priority of the components is that they will be able to withstand well over the predicted shock force loading. The parachutes are connected to the rocket via ½ inch (1.27 cm) tubular Kevlar cords, rated to withstand 30,027.2 N, with both the main and drogue bays utilizing 12.192 m length tethers. The 12.192 m drogue tether was chosen because of the added volume to the drogue bay itself, and to limit nosecone spinning when descending under drogue. It is generally reported that a tether length three to four times the length of the body of the rocket is recommended. This allows the parachute time to open before the shock cords are pulled downwards from the weight of the rocket. Ground testing further supports this decision, as it was evident that the cords provided sufficient length for full removal of each parachute, while still allowing extra length for the parachutes to open before any downward forces occurred. Kevlar cords were chosen over nylon cords for their high strength, low elasticity, and fire-resistant properties. The cords have three sewn loops, with one located approximately one-third of the length from one end. This middle loop will be our parachute attachment points for both the drogue and the main.

Each connection point will have a ¼ inch (0.635 cm) stainless steel quick link, which is rated at ~4450 N. The main and drogue parachutes are connected by stainless steel swivel links, which are rated at 6670 N. and 4448.22 N, respectively. Quick links were chosen because they serve as easy connection points for each component of our system. These components allow for rapid changes if there were to be an oversight in our initial design. These links will also allow for the quick removal of our entire recovery subsystem for transportation. Our tether system will be connected to the rocket via two 5/16" (5.87375 cm) forged steel eyebolts, located on the bottom of the payload and aft bulkheads, which are rated at 4000 N.

Instead of opting for traditional eyebolts or eye nuts, a custom recovery bar, also known as the "yoke," was designed by the team and professionally machined. This decision was made after considering all options and completing a decision matrix to narrow it down. This matrix is shown below in Table H-4. This hardware serves as connection points for both sets of tethers and the main deployment bag to be secured to each side of the avionics bay. This custom piece allows for efficient assembly of recovery components. Simulations were conducted on this custom hardware, and stainless steel 316L was chosen to be the primary material of the machined part. A factor of safety of greater than 1.5 was obtained through these simulations, and further ground testing was conducted to ensure the strength of the yoke during drogue and main deployment.

Table H-4: Primary Recovery Hardware Decision Matrix

Options:		**Forged Steel Eyebolt 1/4 in**			**Galvanized U bolt 3/8 in**			**Custom Tether Bar (Yoke)**		
Wants: Criteria	Weight Factor:	Rating	Value	Score	Rating	Value	Weighted Score	Rating	Value	Weighted Score
Cost	4	6	$3.00	24	5	$25	20	5	~$40	20
Weight	5	7	25g	35	5	120g	25	6	83.4g	30
Working Load	8	6	2225 N	48	8	6228 N	64	8	11000 N	64
Total Score:	-	-	-	107	-	-	109	-	-	114

D. Fiberglass Bulkhead Analysis

G10 Fiberglass is a composite material, which is a material produced from two or more constituent materials. These constituents have notably dissimilar chemical/physical properties and are merged to create a material with properties unlike the individual elements. Within the finished structure, the individual elements remain separate and distinct, unlike mixtures. This material serves as our bulkheads in the rocket, which are crucial structural points for our recovery system. The purpose of this report is to analyze the material properties of COTS G10 Fiberglass to perform better and more accurate simulations for future recovery designs.

Sample Sizing and Cutting

A 12" x 24", 1/4" Thick, Blue Flame-Retardant Garolite G-10/FR4 Sheet was bought from McMaster-Carr (85345K424). To gain the most accurate data possible, our team utilized two ASTM standards, D3039/D3039M – 17 standards for tensile testing and ASTM D790-17 for three-point bend testing. Due to the high risk of delamination that can come with manually cutting samples, our team utilized an OMAX Waterjet to cut out 4" x ½" x ¼" samples. Initially, the wrong thickness was input into the machine, resulting in three poor samples. These samples had extremely rough edges. We utilized these samples in initial testing to ensure the correct setup was used before we started to record data. There were three orientations of samples we had cut out of the G-10 fiberglass: 0°, 45°, and 90°. A visual of these orientations can be shown below.

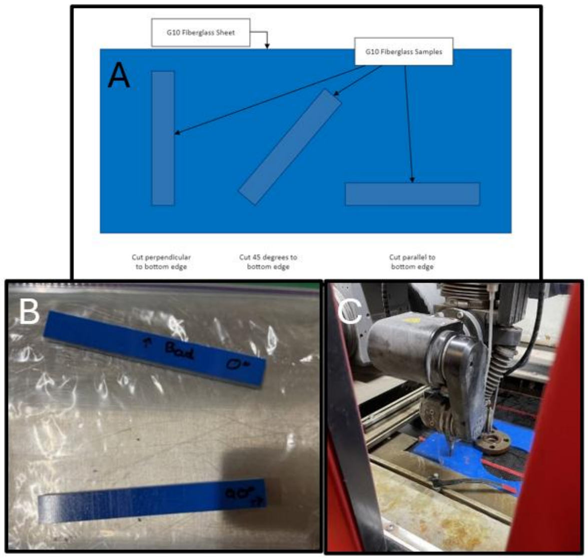


Fig. H-9: A) Orientation of Fiberglass samples. B) A bad sample (top) and a good sample (bottom). C) Samples being cut out with the waterjet

Tensile Testing

All tensile testing was done with an Instron 5982 load frame using two 2716-003 Manual Wedge Action grips for tensile testing, alongside ASTM D3039/D3039M – 17 standards. A crosshead rate of 2 mm/min was used for all tensile tests. Initially, a preliminary test was performed on a sacrificial sample to ensure optimal grip tension on the fiberglass surface. Despite our efforts, the sample fractured approximately 2mm below the top grip. Subsequent tests were conducted with varying grip strengths and alignment adjustments, yet yielded consistent outcomes. Due to concerns regarding potential stress concentration at the grips and its impact on data integrity, the results from these trials were excluded from all future analysis. Upon careful reflection on the overlooked issues detailed in the results section, our team decided to transition to a three-point bend test. This approach was deemed more reflective of the real-world stresses our bulkheads would endure within our recovery system. Moreover, it offers a more accurate simulation of the forces acting upon our bulkheads.

Three Point Bend Testing

The same Instron 5982 load frame was utilized alongside a 100 kN flexure fixture. In addition, the same sample sizing was utilized for simplicity and consistency for future testing. The flexure fixture had a separation distance of 100 mm, and the samples were placed centered on the center of the fixture. A maximum condition of 100 kN was applied; the samples tested never exceeded that value. Then, by using the control panel, the machine was moved until the top metal cylinder rested very lightly on the specimen. Subsequently, the machine's displacement was set to zero. After that, the test was started, and the machine measured variables such as flexural stress, strain, force, and displacement.

1. *Characterization*

A few selected samples, following the compression tests, were wet ground by hand using progressively finer SiC abrasive papers, culminating with 1200 grit. A Keyence VK-15000 optical microscope was used to image the sample surface and assess fiber structure and orientation in the post-bending test.

2. *Results*

Mechanical Property Comparison of Fiber Orientations

The following data was graphed to show the comparison of the sets of samples. Due to little variation in each test conducted, as shown in Fig. H-10 below, comparison between fiber orientations will be simplified to one sample in each orientation.

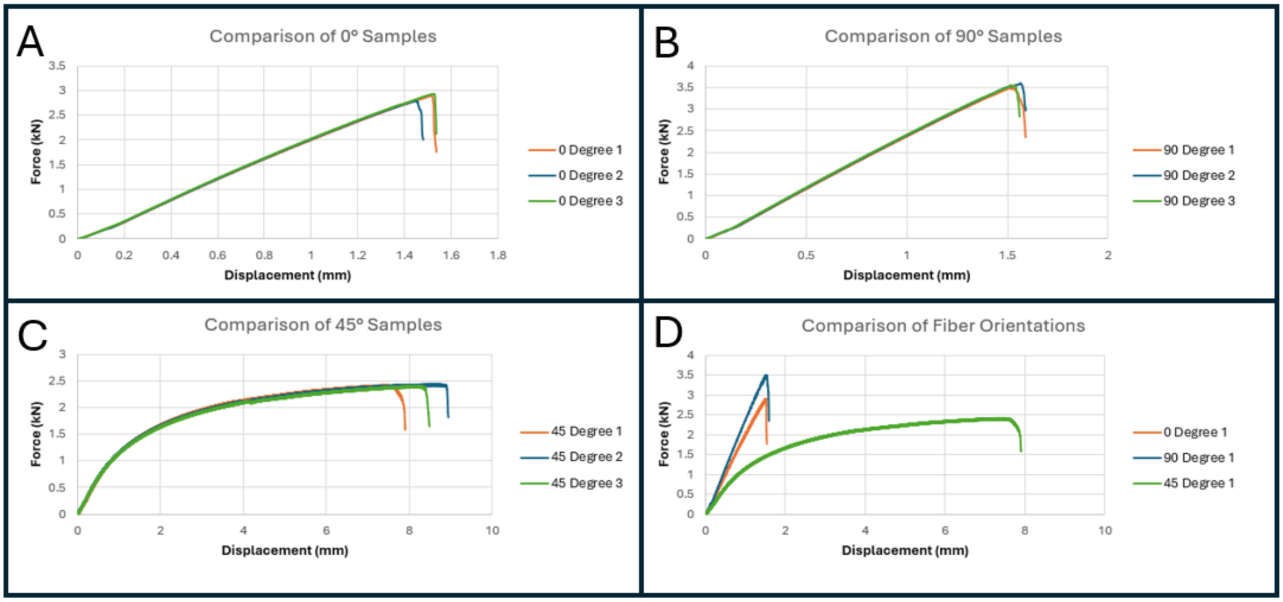


Fig. H-10: A) Comparison of 0-degree orientation; B) Comparison of 90-degree orientation; C) Comparison of 45-degree orientation; D) Comparison of all orientations

The specimens oriented at 0° exhibited an average displacement of approximately 1.4933 +/- 0.00 mm. Similarly, samples oriented at 90° displayed a comparable pattern, with an average displacement of 1.5385 +/0.026 mm. Notably, the 90° orientation demonstrated a significant increase in maximum force, measuring around 3.5307 +/- 0.028 mm. This represents a 23.36% rise from the maximum force recorded for the 0° orientation, which stood at 2.8622 kN. Conversely, specimens oriented at 45° showcased a substantial deviation from both the 0° and 90° samples. These exhibited a maximum force of only 2.4019 kN and a corresponding displacement of 8.0135 mm, representing a 16.0% decrease in maximum force and a 55.94% increase in displacement compared to the 0° orientation.

Table H-5: Comparison of Maximum forces and corresponding displacement for each sample tested

Orientation	Maximum Force (kN)	Corresponding Displacement (mm)
0° Sample 1	2.8921	1.5116
0° Sample 2	2.7752	1.4443
0° Sample 3	2.9194	1.5254
90° Sample 1	3.4882	1.5229
90° Sample 2	3.5791	1.5672
90° Sample 3	3.5248	1.534
45° Sample 1	2.4001	7.3579
45° Sample 1	2.4191	8.5577
45° Sample 1	2.3866	8.1249

The high variation between 0° and 90°, and 45° samples can be attributed to both the fiber structure and orientation of the fibers. As we can see below, the orientation of the fibers follows a 0° and 90° pattern, or a bidirectional pattern. Therefore, when the force was applied on the 0° and 90° orientations, the force was placed directly on the fibers, which caused a sudden break. On the other hand, the force applied to the 45° orientation did not exert direct pressure on the fibers, affording greater flexibility and leading to a significantly higher displacement, albeit with a lower overall maximum force.

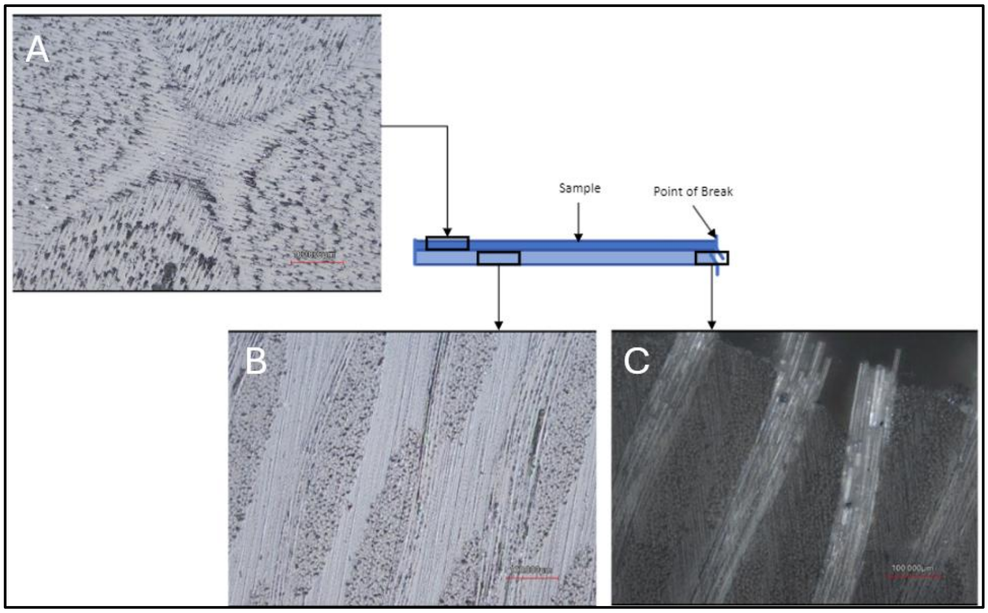


Fig. H-11: A) Polished top view of sample; B) Cross section view of 0° sample; C) Cross-sectional view of 0° sample break point

Our group also plotted our stress-strain curves for each sample. First, we calculated theoretical stress and strain values for our samples. Assuming we have a cross-sectional area of $A_o = 0.000080645m^2$ and an applied force of $F = 2862.233N$. We utilized the following equations to find the flexural stress of $\sigma_f = 396.39MPa$ and a flexural strain of $\epsilon_f = 0.025$.

$$\sigma_f = \frac{3FL}{2bd^2} \tag{17}$$

$$\epsilon_f = \frac{6Dd}{L^2} \tag{18}$$

The stress for all three orientations had a percent difference of ~1.35%, and the strain had a percent difference of only 0.695%.

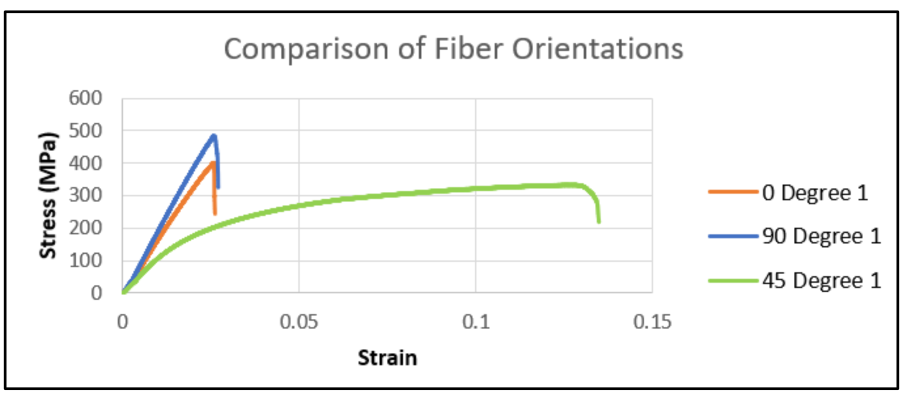


Fig. H-12: Comparison of all fiber orientations

Finding SolidWorks Data

With this information in mind, the data was analyzed using SolidWorks. SolidWorks Finite Element Analysis (FEA) simulations, on average, need three parameters for the simulation to function properly: Young's Modulus, Poisson's Ratio, and Yield strength [29]. Young's Modulus is the tensile or compressive stiffness when a force is applied lengthwise on a material. Poisson's ratio is the negative of the ratio of transverse strain to lateral or axial strain.

The yield strength represents the upper limit to forces that can be applied without producing permanent deformation in a material. For brittle materials, the yield strength is also defined to be where the sample breaks, since most brittle materials exhibit little to no deformation before failure. First, the raw data from the Instron was exported to Excel, so that it may be utilized for these calculations.

Young's Modulus

Due to our testing being accomplished via three-point bend tests, Young's Modulus has several equations associated with this test to account for the central deflection created. Young's Modulus can be calculated via the following equations, obtained from [30].

$$w_0 = \frac{PL^3}{48EI} \tag{19}$$

$$I = \frac{a^3 b}{12} \tag{20}$$

Where E is Young's Modulus, I is the second moment, P is the applied force, and L is the separation distance between the fixture points. By plugging in our values from our tests, Young's Modulus was found to be $E_0 = 15.32GPa$, $E_{90} = 17.55GPa$, and $E_{45} = 2.30GPa$.

Poisson's Ratio

Poisson's Ratio is calculated from tensile tests, using extensometers. Since the samples failed during tensile testing, Poisson's Ratio will be obtained from other research data. There were three main articles viewed to determine the best Poisson's ratio to use. Kasen et. Al. [31], Vogel et al. Al. [32], and Steinburg's novel [33], all tested the mechanical properties of woven G10 fiberglass, or fiberglass composites. [31] found a range of values, from 0.144 to 0.215. These values were taken as a comparison of temperatures, so it is best to choose the ratio found at room temperature, or 0.144. Values from [36] ranged from around 0.1 to a little under 0.5. Finally, Steinburg's novel [33] found Poisson's ratio to be 0.12. Because of the similarity found between the tested data in [31] and our data, .144 was chosen to be Poisson's ratio in the simulations. The minimum value was chosen as well, because if the simulations showed the bulkheads withstanding the forces at a lower Poisson's ratio, then it can be guaranteed they will withstand at higher rates.

Yield Strength

Yield strength occurred at the maximum stress in each sample, $\sigma_0 = 392.32MPa$, $\sigma_{90} = 470.30MPa$, and $\sigma_{45} = 320.02MPa$.

SolidWorks Modeling

The SolidWorks model included a very basic design and setup to prevent any major complications. There were several iterations that we went through to compare and provide the most accurate model. To start, a simple eyebolt in a bulkhead was utilized, and the system was slowly built into a full, accurate model. Throughout this iterative process, simulations were conducted with certain assumptions to streamline the model, deliberately omitting various factors. These assumptions will be rationalized as necessary. All bulkheads had a fixture on the outer diameter of the bulkhead, reflecting how our bulkhead is secured with epoxy inside the rocket. And all simulations were accomplished with SolidWorks Static simulations.

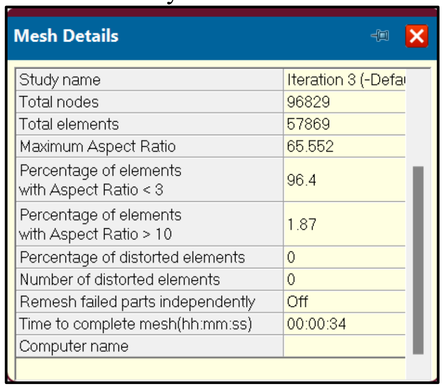
Shock Force

The force utilized in our simulations is also known as the shock force in rocketry. All simulations utilized a force of 2200N (or 500lbf). The shock force chosen was the average force across different parachute designs that was experienced over the years. This is what we had initially calculated from our shock force calculations for our main parachute. It was determined later that the actual force was lower than the initial calculation, meaning the bulkheads chosen are well within standards. The shock force can be calculated using the following equation:

$$F = \frac{1}{2} C_k (C_D S_0) \rho v^2 \tag{21}$$

Meshing

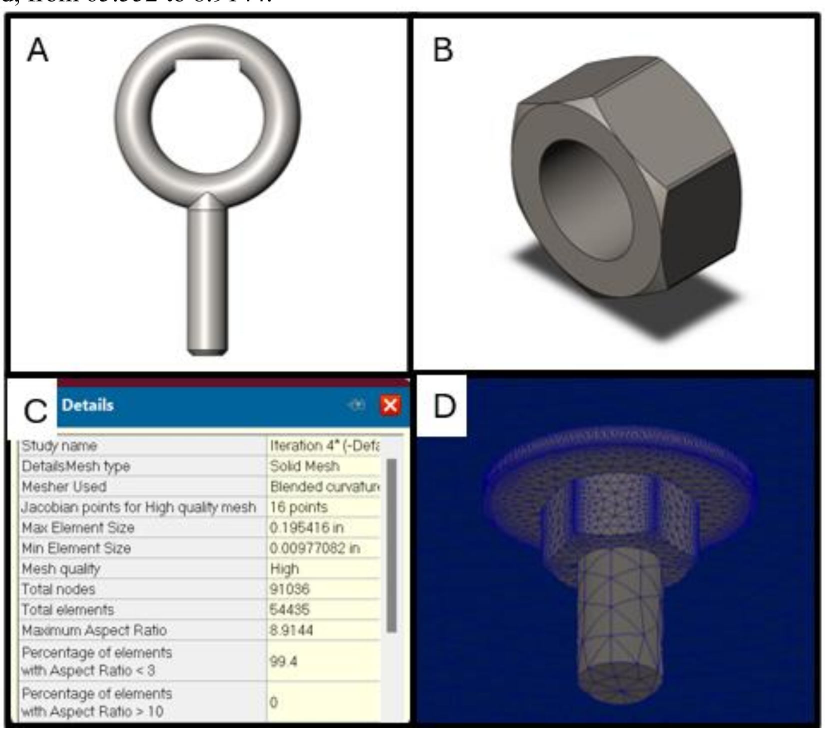
Initially, the eyebolt had a threaded part built into the model. However, this model had several elements with high aspect ratios. According to the SolidWorks website, it is advised for complex models to have a low maximum aspect ratio, alongside a percentage of at least 90% for the percentage of elements with an aspect ratio < 3. As seen in the image below, while there is a high percentage of elements with a low aspect ratio, the maximum aspect ratio was extremely high. This high ratio can affect the accuracy of the simulations to follow.


Mesh Details

Study name	Iteration 3 (-Defa
Total nodes	96829
Total elements	57869
Maximum Aspect Ratio	65.552
Percentage of elements with Aspect Ratio < 3	96.4
Percentage of elements with Aspect Ratio > 10	1.87
Percentage of distorted elements	0
Number of distorted elements	0
Remesh failed parts independently	Off
Time to complete mesh(hh:mm:ss)	00:00:34
Computer name	

Fig. H-13: Meshing details

To lower the maximum aspect ratio, it is often advised that certain structures of the model should be reduced or simplified. Structures such as fillets, chamfers, and even some threading can be removed to further simplify the model. As a test to determine the significance of threading, simulations were performed both with and without the threads involved. The models, after removing the threads, are shown below. It is important to note that the maximum mesh significantly decreased, from 65.552 to 8.9144.

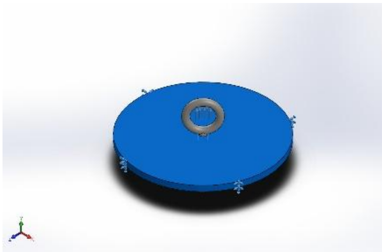
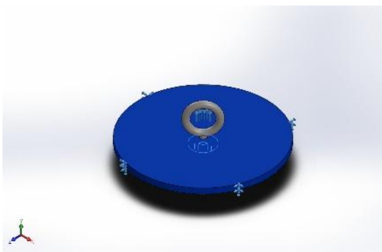
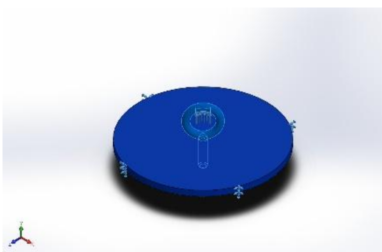


Study name	Iteration 4* (-Defa
DetailsMesh type	Solid Mesh
Mesher Used	Blended curvatur
Jacobian points for High quality mesh	16 points
Max Element Size	0.195416 in
Min Element Size	0.00977082 in
Mesh quality	High
Total nodes	91036
Total elements	54435
Maximum Aspect Ratio	8.9144
Percentage of elements with Aspect Ratio < 3	99.4
Percentage of elements with Aspect Ratio > 10	0

Fig. H-14: A) Eyebolt model; B) Eyenut model; C) Updated meshing; D) Meshing

The model of the forward bulkhead system consisted of an AISI 304 steel eyebolt, an alloy stainless steel washer and hex nut, and the custom G10 fiberglass bulkhead material. Due to the shape of the eyebolt, a small, flat section was cut into the underside of the bulkhead, where the force was applied. Variations of this cut section were made to test the different possibilities of how our force is applied in the system.

Table H-6: Material Details of Simulation Mode

Model Reference	Properties
	Name: G10 Fiberglass 0 Degree Model type: Linear Elastic Isotropic Yield strength: 3.92317e+08 N/m^2 Elastic modulus: 7.99e+09 N/m^2 Poisson's ratio: 0.12 Mass density: 1,851.09 kg/m^3
	Name: Alloy Steel (SS) Model type: Linear Elastic Isotropic Yield strength: 6.20422e+08 N/m^2 Tensile strength: 7.23826e+08 N/m^2 Elastic modulus: 2.1e+11 N/m^2 Poisson's ratio: 0.28 Mass density: 7,700 kg/m^3
	Name: AISI 304 Model type: Linear Elastic Isotropic Yield strength: 2.06807e+08 N/m^2 Tensile strength: 5.17017e+08 N/m^2 Elastic modulus: 1.9e+11 N/m^2 Poisson's ratio: 0.29 Mass density: 8,000 kg/m^3

The first simulation, with the force facing directly upward, yielded a displacement of 1.067mm with a minimum FoS of 0.49. While the FoS may seem extremely low, it is important to note that this occurs over an extremely small section of the eyebolt, as shown below. The stress and strain were also quantified. While it seems like the bulkhead will have an extremely high displacement, we can see through the stress and strain that this is not the case. In fact, the system experiences very little stress and strain, meaning it can easily withstand the force applied.

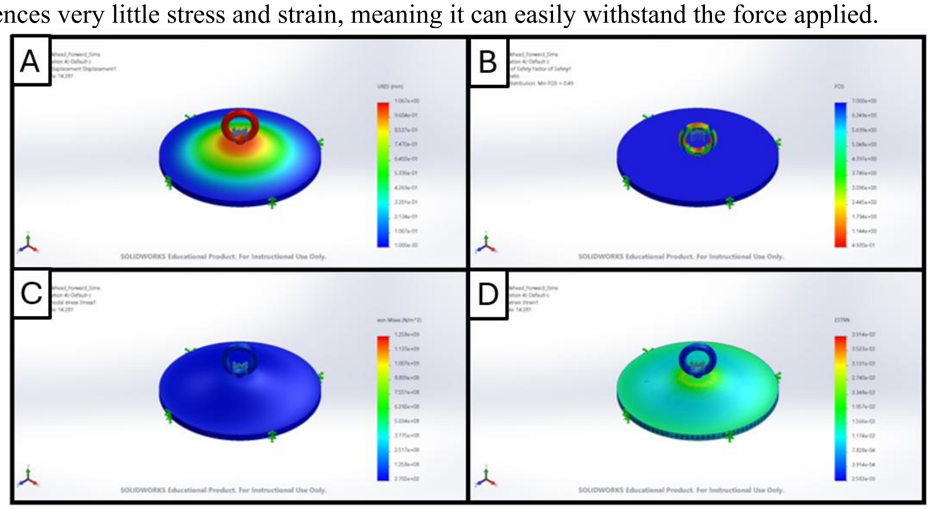


Fig. H-15: A) Displacement contour; B) Factor of safety contour; C) Stress contour; D) Strain contour

While this initial simulation was done with a perfectly vertical force, it is more realistic to assume the force can vary around the eyebolt. Therefore, to determine the worst-case scenario, the angle of force application was tested at additional angles of 45 degrees and 90 degrees (parallel to the bulkhead). This helped assess the severity of several situations and allowed our team to more accurately predict what forces our system experiences under all recovery events. The worst force possible occurred when the force was applied at a 90-degree angle from the bulkhead. The FoS on this simulation was 0.14 at a minimum and occurred at the bottom of the eyebolt.

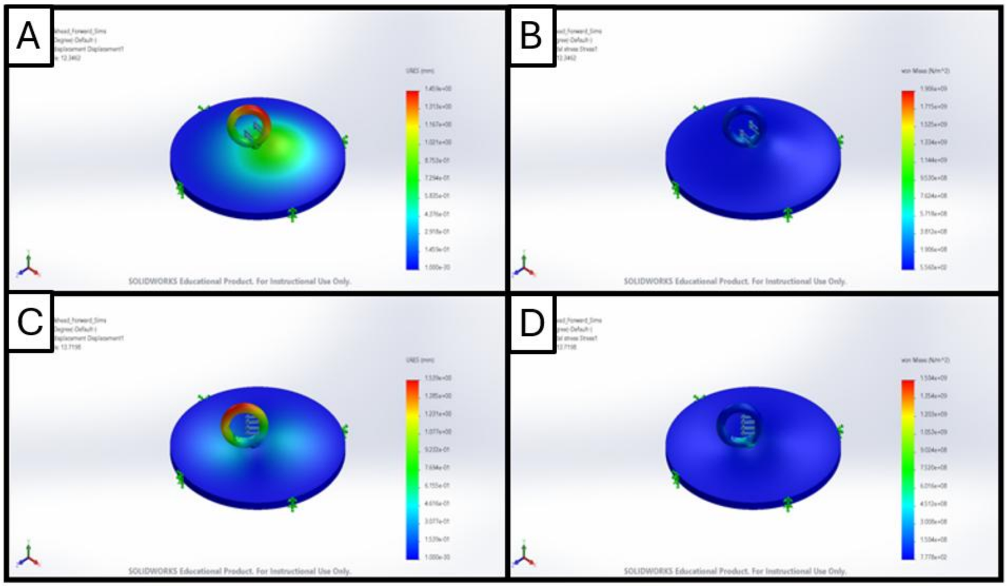


Fig. H-16: A) 45 degree displacement; B) 45 degree stress; C) 90 degree displacement; d) 90 degree stress

While this is an extremely low FoS, it is important to note one item. A high force occurring on one singular point in one singular direction is highly unlikely, as the rocket has free mobility in all planes of direction. Meaning, when the force is applied, the rocket will compensate accordingly. Instead, it is more realistic to assume the force will occur over time, over the entire circular area of the eyebolt. Therefore, the likelihood of the eyebolt ever experiencing a 90-degree force is nearly impossible.

Discussion

The small factor of safety occurs in small sections of the model and, notably, does not occur on the bulkheads themselves. While this may seem like a major potential concern, it is important to note that the eyebolts can easily be swapped out for higher durability/strength. The bulkheads, which are the focus of this paper, are permanently installed in the rocket and cannot be easily removed. The FEA simulations also assume that the bulkhead itself is permanently fixed in an upward position. However, realistically, the nosecone (from where the bulkhead is attached) has freedom in all directions. So as a force is applied, the nosecone will act accordingly, resulting in a much lower application of force on the eyebolt. Finally, while 90 degrees has the highest factor of risk, it is also impossible for this to occur in our subsystem, as the airframe is in the way, and thus the force can never directly be applied at this point. Based on these simulations, along with matching data from other sources, we can definitively say that we have confidence in our decreased bulkhead thickness and that the bulkheads will perform nominally in all scenarios.

Further Research

The simulation results offer a substantial degree of confidence in the structural integrity of our bulkhead against the forces exerted by our recovery system. However, additional testing is slated for the steel eyebolts to validate their capacity to withstand the recovery forces.

A notable aspect of this study pertains to the absence of experimentation with Poisson's ratio and the assumption of isotropic material. To enhance the quality of our simulations, comprehensive testing of all material properties is imperative. Given the diverse fabrication methods employed in composite materials, sourcing material properties from external references may introduce inaccuracies. Moreover, the presence of anisotropic properties can significantly influence experimental outcomes and necessitate meticulous consideration during testing. While SolidWorks enables simulation of anisotropic materials, achieving precise results across the x, y, and z directions demands extensive testing and experimentation.

An additional study for exploration involves conducting a tension test on a circular diameter section of G10, similar to our rocket bulkheads. This test aims to evaluate how a circular cross-section deforms under gradual force application. Building upon the earlier findings, it is hypothesized that the force initially acts directly upon the fibers, like the behavior observed in the 0° and 90° samples, before transitioning to a distributed force pattern resembling the behavior noted in the 45° samples.

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